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Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Entwurf und Implementierung eines Echtzeit-Impedanzreglers für einen 7-achsigen kollaborativen Roboter

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Declaration of Authorship

I hereby declare that this master's thesis entitled

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Munich, 30.09.2026 Heykel Khadhraoui

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ORANGE The argument and presentation are sufficient for the present thesis. Minor polishing may still be possible.

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The clearance gate holds the pose reached by the tool until operator confirmation. **comment:** "verify whether the same timing rule applies to the later gate"

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Abstract

ORANGE — sufficient. The abstract states the problem, implemented mechanism, principal measured ordering, null-space result, and scope limits without overloading the reader with the full parameter matrix.

Angular misalignment between a grinding tool and a surface can lead to edge-first contact and increased contact loads. In applications where the surface pose and tool mounting are subject to tolerances, the direction and magnitude of this misalignment are generally unknown before contact. This thesis investigates a Cartesian impedance control approach that uses the contact interaction itself to support the rotational alignment of the tool with a planar surface.

A real-time Cartesian impedance controller was implemented for a seven-degree-of-freedom (DOF) Franka Emika Panda. Translational and rotational compliance are defined relative to the surface geometry. The controller allows the virtual centre of compliance to be shifted away from the tool centre point (TCP). This shift introduces translation–rotation coupling into the Cartesian impedance, so that the commanded normal press can generate an additional moment that supports the required alignment rotation without prescribing an additional rotational trajectory.

The experimental investigation examined the influence of directional Cartesian stiffness and of the position, direction, definition frame, and magnitude of the virtual compliance-centre displacement. The compliance-centre position produced the largest response variation over the tested parameter ranges. An assisting tangential displacement lies perpendicular to the required rotation direction. Reversing the sign of the initial angular deviation requires the lever to be placed on the opposite side of the TCP, while changing the required tangent-plane rotation direction requires a corresponding change in the lever direction. This relation was also supported by tests at intermediate directions in the surface plane. About the weaker of the two principal alignment directions, a selected 40 mm

displacement changed the measured set-up rotation from $-1.63°$ to $+4.43°$.

Within the tested conditions, every non-zero tangential displacement that assisted alignment was therefore specific to the initial misalignment direction and sign. The TCP-centred condition introduces no preferred tangential direction and is the direction-independent fixed centre of compliance among the tested positions for the investigated task. A displaced centre can instead be selected when the required alignment direction is known and additional rotational alignment authority is needed.

A complementary null-space study showed that projected damping reduced redundant joint motion. Singular-value conditioning was active while the commanded disturbance was applied and strongly suppressed the resulting net redundant displacement under the tested disturbance direction. This response therefore represents disturbance rejection during its application rather than a subsequent return of the redundant configuration.

Kurzfassung

ORANGE — sufficient. The German summary remains aligned with the English abstract in structure, certainty, findings, and limitations.

Eine Winkelabweichung zwischen einem Schleifwerkzeug und einer Fläche kann zu Kantenkontakt und erhöhten Kontaktbelastungen führen. Bei Anwendungen, in denen die Lage der Fläche und die Werkzeugaufnahme Toleranzen aufweisen, sind Richtung und Größe dieser Winkelabweichung vor dem Kontakt in der Regel nicht bekannt. Diese Arbeit untersucht einen kartesischen Impedanzregelungsansatz, bei dem die Kontaktinteraktion selbst die rotatorische Ausrichtung des Werkzeugs an einer ebenen Fläche unterstützt.

Für einen 7-achsigen Franka Emika Panda wurde ein kartesischer Echtzeit-Impedanzregler implementiert. Die translatorische und rotatorische Nachgiebigkeit wird relativ zur Flächengeometrie definiert. Das virtuelle Nachgiebigkeitszentrum kann gegenüber dem Werkzeugbezugspunkt (TCP) verschoben werden. Dadurch entsteht eine Kopplung zwischen Translation und Rotation innerhalb der kartesischen Impedanz, sodass die kommandierte normale Anpressbewegung ein zusätzliches Moment erzeugen kann, das die erforderliche Ausrichtungsdrehung unterstützt, ohne eine zusätzliche rotatorische Trajektorie vorzugeben.

Experimentell wurden der Einfluss der richtungsabhängigen kartesischen Steifigkeit sowie Lage, Richtung, Bezugsrahmen und Größe der Verschiebung des virtuellen Nachgiebigkeitszentrums untersucht. Innerhalb der geprüften Parameterbereiche führte die Lage des Nachgiebigkeitszentrums zur größten Variation der Ausrichtungsreaktion. Eine unterstützende tangentiale Verschiebung liegt senkrecht zur erforderlichen Drehrichtung. Wird das Vorzeichen der anfänglichen Winkelabweichung umgekehrt, muss der Hebel auf die gegenüberliegende Seite des TCP gelegt werden. Ändert sich die erforderliche Drehrichtung in der Tangentialebene, muss sich entsprechend auch die Richtung des Hebels ändern. Dieser

Zusammenhang wurde ebenfalls für dazwischenliegende Richtungen in der Flächenebene bestätigt. Um die schwächere der beiden Hauptausrichtungsrichtungen änderte eine gewählte Verschiebung von 40 mm die gemessene Drehung während des Kontaktaufbaus von $-1{,}63°$ auf $+4{,}43°$.

Unter den untersuchten Bedingungen war damit jede nicht verschwindende tangentiale Verschiebung, die die Ausrichtung unterstützte, von Richtung und Vorzeichen der anfänglichen Winkelabweichung abhängig. Die TCP-zentrierte Einstellung gibt keine bevorzugte tangentiale Richtung vor und stellt unter den getesteten Lagen das richtungsunabhängige feste Nachgiebigkeitszentrum für die untersuchte Aufgabe dar. Ein verschobenes Nachgiebigkeitszentrum kann dagegen gezielt eingesetzt werden, wenn die erforderliche Ausrichtungsrichtung bekannt ist und zusätzliche rotatorische Ausrichtungswirkung benötigt wird.

Eine ergänzende Nullraumuntersuchung zeigte, dass die projizierte Dämpfung redundante Gelenkbewegungen reduzierte. Die Singularwert-Konditionierung war während der kommandierten Störung aktiv und unterdrückte unter der untersuchten Störungsrichtung die daraus resultierende Netto-Verschiebung im Nullraum deutlich. Dieses Verhalten beschreibt somit eine Unterdrückung der Störwirkung während ihrer Einwirkung und keine nachträgliche Rückführung der redundanten Konfiguration.

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List of Abbreviations

ORANGE — sufficient. The abbreviations are separated from the symbol list and their printed forms are controlled centrally.

CoC	Centre of Compliance
CSV	Comma-Separated Values
DH	Denavit–Hartenberg
DOF	Degree of Freedom
EE	End-Effector
E-stop	Emergency Stop
FCI	Franka Control Interface
FCU	Franka Control Unit
F/T	Force/Torque
IP	Internet Protocol
LDLT	Lower–Diagonal–Lower-Transpose Factorisation
OS	Operating System
PREEMPT_RT	Real-Time Preemption Patch
ROS	Robot Operating System
SE(3)	Special Euclidean Group in three dimensions
SO(3)	Special Orthogonal Group in three dimensions
SVD	Singular Value Decomposition
TCP	Tool Centre Point

List of Symbols

ORANGE — sufficient. The notation is extensive but well organised, and the unit column makes the mixed translational and rotational quantities traceable.

Frames and Kinematics

$\{0\}$	Robot base frame	[-]
$\{i\}$	Joint frame i	[-]
$\{\mathrm{EE}\}$	End-effector frame used by the robot model	[-]
$\{d\}$	Desired end-effector and TCP reference frame	[-]
$\{S\}$	Surface task frame, with axes t_1, t_2 and n_s	[-]
n	Number of joints ($n = 7$)	[-]
$q,\ \dot{q}$	Joint position / velocity vector	[rad], [rad/s]
q_{init}	Initial joint configuration used by the controller and pose-hold study	[rad]
p	Position / translation vector	[m]
R	Rotation matrix	[-]
$R_x,\ R_y,\ R_z$	Elementary rotations about x, y, z	[-]
$\alpha,\ \beta,\ \gamma$	Generic rotations about the x, y, z axes	[rad]
$\alpha_s,\ \beta_s$	Configured surface rotations about the base-frame x- and y-axes	[rad]

e_z	Base-frame unit vector along the z-axis	[-]
T	Homogeneous transformation matrix	[-]
$J(q)$	Geometric Jacobian; linear rows above angular rows	[m/rad], [-]
z_{i-1}	Rotation axis of joint i, expressed in the base frame	[-]
p_{i-1}	Point on the axis of joint i, expressed in the base frame	[m]
$x_{\mathrm{EE}}, y_{\mathrm{EE}}, z_{\mathrm{EE}}$	End-effector frame axes (base frame)	[-]

Pose and Error

$p_{\mathrm{EE}},\ p_{\mathrm{TCP}}$	End-effector / tool-centre-point position; $p_{\mathrm{TCP}} = p_{\mathrm{EE}}$ in the implemented model	[m]
$R_{\mathrm{EE}},\ T_{\mathrm{EE}}$	End-effector orientation / transformation	[-]
$p_d,\ R_d,\ T_d$	Desired position / orientation / transformation	[m], [-]
T_{err}	Relative (error) transformation	[-]
ΔR	Current-to-desired relative rotation, $R_{\mathrm{EE}}^{\top} R_d$	[-]
ϕ	Orientation-error angle	[rad]
$u,\ [u]_{\times}$	Current-EE-frame orientation-error axis / its skew matrix	[-]
e_p	Translational (position) error	[m]
e_R	Base-frame rotational error used by the command	[rad]
$e_{R,\mathrm{EE}}$	Rotational error expressed in the current EE frame	[rad]
$e_{p,c},\ e_c$	Compliance-centre position error / stacked error	[m], [rad]
e	Stacked Cartesian error	[m], [rad]
$\dot{x}$	Cartesian velocity (twist)	[m/s], [rad/s]
$\dot{p}_{\mathrm{EE}},\ \dot{p}_d,\ \omega_{\mathrm{EE}}$	Measured linear end-effector velocity, desired linear velocity, and angular end-effector velocity	[m/s], [m/s], [rad/s]

Cartesian Impedance Gains

$K_p,\ D_p$	Generic translational stiffness / damping block; a frame or phase subscript is added when needed	[N/m], [N s/m]
$K_R,\ D_R$	Generic rotational stiffness / damping block; a frame or phase subscript is added when needed	[N m/rad], [N m s/rad]
$K_{p,t_1},\ K_{p,t_2},\ K_{p,n}$	Translational stiffness entries along t_1, t_2, and n_s	[N/m]
$K_{R,t_1},\ K_{R,t_2},\ K_{R,n}$	Rotational stiffness entries about t_1, t_2, and n_s	[N m/rad]
$R_{\text{task}},\ T_R$	Generic task-frame orientation and corresponding block rotation; $R_{\text{task}} = R_{\text{surface}}$ for the implemented task	[-]
$K_{p,\text{task}},\ D_{p,\text{task}}$	Local translational stiffness / damping in the chosen task frame	[N/m], [N s/m]
$K_{R,\text{task}},\ D_{R,\text{task}}$	Local rotational stiffness / damping in the chosen task frame	[N m/rad], [N m s/rad]
$K_{p,0},\ D_{p,0}$	Translational stiffness / damping used in the base-frame command	[N/m], [N s/m]
$K_{R,0},\ D_{R,0}$	Rotational stiffness / damping used in the base-frame command	[N m/rad], [N m s/rad]
$K_0,\ D_0$	Full 6×6 stiffness / damping used in the base-frame command	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$G_{\text{surface}},\ G_0$	Generic diagonal directional gain block along $[t_1, t_2, n_s]$ / its base-frame representation	[N/m] or [N m/rad]

Dynamics and Torques

F	Cartesian wrench	[N], [N m]
$f,\ m$	Commanded Cartesian force / moment	[N], [N m]
$f_K,\ m_{c,K}$	Elastic commanded force component, $f_K = K_p e_p$ / its compliance-centre coupling moment, $m_{c,K} = r_c \times f_K$	[N], [N m]
$K,\ D$	Full Cartesian stiffness / damping matrices	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$\Lambda_0(q),\ \Lambda_{\text{task}}(q)$	Operational-space (Cartesian) inertia matrix in the base frame / resolved in the active gain frame	[kg], [kg m], [kg m^2]
$M(q)$	Joint-space mass matrix	[kg m^2]
ζ	Directional coefficient in the inertia-scaled damping calculation	[-]
τ	Joint torque vector	[N m]
$\tau_c,\ \tau_g$	Coriolis / gravity compensation torque	[N m]
τ_{cart}	Cartesian impedance torque, $J^\top(q)F$	[N m]
τ_{cmd}	Complete commanded joint torque	[N m]
$\tau_{\text{guide}},\ d_{\text{guide}}$	Manual-guidance joint torque / its joint-space damping coefficient	[N m], [N m s/rad]

Surface and Contact Geometry

n_s	Calibrated surface-plane normal copied into the controller and used in the evaluation	[-]
$t_1,\ t_2$	Surface tangent directions (unit)	[-]
$a_s,\ \tilde{t}_1$	Configured base-frame reference direction fixing t_1 within the surface plane / projected tangent before normalisation	[-]
R_{surface}	Surface task frame $[\,t_1\ \ t_2\ \ n_s\,]$ in the base frame	[-]
$p_{\text{cal}},\ n_{\text{cal}}$	Tool-face centre and surface normal obtained from a seated pose, both in the base frame	[m], [-]
$R_{\text{cal},i}$	Rotation of a captured tool-calibration pose	[-]
$\bar{n}_B$	Invariant base-frame direction in the tool-normal calibration	[-]
$\varepsilon_{\text{plane}},\ \varepsilon_g$	Surface-plane re-seating residual / tool-face feature-selection tolerance	[m]
ψ_{plane}	Angle between the re-seated and the stored surface normal	[rad]
$\Delta\alpha_s,\ \Delta\beta_s$	The same re-seating comparison read per axis, as differences from the stored α_s and β_s	[rad], [rad]
$n_{\text{Tool,EE}},\ n_{\text{Tool,0}},\ n_d$	Tool-face normal expressed in the EE frame, fixed by calibration / the same direction expressed in the base frame, $n_{\text{Tool,0}} = R_{\text{EE}} n_{\text{Tool,EE}}$ / signed controller target direction	[-]

θ_{tilt}, ϕ_{tilt}, u_{tilt}	Commanded orientation-offset rotation vector $\theta_{t_1} t_1 + \theta_{t_2} t_2$ / its magnitude / its unit axis	[rad], [rad], [-]
R_{tilt}	Rotation built from $(u_{\text{tilt}}, \phi_{\text{tilt}})$ and applied to n_s to give n_d	[-]
s_a	Tool-axis target sign, with $n_d = s_a n_s$	[-]
p_s	Point on the configured surface	[m]
p_c	Centre of compliance	[m]
r_c	TCP-to-compliance-centre lever, $p_c - p_{\text{TCP}}$, as it enters the point-shift transformation and as the experimental plots and tables report it	[m]
r_{c,t_1}, r_{c,t_2}, $r_{c,n}$	Surface-frame components of r_c along t_1, t_2 and n_s	[m]
$r_{\text{face,EE}}$	Tool-face centre offset in the EE frame	[m]
$r_{\text{width,EE}}$, $r_{\text{length,EE}}$	Tool face half-width / half-length vector in the EE frame	[m]
$r_{T,\text{EE}}$, p_T	Selected tool-point offset from the TCP, expressed in the EE frame / selected tool-point position in the base frame	[m]
r_T	Selected tool-point offset from the TCP, $p_T - p_{\text{TCP}}$	[m]
R_{clr}, $r_{T,\text{clr}}$	Orientation captured at the clearance transition and held through set-up / the stored tool-point offset expressed in the base frame at that orientation	[-], [m]

$p_{T,0},\ p_{T,d}(t),\ s_{\mathrm{set}}(t)$	Projection of the selected tool point onto the calibrated plane / its commanded position during set-up / the commanded press coordinate	[m], [m], [m]
m_0	Commanded rotational impedance moment at the TCP, independent of r_c	[N m]
m_{r_T}	Geometry-based force-lever moment about the TCP, reconstructed as $r_T \times \Delta\hat{f}_{\mathrm{ext}}$	[N m]

Compliance and Alignment

$K_{\text{task}},\ D_{\text{task}}$	Full local Cartesian stiffness / damping in the chosen task frame	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$\mathrm{Ad}(r_c)$	TCP-to-compliance-centre point-shift adjoint	[-]
K_c, D_c	Block-diagonal stiffness / damping at the centre of compliance	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$F_c,\ F_{\text{TCP}}$	Cartesian wrench at the centre of compliance / corresponding wrench at the TCP	[N], [N m]
$K_{\text{TCP}}, D_{\text{TCP}}$	Centre-of-compliance gains mapped by congruence to the TCP	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$\theta_{\text{align}},\ \Delta\theta_{\text{align}}$	pose-based tool-axis alignment angle / before–after improvement; the indices before and after denote the start and end of set-up	[rad]
$\theta_{t_1},\ \theta_{t_2}$	Surface-frame components of the commanded tool orientation offset	[rad]
$\Delta\theta_{\text{set}},\ \Delta\theta^{S}_{\text{set}}$	Finite set-up rotation in the base frame / resolved in the surface frame; $\Delta\theta_{\text{set},t_i}$ is its component about t_i	[rad]
$\phi_{\text{set}},\ u_{\text{set}}$	Set-up rotation angle / unit rotation axis	[rad], [-]
$R_{\text{before}},\ R_{\text{after}}$	End-effector orientation at the clearance transition / at the end of set-up	[-]
$r_{c,t},\ \rho_c$	Selected tangential compliance-centre lever / its magnitude	[m]
$\Delta p_T,\ s_T$	Selected tool point displacement / its magnitude	[m]

$\Delta p_{\mathrm{TCP}},\ s_{\mathrm{TCP}}$	TCP displacement / its magnitude	[m]
$\theta_{\mathrm{dev}},\ \theta^S_{\mathrm{dev}}$	Rotation vector taking $n_{\mathrm{Tool},0}$ onto the flat target $-n_s$ / the same vector resolved on the surface axes; $\lVert\theta^S_{\mathrm{dev}}\rVert = \theta_{\mathrm{align}}$	[rad], reported in [deg]
$\theta_{\mathrm{dev},t_i,\mathrm{before}}$	Signed surface-frame component of θ^S_{dev} at set-up entry; opposite in sign to the commanded orientation offset	[rad], reported in [deg]
F_n	Positive normal magnitude of the model-estimated external-force change from the clearance transition, $\lvert n_s^\top \Delta\hat{f}_{\mathrm{ext}}\rvert$	[N]
$F_{n,\mathrm{cmd}},\ M_{t_i,\mathrm{cmd}}$	Signed commanded normal force / commanded TCP moment about surface tangent t_i	[N], [N m]
$F_{K,n}$	Magnitude of the commanded elastic normal component, $F_{K,n} = -n_s^\top f_K$	[N]
$F_{n,\mathrm{block}}$	Fully blocked normal virtual-spring command scale	[N]
$\hat{f}_{\mathrm{ext}},\ \hat{f}_{\mathrm{ref}}$	Model-estimated external force / clearance-transition reference	[N]
$\Delta\hat{F}_{\mathrm{ext}}$	Model-estimated external-wrench change relative to the clearance-transition reference	[N], [N m]
$\Delta\hat{f}_{\mathrm{ext}},\ \Delta\hat{m}_{\mathrm{ext}}$	Force part / moment part of $\Delta\hat{F}_{\mathrm{ext}}$	[N], [N m]
f_D	Damping commanded force component, $f = f_K + f_D$	[N]
$m_{c,D}$	Coupling moment of the damping component, $r_c \times f_D$	[N m]

$m_{c,\mathrm{trans}}$	Complete translational coupling moment, $m_{c,K} + m_{c,D} = r_c \times f$	[N m]

Quasi-Static Stiffness and Damping Design

k_i	Directional translational or rotational stiffness	[N/m] or [N m/rad]
$\lambda_i(q)$	Directional Cartesian inertia used for damping selection, $[\Lambda_{\mathrm{task}}(q)]_{ii}$	[kg] or [kg m^2]
$d_i(q)$	Directional damping coefficient, $2\zeta\sqrt{\lambda_i(q)k_i}$	[N s/m] or [N m s/rad]
$Y(q)$	Solution matrix of $M(q)Y(q) = J^\top(q)$, used instead of forming $M(q)^{-1}$	[kg^{-1} m/rad], [kg^{-1}]
ε	Regularisation added before inverting the operational-space inertia (10^{-9})	[-]

Null-Space and Singular Value Decomposition

J^+	Thresholded-SVD Moore–Penrose pseudoinverse	[rad/m], [-]
$N_q,\ N_\tau$	Joint-velocity / torque null-space projector	[-]
$\rho,\ \sigma_{\text{thr}}$	Relative SVD tolerance / resulting singular-value threshold; the threshold follows the scaling of σ_i below	[-], [m/rad], [-]
$\sigma_i,\ \sigma_{\min}$	Jacobian singular value / σ_6 conditioning indicator. The geometric Jacobian mixes linear and angular rows, so a singular value carries no single physical unit and its numerical value depends on the chosen Jacobian representation and length unit; it is used as a relative indicator under one fixed convention	[m/rad], [-]
$U_J,\ \Sigma_J,\ V_J$	Singular value decomposition factors of J; Σ_J follows the scaling of σ_i	[-], [m/rad] and [-], [-]
$c,\ c_i$	Joint-velocity coefficients in the right-singular-vector basis / coefficient for direction i	[rad/s]
$u_i,\ v_i$	Left / right singular vectors	[-]
$v_6,\ v_7$	σ_6-associated direction / structural null direction	[-]
$\dot{q}_{\text{null}}$	Projected joint velocity in the numerical null space	[rad/s]
F_d	Commanded point-force-equivalent disturbance magnitude	[N]

$r_p,\ J_p(q)$	Point on the robot link used for the commanded disturbance / its translational Jacobian	[m], [m/rad]
$u_f,\ f_d(t),\ s_d(t)$	Fixed disturbance-force direction / commanded disturbance-force vector / dimensionless time profile	[-], [N], [-]
τ_{dist}	Joint-torque-equivalent commanded disturbance	[N m]
d_{null}	Null-space damping	[N m s/rad]
k_σ	Sigma-conditioning torque magnitude	[N m]
α_{probe}	Null-direction sampling angle	[rad]
$q_+,\ q_-$	Joint configurations sampled at $q \pm \alpha_{\text{probe}} v_7$	[rad]
$\sigma_+,\ \sigma_-$	Minimum Jacobian singular values evaluated at q_+ and q_-	[m/rad], [-]
$\delta_\sigma,\ \Delta_\sigma$	Sigma-comparison deadband / sampled difference $\sigma_+ - \sigma_-$ used by the controller's direction selector	[m/rad], [-]
$\Delta\sigma_{\text{min,dist}}$	Experimental change in σ_{min} over the disturbance interval, $\sigma_{\text{min}}(9\,\text{s}) - \sigma_{\text{min}}(5\,\text{s})$; distinct from the controller's Δ_σ	[m/rad], [-]
s_σ	Deadbanded singular-value-conditioning direction selector	[-]
$\tau_d,\ \tau_\sigma,\ \tau_{\text{null}}$	Projected damping / sigma / total null-space torque	[N m]

E_N	Cumulative projected null-space motion; integrated projected joint-velocity magnitude during the disturbance interval, reported in degrees	[rad]
v_{ref}	Fixed redundant comparison direction, taken from the condition without null-space torque	[-]
$\Delta q_{\mathrm{null}},\ \Delta\eta_{\mathrm{dist}}$	Net projected joint displacement over the disturbance interval / its signed component along v_{ref}	[rad]

Miscellaneous

$I,\ I_3,\ I_4,\ I_6,\ I_7$	Identity matrices	[-]

Chapter 1

Introduction

1.1 Motivation

Industrial manipulators commonly use position control to track commanded trajectories. In surface-finishing applications, placement tolerances can cause the physical surface pose to differ from its programmed pose. The surface is positioned manually, and the tool is held in a gripper. Their residual angular misalignment is therefore unknown before contact. Rigid position control resists the corresponding corrective motion and can generate high contact loads.

Torque-controlled collaborative robots allow contact compliance to be specified directly and account for a growing share of industrial deployments [14, 26]. Joint torque sensing and a torque-level command interface determine how the robot yields under contact as well as where it moves. In surface finishing, this capability can reduce the loads caused by angular tool–surface misalignment. It can also permit the tool face to rotate towards the surface while maintaining the commanded normal press. The resulting alignment is evaluated relative to the physical surface.

Impedance control specifies this yielding behaviour as a dynamic relation between motion and interaction force [13]. Cartesian impedance control defines the relation for the end-effector pose. Its translational and rotational compliance can therefore be assigned independently along selected task directions. During edge-first contact, the wrench contains both force and moment components. Gains defined from the contact geometry allow part of this moment to rotate the tool towards the surface. The stiffness and damping matrices consequently determine both the contact response and the free-space tracking behaviour.

1.2 Related Work

The relevant literature comprises three areas: Cartesian impedance control, real-time implementation on torque-controlled robots, and the placement of the centre of compliance (CoC). The first area is Cartesian impedance control itself. The controller uses standard serial-manipulator kinematics [23, 4, 19]: homogeneous transformations for representing the end-effector pose and the Jacobian for relating joint motion to Cartesian motion. Khatib established the operational-space formulation, in which a Cartesian wrench F is mapped to joint torques τ through $J^\top$ [15]; this mapping is adopted in the controller. Hogan defined the impedance relation between end-effector motion and interaction force [13]. Albu-Schäffer et al. implemented Cartesian impedance control using joint-torque sensing [1], which corresponds to the torque-controlled interface used in this thesis.

The second area concerns the real-time implementation of Cartesian impedance control on torque-controlled robots. Mayr et al. implemented a modular Cartesian impedance controller for the Franka Emika Panda that combines real-time model access, null-space control, signal treatment, and command limiting [17]. The controller developed in this thesis uses the libfranka model interface to obtain the required robot state and model quantities during the real-time control loop. It also includes a selectable null-space torque τ_{null} for influencing the redundant joint motion without changing the commanded Cartesian task.

Recent work has applied compliant and force-controlled strategies directly to robotic surface finishing. Alt et al. combined perception, planning, and force-controlled execution for robotic surface treatment [2]. Min et al. integrated demonstrated motion and force skills with impedance control on a 7-DOF collaborative robot [18]. Wu et al. used adaptive variable impedance control to regulate the contact force during robotic grinding [29]. These studies address force-controlled surface treatment. The present work instead examines contact-induced angular alignment through directional Cartesian stiffness and a virtual CoC.

The third area concerns the location of the CoC. Fasse and Broenink formulated spatial impedance about a selected point, so that translation and rotation become coupled through the position of this point [7]. This formulation provides the theoretical basis for the compliance-centre shift used in this thesis. The same principle appears mechanically in remote-centre-compliance devices, in which elastic elements are arranged so that the tool behaves as if it rotates about a virtual point located away from the mechanism itself [20, 28]. More recent mechanically compliant and variable-compliance-centre strategies apply this principle to peg-in-hole insertion [27, 12]. In the mechanically compliant mechanisms, elastic bending produces the alignment motion rather than software control alone. In these approaches, the compliance is arranged so that contact forces reduce misalignment instead of increasing it. Most of these studies focus on insertion tasks and define the compliance centre through either the mechanical design or a strategy adapted to the hole geometry. In this thesis, an equivalent effect is created in software by shifting the virtual CoC away from the TCP. The independent effects of this position and the axis-specific stiffness during planar tool alignment are then investigated on a torque-controlled robot.

Suomalainen et al. experimentally investigated how compliance-centre placement and compliant-axis selection influence alignment in an assembly task [25]. Friedrich et al. subsequently proposed an active remote CoC that combines passive compliance with active force control [11]. These approaches further illustrate the role of compliance location in contact alignment. The present work examines a software-defined virtual compliance centre for planar tool–surface alignment.

1.3 Problem Statement

The angular relation between the tool and the surface plane is uncertain before contact because the surface placement and tool mounting both carry tolerances. The sign of the initial angle and its tangent-plane rotation direction are therefore unknown when a run starts. Under rigid position control, the contact geometry can generate high moments while the controller resists the corrective motion. In contrast, finite rotational compliance allows contact-induced rotation, whereas a displaced CoC adds a selectable coupling between translation and rotation.

The lever r_c is the displacement from the TCP to the virtual CoC. A tangential lever enters the commanded wrench through a cross product with the translational force and thereby selects an additional rotational moment. The direction and sign of this moment depend on the lever direction. A displaced centre chosen before contact therefore encodes a preferred alignment direction.

The investigation evaluates a fixed CoC for conditions in which the sign and tangent-plane direction of the initial angular misalignment are unknown. In this thesis, a direction-independent fixed centre denotes a centre position that can be selected without this prior information. The assessment is limited to the investigated task, parameter range, and tested centre positions. The term describes direction independence of the selected reference and carries no claim of global optimality.

Contact-induced alignment forms the set-up stage of the intended surface task. The tool approaches the surface with an angular offset, establishes contact, and rotates under the compliant interaction. Sustained tangential motion would follow after alignment. However, the reported measurements cover the contact set-up stage and end at the pre-grinding gate.

The alignment response depends on the directional stiffness, contact geometry, lever direction, robot configuration, and compliance of the tool mounting. The experiments therefore vary these controller parameters under otherwise matched conditions. A commanded tool orientation offset represents the initial angular misalignment between the tool and the surface. Chapters 2 and 3 introduce the frame conventions and controller formulation used in the experiments.

The experimental sequence follows four steps:

1. The TCP-centred condition is measured first. Here $r_c = 0$ and the virtual point shift contributes no coupling moment.
2. The rotational stiffness about the commanded rotation axis and the translational stiffness perpendicular to it are varied to quantify their influence on the direction-dependent response.
3. The direction, sign, and magnitude of the tangential compliance-centre lever are varied across commanded rotation axes, directions, and offset angles.
4. The geometric lever-selection rule is evaluated at tangent-plane directions between the two principal surface tangents and compared with one fixed lever.

This sequence separates the response already present at the TCP from the additional effect introduced by a displaced centre and then tests the geometric dependence of that displacement.

1.4 Scope and Contributions

The experiments evaluate the influence of the commanded tool orientation offset, surface-frame stiffness, and virtual CoC on contact-induced alignment. The controller contribution and the experimental contribution are treated separately.

The implemented controller is a phase-scheduled Cartesian impedance controller written in C++ using libfranka and Eigen. It runs in the 1 kHz torque-control loop of the Franka Emika Panda. The commanded Cartesian wrench F is mapped to the Cartesian impedance torque τ_{cart} through $J^{\top}$, and the Coriolis torque τ_c from the robot model is added to the command. Each control phase uses phase-specific Cartesian stiffness values, while the damping is calculated from an estimate of the Cartesian inertia Λ.

A surface-fixed coordinate frame is constructed from the calibrated surface geometry. Its axes are the unit surface normal n_s and the two orthogonal tangent directions t_1 and t_2. The same frame is used for the orientation offsets, directional gains, compliance-centre coordinates, and surface-resolved evaluation quantities.

Each run follows a fixed control sequence. The tool is first brought to the commanded orientation and approaches the surface to a geometric clearance. The controller then selects the leading tool feature from the rectangular face geometry and stores its offset relative to the TCP. During set-up, the desired position of this selected tool point advances toward the surface and slightly beyond the plane as a virtual spring reference. The physical contact limits the motion and generates the normal pressing action. The desired orientation is held at the value reached at the clearance transition, so the measured rotation during set-up arises from the compliant contact response.

During the shifted set-up conditions, the 6×6 stiffness and damping matrices defined at the virtual CoC are transformed to the TCP by an adjoint congruence transformation. The resulting off-diagonal blocks couple translation and rotation and allow the normal pressing action to produce an additional commanded alignment moment. The reported contact runs end at the pre-grinding gate. The implemented grinding phase therefore lies outside the quantitative evaluation.

The implementation contribution consists of the real-time integration of this point-shifted Cartesian impedance with the surface-relative task frame, contact sequence, null-space controller, safety handling, and data recording. The experimental methodology defines the physical surface and tool geometry independently so that the measured end-effector response can be evaluated relative to the calibrated surface.

The first experimental result concerns the fixed centre. Every tested non-zero tangential displacement showed a dependence on the sign or direction of the initial misalignment. The assisting side of the TCP reversed with the sign of the commanded rotation, and the displacement used for a command about t_1 differed from that used about t_2. Changes in rotational and cross-axis translational stiffness modified the response magnitude while preserving this directional dependence. Within the investigated task and tested centre positions, the TCP-centred condition is therefore the direction-independent fixed reference because it requires no prior selection of misalignment direction or sign.

The second experimental result concerns the additional alignment authority of a displaced centre. A tangential lever contributes the commanded moment $m_{c,K} = r_c \times f_K$ and thereby converts part of the normal pressing action into rotation of a selected sense. About t_1, the TCP-centred condition already produced substantial alignment-directed rotation. About t_2, a selected 40 mm tangential lever changed the signed set-up rotation from $-1.63°$ to $+4.43°$. The displaced centre therefore serves as a condition-dependent means of adding rotational alignment authority.

The third experimental result is the geometric lever-selection rule. The assisting tangential lever lies perpendicular to the commanded rotation direction, with its sign chosen so that the induced moment acts in the required alignment direction. Case H evaluated this relation at four tangent-plane directions using one common 40 mm magnitude. The diagonal selected-versus-fixed comparisons differed by 0.07° and 0.05°, while the t_2 comparison separated $-1.61°$ from $+4.43°$.

Projected joint damping and two-point minimum-singular-value $\sigma_{\min}$ conditioning were also implemented for the redundant seventh joint. Complementary Cartesian pose-hold trials evaluated both terms under an internally generated joint-torque disturbance equivalent to a point force at an offset point on link 3. These trials provide a separate assessment of the secondary null-space controller.

1.5 Thesis Structure

Chapter 2 establishes the impedance law, frame conventions, and compliance-centre shift. Their real-time implementation on the Franka Emika Panda is described in Chapter 3. The experimental method and results follow in Chapters 4 and 5. Chapter 6 states the resulting conclusions and future work.

Chapter 2

Theoretical Background

This chapter presents the mathematical basis of the Cartesian impedance controller. The required frame conventions and pose errors are introduced first. The subsequent sections derive the kinematic mappings, the impedance wrench F, the compliance-centre transformation, and the null-space torque τ_{null}.

2.1 Reference Frames

The following right-handed coordinate frames are used in the theoretical formulation and the controller [23, 4].

- **Base frame** $\{0\}$**:** The base frame is fixed to the robot mounting surface. Its origin is the robot base reference point used by the kinematic model. For the upright Panda configuration used in this thesis, the z_0-axis is normal to the mounting surface, while the x_0- and y_0-axes lie in the mounting plane. Unless stated otherwise, Cartesian positions, directions, velocities, forces, and moments are expressed in this frame.
- **Joint frames** $\{1\}, \{2\}, \ldots, \{n\}$**:** These frames are attached to the successive links of the kinematic chain according to the Denavit–Hartenberg convention [5, 4]. Joint i rotates about z_{i-1}, and p_{i-1} denotes a point on this axis. The joint frames define the transformations between consecutive links and the indexing used in the forward kinematics and geometric Jacobian $J(q)$.
- **End-effector frame** $\{\text{EE}\}$**:** The end-effector frame is attached to the controlled TCP and moves with the tool. In the implemented robot model, the end-effector

origin and TCP are identified, so $p_{\mathrm{TCP}} = p_{\mathrm{EE}}$. The physical tool face is represented separately by its calibrated geometric offset from this frame. The position p_{EE} and orientation R_{EE}, both expressed relative to the base frame, define the current Cartesian pose of the robot. The axes x_{EE}, y_{EE}, and z_{EE} describe the current end-effector orientation. The configured tool approach axis is defined in this frame and is transformed to the base frame using R_{EE}.

- **Desired end-effector frame $\{d\}$:** The desired frame represents the commanded TCP pose. Its origin is p_d, and its orientation is R_d, both expressed relative to the base frame. The difference between the desired and current end-effector frames provides the translational and rotational errors used by the impedance controller.
- **Surface task frame $\{S\}$:** The surface task frame is tied to the configured surface rather than to the moving tool. Its origin is placed at the configured surface point p_s. Its orientation R_{surface} is defined by the two surface tangents t_1 and t_2 and the surface normal n_s:

$$R_{\mathrm{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.1}$$

The directions t_1 and t_2 lie in the surface plane and are mutually orthogonal. The unit normal n_s is defined to point outward from the surface towards the free-space region above the plate. This sign convention is kept throughout the contact sequence, so motion towards the surface is along $-n_s$. The surface frame is used to describe the commanded tool orientation offset and to assign stiffness and damping values according to the two tangential directions and the normal direction. For the contact task, the task-frame orientation R_{task} equals the surface-frame orientation:

$$R_{\mathrm{task}} = R_{\mathrm{surface}}. \tag{2.2}$$

Figure 2.1 shows these frames together with the pose error between the current and the desired end-effector frame.

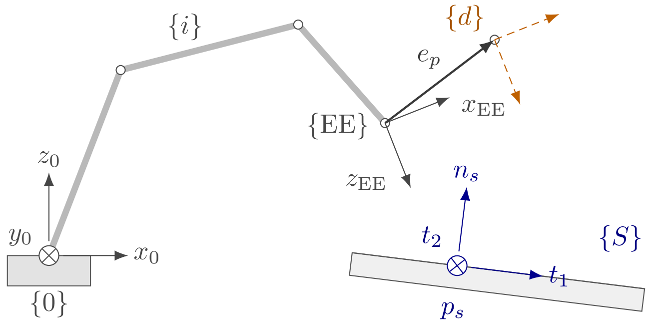


Figure 2.1: Frames used in the controller and the end-effector pose error.

2.2 Rigid-Body Transformations

The positions and orientations of the frames introduced in Section 2.1 are related through rotation matrices and homogeneous transformations.

2.2.1 Rotation Matrices

An orientation is represented by a rotation matrix $R \in SO(3)$, satisfying

$$R^\top R = I, \qquad \det R = 1, \qquad R^{-1} = R^\top. \tag{2.3}$$

Here, I is the 3×3 identity matrix and $\det(\cdot)$ denotes the determinant.

The notation ${}^bR_a$ denotes the orientation of frame $\{a\}$ expressed in frame $\{b\}$. Its columns contain the axes of frame $\{a\}$ expressed in frame $\{b\}$. A vector v is transformed from frame $\{a\}$ to frame $\{b\}$ according to

$${}^bv = {}^bR_a\,{}^av. \tag{2.4}$$

The elementary right-handed rotations use the rotation angles α, β, and γ about the x-, y-, and z-axes, respectively:

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}, \quad R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}, \quad R_z(\gamma) = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2.5}$$

2.2.2 Homogeneous Transformations

A position is represented by a vector $p \in \mathbb{R}^3$. The homogeneous transformation ${}^bT_a$ combines the rotation and translation from frame $\{a\}$ to frame $\{b\}$ [4, 23]:

$$ {}^bT_a = \begin{bmatrix} {}^bR_a & {}^bp_a \\ 0^\top & 1 \end{bmatrix}, \tag{2.6} $$

where ${}^bp_a$ is the position of the origin of frame $\{a\}$ expressed in frame $\{b\}$. A point expressed in frame $\{a\}$ is transformed to frame $\{b\}$ according to

$$ {}^bp = {}^bR_a\, {}^ap + {}^bp_a. \tag{2.7} $$

Successive transformations are composed by matrix multiplication:

$$ {}^cT_a = {}^cT_b\, {}^bT_a. \tag{2.8} $$

2.2.3 End-Effector Pose

The controlled pose consists of the end-effector position

$$ p_{\mathrm{EE}} = \begin{bmatrix} p_{x,\mathrm{EE}} & p_{y,\mathrm{EE}} & p_{z,\mathrm{EE}} \end{bmatrix}^\top \in \mathbb{R}^3 \tag{2.9} $$

and the orientation $R_{\mathrm{EE}} \in SO(3)$. Both are expressed in the base frame and combined in the end-effector transformation T_{EE}:

$$ T_{\mathrm{EE}} = \begin{bmatrix} R_{\mathrm{EE}} & p_{\mathrm{EE}} \\ 0^\top & 1 \end{bmatrix}. \tag{2.10} $$

In the implementation, T_{EE} is obtained directly from the reported robot state, and the orientation is not converted to an Euler-angle representation.

2.2.4 Cartesian Pose Error

The desired end-effector transformation T_d is

$$T_d = \begin{bmatrix} R_d & p_d \\ 0^\top & 1 \end{bmatrix}, \tag{2.11}$$

where p_d and R_d are the desired TCP position and orientation, respectively. The current end-effector pose is denoted by T_{EE}.

The relative pose-error transformation T_{err} maps the current end-effector frame to the desired frame:

$$T_{\text{err}} = T_{\text{EE}}^{-1} T_d = \begin{bmatrix} R_{\text{EE}}^\top R_d & R_{\text{EE}}^\top (p_d - p_{\text{EE}}) \\ 0^\top & 1 \end{bmatrix}. \tag{2.12}$$

When the current and desired poses coincide, $T_{\text{err}} = I_4$. The rotational and translational blocks of Equation (2.12) therefore both represent zero error in this case.

Position error. The translational block of Equation (2.12) expresses the position error e_p in the current end-effector frame. The position error e_p used by the Cartesian impedance law is expressed in the base frame:

$$e_p = p_d - p_{\text{EE}} \in \mathbb{R}^3. \tag{2.13}$$

A positive position error e_p therefore points from the current TCP position towards the desired TCP position.

Orientation error. The relative rotation ΔR is the rotational block of Equation (2.12):

$$\Delta R = R_{\text{EE}}^\top R_d. \tag{2.14}$$

It represents the rotation required to carry the current end-effector orientation to the desired orientation. If both orientations coincide, then $\Delta R = I$.

For use in the impedance controller, ΔR is represented by a rotation angle ϕ and a unit rotation axis $u \in \mathbb{R}^3$ [23, 19]. The rotation-matrix and axis–angle representations can be converted in both directions. Given a unit axis u and an angle ϕ, the corresponding rotation matrix is obtained from Rodrigues' formula,

$$R(u, \phi) = I + \sin\phi\,[u]_\times + (1 - \cos\phi)[u]_\times^2. \tag{2.15}$$

For the relative rotation considered here, $R(u, \phi) = \Delta R$. The skew-symmetric matrix of the rotation axis is

$$[u]_\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}, \qquad [u]_\times^\top = -[u]_\times. \tag{2.16}$$

Conversely, when the rotation matrix ΔR is known, the angle follows from

$$\mathrm{tr}(\Delta R) = 1 + 2\cos\phi, \tag{2.17}$$

and therefore

$$\phi = \arccos\left(\frac{\mathrm{tr}(\Delta R) - 1}{2}\right), \qquad \phi \in [0, \pi]. \tag{2.18}$$

Subtracting the transpose of ΔR isolates its skew-symmetric part:

$$\frac{1}{2}\left(\Delta R - \Delta R^\top\right) = \sin\phi\,[u]_\times. \tag{2.19}$$

For $\sin\phi \neq 0$, the rotation axis is obtained from

$$u = \frac{1}{2\sin\phi}\begin{bmatrix} \Delta R_{32} - \Delta R_{23} \\ \Delta R_{13} - \Delta R_{31} \\ \Delta R_{21} - \Delta R_{12} \end{bmatrix}. \tag{2.20}$$

Multiplying the unit axis by the angle gives the current-frame orientation error $e_{R,\mathrm{EE}}$. Rotating it into the base frame gives the controller orientation error e_R:

$$e_{R,\mathrm{EE}} = \phi u, \qquad e_R = R_{\mathrm{EE}} e_{R,\mathrm{EE}} \in \mathbb{R}^3. \tag{2.21}$$

The direction of e_R defines the axis of the required correction, while its magnitude is the angular error in radians. When the current and desired orientations coincide, $\phi = 0$ and

$e_R = 0$. Together, $e_p = p_d - p_{\mathrm{EE}}$ and e_R are expressed in the base frame and point in the translational and rotational directions required to reduce the pose error.

The complete Cartesian error e stacks the position error e_p and orientation error e_R used by the impedance controller:

$$e = \begin{bmatrix} e_p \\ e_R \end{bmatrix} \in \mathbb{R}^6. \tag{2.22}$$

2.3 Kinematics of a 7-DOF Manipulator

Having defined the end-effector pose as the controlled variable, its relation to the joint configuration is introduced next.

2.3.1 Forward Kinematics

Forward kinematics maps the joint-configuration vector q

$$q = \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}^\top \in \mathbb{R}^n \tag{2.23}$$

to the end-effector pose. Using the joint frames introduced in Section 2.1, the transformation from the base frame $\{0\}$ to the final link frame $\{n\}$ is

$$^{0}T_n(q) = {}^{0}T_1(q_1)\,{}^{1}T_2(q_2)\cdots{}^{n-1}T_n(q_n). \tag{2.24}$$

In the standard Denavit–Hartenberg convention [5, 4, 23], joint i rotates by q_i about the axis z_{i-1}. The transformation between two consecutive joint frames is

$$^{i-1}T_i(q_i) = R_z(q_i)\,T_z(d_i)\,T_x(a_i)\,R_x(\alpha_i), \tag{2.25}$$

where a_i, d_i, and α_i describe the fixed geometry between the consecutive joint frames, while q_i is the revolute-joint coordinate.

The complete transformation gives the end-effector pose in the base frame:

$$
{}^{0}T_n(q) = T_{\mathrm{EE}}(q) = \begin{bmatrix} R_{\mathrm{EE}}(q) & p_{\mathrm{EE}}(q) \\ 0^\top & 1 \end{bmatrix}. \tag{2.26}
$$

This convention establishes the standard kinematic relation and the joint and frame indexing used in the subsequent Jacobian $J(q)$ formulation. The real-time controller does not reconstruct the Panda pose from these Denavit–Hartenberg transformations. Instead, the current end-effector pose is read from the libfranka robot state, while the Jacobian is obtained from the libfranka model interface. The Denavit–Hartenberg formulation is therefore used here as the theoretical kinematic basis rather than as a second real-time kinematic implementation.

2.3.2 Differential Kinematics

Differential kinematics maps the joint-velocity vector $\dot{q}$ to the end-effector twist $\dot{x}$:

$$
\dot{x} = \begin{bmatrix} \dot{p}_{\mathrm{EE}} \\ \omega_{\mathrm{EE}} \end{bmatrix} = J(q)\dot{q}, \tag{2.27}
$$

where $\dot{p}_{\mathrm{EE}} \in \mathbb{R}^3$ and $\omega_{\mathrm{EE}} \in \mathbb{R}^3$ are the linear and angular velocities of the end-effector, respectively. The joint-velocity vector is

$$
\dot{q} = \begin{bmatrix} \dot{q}_1 & \dot{q}_2 & \cdots & \dot{q}_n \end{bmatrix}^\top \in \mathbb{R}^n, \tag{2.28}
$$

and $J(q) \in \mathbb{R}^{6\times n}$ is the geometric Jacobian. For the 7-DOF Panda, $n = 7$. The end-effector twist and the Jacobian used in the controller are expressed in the base frame. This relation provides the Cartesian velocity $\dot{x}$ required by the damping term of the impedance controller.

2.3.3 Geometric Jacobian

The geometric Jacobian $J(q)$ is written column-wise as

$$
J(q) = \begin{bmatrix} J_1 & J_2 & \cdots & J_n \end{bmatrix} \in \mathbb{R}^{6\times n}, \tag{2.29}
$$

where J_i describes the contribution of joint i to the end-effector twist. Since all joints of the Panda are revolute, each column is given by

$$J_i = \begin{bmatrix} z_{i-1} \times (p_{\mathrm{EE}} - p_{i-1}) \\ z_{i-1} \end{bmatrix} \in \mathbb{R}^6, \tag{2.30}$$

where z_{i-1} is the axis of joint i, and p_{i-1} is a point on that axis. Both are expressed in the base frame. The upper block describes the linear velocity generated at the end-effector, while the lower block describes the corresponding angular velocity [23, 4].

Because the joint axes and their positions depend on the current configuration, the Jacobian also depends on q. In the implemented controller, $J(q)$ is obtained directly from the libfranka model interface.

2.4 Cartesian Impedance Control

The geometric Jacobian $J(q)$ is also used through its transpose to map a Cartesian wrench F to joint torques τ [15]. Before introducing this mapping, the desired Cartesian interaction behaviour is defined.

In this thesis, Cartesian impedance control is formulated as a spring–damper relation between the end-effector motion and the commanded wrench [13, 21, 17]. The Cartesian pose error e, defined in Section 2.2.4, produces a restoring wrench through the stiffness matrix K, while the Cartesian velocity $\dot{x}$ produces an opposing wrench through the damping matrix D. Their entries determine the translational and rotational compliance in the selected task directions [22]. Physically, the stiffness terms determine how strongly a pose error generates a restoring force or moment, while the damping terms oppose the corresponding Cartesian motion.

2.4.1 Impedance Wrench

The impedance wrench is evaluated in the robot base frame. The corresponding translational stiffness and damping matrices are $K_{p,0}, D_{p,0} \in \mathbb{R}^{3\times3}$, while the rotational matrices are $K_{R,0}, D_{R,0} \in \mathbb{R}^{3\times3}$. These matrices are not generally diagonal in the base frame because the direction-dependent gains are defined relative to the task frame. Their base-frame representation therefore changes with the task-frame orientation, even when the local gain values remain unchanged.

The commanded Cartesian force f is given by the translational spring–damper law [13, 17]

$$f = K_{p,0} e_p + D_{p,0} \left(\dot{p}_d - \dot{p}_{\mathrm{EE}} \right) \in \mathbb{R}^3, \tag{2.31}$$

where $\dot{p}_d$ and $\dot{p}_{\mathrm{EE}}$ are the desired and measured linear velocities expressed in the base frame. For a fixed position setpoint, $\dot{p}_d = \mathbf{0}$, and the damping term becomes $-D_{p,0}\dot{p}_{\mathrm{EE}}$.

The commanded Cartesian moment m is

$$m = K_{R,0} e_R - D_{R,0} \omega_{\mathrm{EE}} \in \mathbb{R}^3, \tag{2.32}$$

where ω_{EE} is the measured end-effector angular velocity expressed in the base frame. Since no angular reference velocity is commanded, the rotational damping acts directly against ω_{EE}.

The Cartesian wrench F stacks the force and moment:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \begin{bmatrix} K_{p,0} e_p + D_{p,0} \left(\dot{p}_d - \dot{p}_{\mathrm{EE}} \right) \\ K_{R,0} e_R - D_{R,0} \omega_{\mathrm{EE}} \end{bmatrix} \in \mathbb{R}^6. \tag{2.33}$$

The wrench is ordered as force followed by moment, and all its components are expressed in the base frame. It is mapped to joint torques τ_{cart} through the Jacobian transpose in the following subsection.

2.4.2 Joint-Torque Control Law

By the principle of virtual work, the commanded Cartesian wrench F is mapped to joint torques τ_{cart} through the transpose of the geometric Jacobian $J(q)$ [15]:

$$\tau_{\mathrm{cart}} = J^{\top}(q) F, \tag{2.34}$$

where $\tau_{\mathrm{cart}} \in \mathbb{R}^n$ is the Cartesian impedance torque contribution and $J^{\top}(q) \in \mathbb{R}^{n \times 6}$. Substituting Equation (2.33) gives

$$\tau_{\text{cart}} = J^{\top}(q) \begin{bmatrix} K_{p,0}e_p + D_{p,0}\left(\dot{p}_d - \dot{p}_{\text{EE}}\right) \\ K_{R,0}e_R - D_{R,0}\omega_{\text{EE}} \end{bmatrix}. \tag{2.35}$$

The gain matrices in this expression are the base-frame matrices obtained from the selected task-frame gains. A change in the surface-frame orientation therefore changes the Cartesian wrench F and the resulting joint-torque contribution τ_{cart}, even when the local gain values remain unchanged.

2.4.3 Coriolis and Gravity Compensation in libfranka

In a general model-compensated torque controller, the commanded joint torque τ_{cmd} combines Cartesian impedance, gravity, and Coriolis/centrifugal compensation:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_g(q) + \tau_c(q, \dot{q}), \tag{2.36}$$

where $\tau_g(q)$ is the gravity torque and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal torque.

These terms are handled differently by the Franka Control Interface. The Coriolis vector provided by the libfranka model is added explicitly to the command because it is not compensated internally. A separate gravity term is not added. In external joint-torque mode, the robot compensates gravity and motor friction internally, and the commanded torque is interpreted as excluding gravity [9]. Adding $\tau_g(q)$ would therefore compensate gravity twice.

The implemented commanded torque $\tau_{\text{cmd,impl}}$ is consequently

$$\tau_{\text{cmd,impl}} = \tau_{\text{cart}} + \tau_c(q, \dot{q}) = J^{\top}(q)F + \tau_c(q, \dot{q}), \tag{2.37}$$

which follows the official libfranka Cartesian impedance example [10]. The null-space torque τ_{null} introduced later is added to this expression to obtain the complete commanded torque in Equation (2.103).

2.4.4 Commanded and Estimated Wrench

The commanded wrench F and the model-estimated external wrench are distinct controller signals. The commanded wrench follows from Equation (2.33) and is mapped to the joints by $J^\top$. By contrast, the robot estimates the external wrench from the measured joint torques and its dynamic model. Its translational part is denoted by $\hat{f}_{\text{ext}}$.

Contact evaluation uses the force change from the value stored at the clearance transition,

$$\Delta\hat{f}_{\text{ext}} = \hat{f}_{\text{ext}} - \hat{f}_{\text{ref}}. \tag{2.38}$$

Both f and $\Delta\hat{f}_{\text{ext}}$ are expressed in the base frame. Their signed components along the surface normal n_s are therefore obtained by projection. The commanded normal force is $F_{n,\text{cmd}} = n_s^\top f$, whereas the estimated normal-load magnitude is $F_n = |n_s^\top \Delta\hat{f}_{\text{ext}}|$.

During a quasi-static press, the selected sign convention gives $n_s^\top \Delta\hat{f}_{\text{ext}} \approx F_{n,\text{cmd}} < 0$ and hence $F_n \approx -F_{n,\text{cmd}}$. This relation expresses approximate force balance; it does not identify the two signals. Model error, friction, transient motion, and motion within the tool mount can produce differences.

2.5 Surface Task Frame

The contact experiment uses a task frame tied to the surface geometry. This frame is defined before the stiffness and damping matrices are introduced, so that the local gain entries can later be interpreted as normal and tangential values instead of as abstract base-frame directions.

Let the unit surface normal $n_s \in \mathbb{R}^3$, with $\|n_s\| = 1$, be expressed in the robot base frame. Its sign is chosen such that n_s points outward from the surface towards the free-space region above the plate. Consequently, the approach and set-up direction towards the surface is $-n_s$. The normal alone does not determine the frame. One rotational freedom about n_s remains, so the two tangent directions follow from a separate choice rather than from the surface geometry.

That choice is the surface-frame reference direction $a_s \in \mathbb{R}^3$, also expressed in the base frame. It is a configured direction and not a measured property of the surface. It need not lie in the surface plane, because only its component in the plane is used, and the single requirement is that it must not be parallel to n_s: a direction parallel to the normal leaves nothing to project. Two reference directions differing only by a multiple of n_s therefore define the same frame, whereas rotating a_s within the plane rotates t_1 and t_2 with it.

The first tangent vector is obtained by projecting a_s onto the plane orthogonal to n_s, and the second tangent vector is then obtained by a cross product. With $\tilde{t}_1$ denoting the unnormalised projected tangent, the construction is a Gram–Schmidt orthonormalisation step applied to the surface normal n_s and the surface-frame reference direction [23, 4]:

$$\begin{aligned} \tilde{t}_1 &= a_s - n_s \left(n_s^\top a_s \right), \\ t_1 &= \frac{\tilde{t}_1}{\|\tilde{t}_1\|}, \\ t_2 &= n_s \times t_1. \end{aligned} \tag{2.39}$$

Here, t_1 and t_2 are unit tangent vectors lying in the surface plane. This frame makes the gain directions physically meaningful: separate translational and rotational compliance can be assigned along the two surface tangents and along or about the surface normal instead of along arbitrary base-frame axes. Since the plane and its normal do not depend on a_s, a different reference direction changes which in-plane direction carries the name t_1 without changing the surface geometry. An axis-specific result therefore holds with respect to the configured reference direction, which for the reported experiments was the base-frame x-axis; the resulting tangents are given in Section 4.2.

The construction requires $\|\tilde{t}_1\| \neq 0$, and the behaviour of the implementation when the configured direction is close to the normal is described in Section 3.2.1. To match the implemented controller and its parameter order, the surface task frame collects the two tangents first and the normal last,

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.40}$$

The column order is fixed: every surface-frame three-vector in the implementation uses $[\,\text{tangent 1}, \text{tangent 2}, \text{normal}\,]^\top$. In the gain representation below, the contact task uses this frame by setting $R_{\text{task}} = R_{\text{surface}}$. This makes stiffness values such as K_{p,t_1}, K_{p,t_2}, and $K_{p,n}$ local surface-frame quantities in that same order.

2.6 Stiffness and Damping Representation

Direction-dependent stiffness and damping gains are most conveniently defined in a task frame whose axes coincide with the desired principal compliance directions. The corresponding matrices are diagonal in that frame but are not generally diagonal when expressed in the robot base frame.

Let $R_{\text{task}} \in SO(3)$ denote the orientation of the task frame relative to the base frame. Its columns contain the task-frame axes expressed in the base frame. The local gains $K_{p,\text{task}}$, $D_{p,\text{task}}$, $K_{R,\text{task}}$, and $D_{R,\text{task}}$ are rotated into the corresponding base-frame gain matrices:

$$\begin{aligned} K_{p,0} &= R_{\text{task}} K_{p,\text{task}} R_{\text{task}}^\top, \quad & D_{p,0} &= R_{\text{task}} D_{p,\text{task}} R_{\text{task}}^\top, \\ K_{R,0} &= R_{\text{task}} K_{R,\text{task}} R_{\text{task}}^\top, \quad & D_{R,0} &= R_{\text{task}} D_{R,\text{task}} R_{\text{task}}^\top. \end{aligned} \tag{2.41}$$

For an impedance without translation–rotation coupling in the task frame, the full task-frame stiffness K_{task} and damping D_{task} are

$$K_{\text{task}} = \begin{bmatrix} K_{p,\text{task}} & 0 \\ 0 & K_{R,\text{task}} \end{bmatrix}, \qquad D_{\text{task}} = \begin{bmatrix} D_{p,\text{task}} & 0 \\ 0 & D_{R,\text{task}} \end{bmatrix}. \tag{2.42}$$

The block rotation matrix T_R applies the task-frame rotation to the translational and rotational blocks:

$$T_R = \operatorname{diag}\left(R_{\text{task}}, R_{\text{task}}\right), \tag{2.43}$$

The corresponding full base-frame stiffness K_0 and damping D_0 are

$$K_0 = T_R K_{\text{task}} T_R^\top, \qquad D_0 = T_R D_{\text{task}} T_R^\top. \tag{2.44}$$

The task-frame rotation changes the matrix representation without changing the physical directions assigned to the gains. For example, $K_{p,n}$ acts along the calibrated surface normal even when that normal is not aligned with a base-frame axis.

For the contact task, the task frame is the surface frame, whose orientation is

$$R_{\text{task}} = R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.45}$$

For example, the local translational stiffness is defined as

$$K_{p,\text{task}} = \text{diag}\left(K_{p,t_1}, K_{p,t_2}, K_{p,n}\right), \tag{2.46}$$

and its base-frame representation is

$$K_{p,0} = R_{\text{surface}} \ \text{diag}\left(K_{p,t_1}, K_{p,t_2}, K_{p,n}\right) R_{\text{surface}}^{\top}. \tag{2.47}$$

The local rotational-stiffness entries K_{R,t_1}, K_{R,t_2}, and $K_{R,n}$ follow the same axis order:

$$K_{R,\text{task}} = \text{diag}\left(K_{R,t_1}, K_{R,t_2}, K_{R,n}\right), \tag{2.48}$$

and the damping matrices are constructed analogously. A change in the configured surface orientation therefore changes the corresponding base-frame gain matrices even when the local gain values remain unchanged. During approach, contact set-up, and grinding, the reported contact runs used this surface-frame construction for both translational and rotational gain blocks. In the reported experiments, the phase-indexed numerical entries of Table 4.1 were assigned to $[t_1, t_2, n_s]$ before transformation to the base frame. The active gain matrices were based on the calibrated surface frame and were not rotated with the current end-effector orientation R_{EE}.

2.7 Centre of Compliance

The task-frame transformation of Equation (2.44) rotates the principal stiffness directions while leaving the impedance reference point at the end-effector origin, identified here with the tool centre point (TCP). Independently of this orientation choice, the virtual spring can be centred at a different point. Shifting the impedance reference from the TCP to a CoC p_c transforms a block-diagonal impedance at p_c into a coupled impedance at the TCP [19, 7, 24, 16]. This construction is applied during contact set-up to generate translation–rotation coupling that supports tool alignment. Its physical purpose is to use the normal pressing action not only to produce a translational contact load, but also to generate a rotational moment. The selected offset between p_c and the TCP defines the lever through which this moment is produced.

Two transformations used by the controller must therefore be distinguished. A task-frame rotation changes the directions along which the impedance acts, whereas the point shift changes the point about which the impedance is defined.

Figure 2.3 shows this geometry for the two signs of the tangential lever.

Three points must be distinguished. The selected tool point p_T is determined from the tool geometry and is used as the contact reference. The tool centre point p_{TCP} is the Cartesian control point of the robot, while p_c is the virtual centre of compliance.

The selected tool point is chosen at the clearance transition and then held constant relative to the tool for the whole of set-up. Once the tool begins to rotate under contact, it is therefore a geometric contact reference rather than the exact point at which the surface force acts at every instant.

Moving p_c moves neither the selected tool point nor the physical contact force. The surface force still acts where the tool touches. The point about which the Cartesian impedance is defined changes, and with it the translation–rotation coupling. The same normal pressing action can then produce a different rotational response.

The geometric consequence follows from where p_c sits relative to the press. A centre lying on the line of action of the press contributes little or no moment. Displacing it sideways gives the compliant response an effective lever: to one side the induced moment assists the required alignment, and to the other side it opposes or reverses it.

Each point is reached from the TCP by its own offset,

$$r_T = p_T - p_{\mathrm{TCP}}, \qquad r_c = p_c - p_{\mathrm{TCP}}, \tag{2.49}$$

and therefore

$$p_T - p_c = r_T - r_c. \tag{2.50}$$

The two vectors have different mechanical roles. The vector r_T describes the physical tool geometry between the TCP and the selected tool point, whereas r_c describes the virtual shift of the Cartesian impedance.

The contact interaction is considered from a commanded and a model-estimated viewpoint. The controller generates the Cartesian wrench of Equation (2.33), whose translational part separates into an elastic and a damping contribution,

$$f = f_K + f_D, \qquad f_K = K_p e_p, \qquad f_D = D_p \left(\dot{p}_d - \dot{p}_{\text{EE}} \right), \tag{2.51}$$

where f_K is the elastic commanded force component and f_D the damping component. The name matters here: f_K is the spring contribution alone and not the complete force pressing against the surface, which is f.

Independently, the robot model provides an estimate of the external wrench. Relative to the value stored at the clearance transition, its change is

$$\Delta \hat{F}_{\text{ext}} = \begin{bmatrix} \Delta \hat{f}_{\text{ext}} \\ \Delta \hat{m}_{\text{ext}} \end{bmatrix}. \tag{2.52}$$

Thus F describes the controller command, whereas $\Delta \hat{F}_{\text{ext}}$ provides a model-based observation of the resulting external loading.

On the commanded side, the rotational impedance of Equation (2.32) already produces a moment at the TCP whatever the CoC is. Writing it as

$$m_0 = K_R e_R - D_R \omega_{\text{EE}}, \tag{2.53}$$

the shifted impedance adds a coupling contribution when $r_c \neq 0$. Both the stiffness and the damping matrices are shifted, so both commanded force components acquire the same lever,

$$m_{c,K} = r_c \times f_K, \qquad m_{c,D} = r_c \times f_D, \tag{2.54}$$

and their sum is the complete translational coupling moment

$$m_{c,\text{trans}} = m_{c,K} + m_{c,D} = r_c \times f. \tag{2.55}$$

The commanded moment is therefore $m \approx m_0 + m_{c,\text{trans}}$ in this simplified picture, with m the moment part of the commanded wrench $F = [f^\top,\ m^\top]^\top$. The approximation is deliberate: when $e_R \neq 0$ the shifted law of Equations (2.61) and (2.62) carries further r_c-dependent terms in its lower-right block, so the sum explains the translational coupling mechanism rather than expanding every term of the shifted law.

The two contributions act over different time scales. The damping term is velocity dependent, so as the commanded reference stops advancing and the robot settles, $\dot{p}_d - \dot{p}_{\text{EE}} \to \mathbf{0}$ and $f_D \to \mathbf{0}$, leaving $m_{c,\text{trans}} \to r_c \times f_K$. The elastic contribution therefore carries the steady alignment tendency, while the damping contribution shapes the transient. The sign analysis below uses the elastic contribution alone, written

$$m_{c,K} = r_c \times f_K,$$

which Equation (2.63) obtains from the point-shifted stiffness blocks.

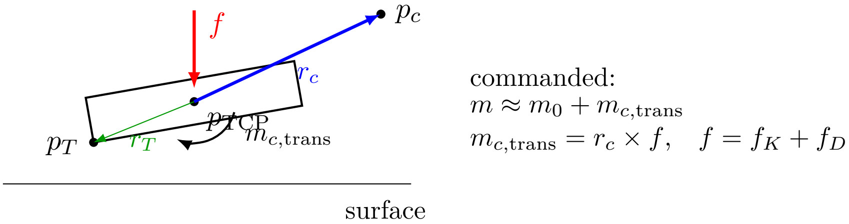
On the observed side, the surface interaction produces an external wrench whose change the robot model estimates as $\Delta\hat{m}_{\text{ext}}$. The tool-geometry lever explains how an external force acting away from the TCP contributes to that moment. Representing the estimated resultant as acting at the selected tool point gives the force-lever contribution

$$m_{r_T} = r_T \times \Delta\hat{f}_{\text{ext}}. \tag{2.56}$$

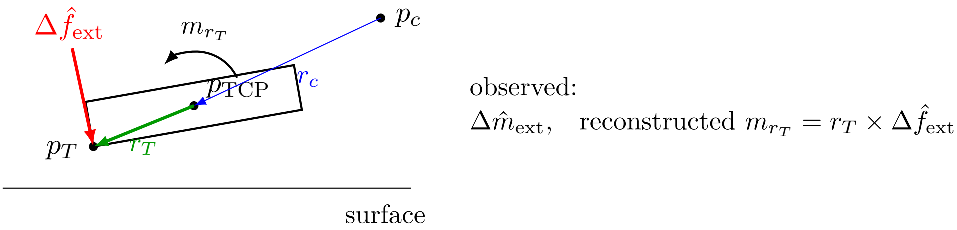
Because the selected tool point is a geometric contact reference and the exact instantaneous force-application point is not measured, m_{r_T} is a geometry-based reconstruction and not an independently measured moment. It is compared with $\Delta\hat{m}_{\text{ext}}$ and never equated to it, and it is not added to m_0 or $m_{c,K}$: those belong to the command, and $\Delta\hat{F}_{\text{ext}}$ is the observation of the resulting interaction. About the CoC the same reconstruction would take the complete lever of Equation (2.50), $(r_T - r_c) \times \Delta\hat{f}_{\text{ext}}$, which the controller never forms.

The zero-lever condition is therefore not a zero-moment condition. Setting $r_c = 0$ makes $\mathrm{Ad}(r_c)$ the identity, so it removes the complete additional coupling introduced by the virtual point shift: the elastic and damping coupling moments $m_{c,K}$ and $m_{c,D}$, their sum $m_{c,\text{trans}}$, and the remaining r_c-dependent terms of K_{TCP} and D_{TCP}, including the off-diagonal blocks and the added rotational entries $[r_c]_\times^\top K_p [r_c]_\times$ and $[r_c]_\times^\top D_p [r_c]_\times$. The ordinary rotational impedance and the physical tool and contact geometry are untouched by it. The commanded rotational moment m_0 remains, the rotational compliance stays finite, and because $r_T \neq 0$ the external force can still contribute m_{r_T}, so contact-induced rotation is possible with the centre of compliance at the TCP. Shifting the CoC adds $m_{c,K}$ to the command, which changes the interaction and with it the estimated $\Delta\hat{f}_{\mathrm{ext}}$ and $\Delta\hat{m}_{\mathrm{ext}}$. The external moment is the result of that closed-loop interaction rather than an algebraic sum of the commanded and reconstructed terms.

Figure 2.2 places the two sides side by side. The model-estimated external wrench, and the form it takes during the experiment, is treated in Section 5.1.3.

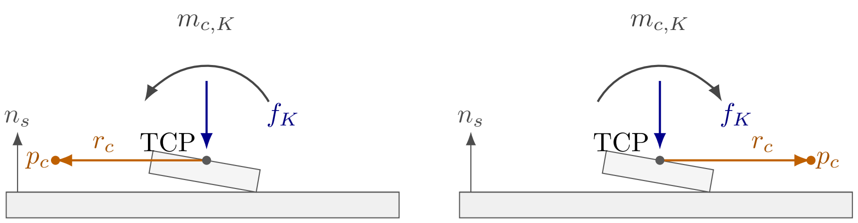


(a) commanded side



(b) observed side

Figure 2.2: Commanded and observed moment relations: (a) the translation-induced coupling carried by the commanded wrench; (b) the model-estimated external interaction with the geometry-based force-lever reconstruction.



(a) lever selected from the tilt (b) lever with the opposite sign

Figure 2.3: Elastic compliance-centre coupling moment for the two tangential lever signs: (a) the lever selected from the orientation offset; (b) the lever with the opposite sign.

For a dual-arm assembly task, Suomalainen et al. experimentally showed that compliance-centre placement and compliant-axis selection affect the alignment response [25]. This task-level result motivates the placement study in this thesis. The point-shift adjoint below remains derived from the spatial-transformation formulation.

The compliance-centre shift r_c from the TCP to the CoC p_c, expressed in the base frame, is

$$r_c = p_c - p_{\mathrm{TCP}} \in \mathbb{R}^3, \qquad p_{\mathrm{TCP}} = p_{\mathrm{EE}}. \tag{2.57}$$

Let $[r_c]_\times$ denote the skew-symmetric matrix satisfying $[r_c]_\times a = r_c \times a$. Under the small-angle approximation, the compliance-centre position error $e_{p,c}$ is

$$e_{p,c} = e_p - [r_c]_\times e_R. \tag{2.58}$$

The negative sign follows from the selected lever convention. The vector from the TCP to the CoC p_c is r_c itself, and therefore

$$e_R \times r_c = -[r_c]_\times e_R.$$

The compliance-centre error e_c stacks the translational error $e_{p,c}$ and the rotational error e_R. The point-shift adjoint $\mathrm{Ad}(r_c)$ maps the corresponding TCP error to this stacked error:

$$e_c = \begin{bmatrix} e_{p,c} \\ e_R \end{bmatrix} = \mathrm{Ad}(r_c) \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \qquad \mathrm{Ad}(r_c) = \begin{bmatrix} I_3 & -[r_c]_\times \\ 0 & I_3 \end{bmatrix}. \tag{2.59}$$

The point-shift adjoint $\mathrm{Ad}(r_c)$ changes the reference point of the impedance. By contrast, $T_R = \mathrm{diag}(R_{\mathrm{task}}, R_{\mathrm{task}})$ in Equation (2.44) changes its principal directions.

After any required task-frame rotation, the decoupled centre-of-compliance stiffness K_c and damping D_c are

$$K_c = \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}, \qquad D_c = \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}. \tag{2.60}$$

All blocks in Equation (2.60) are expressed in the base-frame orientation. The compliance-centre wrench F_c is

$$F_c = K_c e_c.$$

Transforming this power-conjugate wrench gives the corresponding TCP wrench F_{TCP}:

$$F_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top F_c.$$

The resulting TCP stiffness K_{TCP} is therefore

$$K_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top K_c \mathrm{Ad}(r_c) = \begin{bmatrix} K_p & -K_p[r_c]_\times \\ -[r_c]_\times^\top K_p & K_R + [r_c]_\times^\top K_p [r_c]_\times \end{bmatrix}. \tag{2.61}$$

The TCP damping D_{TCP} uses the same point shift:

$$D_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top D_c \mathrm{Ad}(r_c) = \begin{bmatrix} D_p & -D_p[r_c]_\times \\ -[r_c]_\times^\top D_p & D_R + [r_c]_\times^\top D_p [r_c]_\times \end{bmatrix}. \tag{2.62}$$

The off-diagonal blocks introduce translation–rotation coupling. A translational error can therefore generate a restoring moment, while a rotational error can generate a restoring force. This coupling is absent from the block-diagonal impedance defined directly at the TCP.

For a purely translational error, $e_R = 0$, the coupling reduces to a single term. With

$$f_K = K_p e_p,$$

the elastic moment at the TCP is

$$m_{c,K} = -[r_c]_\times^\top f_K = r_c \times f_K. \tag{2.63}$$

A commanded tool orientation offset turns the tool about one of the surface tangents. The lever is chosen perpendicular to that tangent within the tangent plane, so that the moment the press generates opposes the commanded rotation. Reversing the commanded orientation offset reverses the lever with it.

The right-handed surface frame satisfies

$$t_1 \times t_2 = n_s, \qquad n_s \times t_1 = t_2, \qquad n_s \times t_2 = -t_1. \tag{2.64}$$

Let the commanded orientation offset in the surface plane have components θ_{t_1} and θ_{t_2}. For a non-zero offset and a selected tangential lever magnitude $\rho_c > 0$, the lever direction can be written directly from these two components as

$$r_{c,t} = \rho_c \frac{\theta_{t_2} t_1 - \theta_{t_1} t_2}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \tag{2.65}$$

The selected lever is therefore perpendicular to the commanded orientation-offset direction in the surface plane. Its sign changes when the commanded offset is reversed. For the two principal surface tangents, Equation (2.65) reduces to

$$r_{c,t} = -\rho_c t_2 \quad \text{for a positive } t_1 \text{ offset}, \qquad r_{c,t} = +\rho_c t_1 \quad \text{for a positive } t_2 \text{ offset}.$$

Because the set-up motion is commanded primarily along the surface normal, the elastic press is approximated for this sign analysis as

$$f_K \approx -F_{K,n} n_s, \qquad F_{K,n} > 0, \tag{2.66}$$

where $F_{K,n} = -n_s^\top f_K > 0$ is the magnitude of the commanded elastic normal component. It is a commanded quantity and is not the model-estimated external normal-load magnitude F_n. Tangential force components caused by pose error, friction, or the evolving contact geometry are not represented in this first-order argument. Substitution into Equation (2.63) gives

$$m_{c,K} \approx -F_{K,n} \rho_c \frac{\theta_{t_1} t_1 + \theta_{t_2} t_2}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \tag{2.67}$$

Thus, the press-induced moment is directed opposite to the commanded orientation offset. For a positive command about t_1 it reduces to $m_{c,K} \approx -F_{K,n}\rho_c t_1$, and for a positive command about t_2 it reduces to $m_{c,K} \approx -F_{K,n}\rho_c t_2$. Reversing the orientation offset reverses both the selected lever and the generated moment.

The coupling vanishes for $r_c = 0$. Conversely, if K_p is nonsingular, zero coupling implies $r_c = 0$. Since $\mathrm{Ad}(r_c)$ is invertible, the congruence transformation also preserves definiteness:

$$K_c \succeq 0 \implies K_{\mathrm{TCP}} \succeq 0, \qquad K_c \succ 0 \implies K_{\mathrm{TCP}} \succ 0. \tag{2.68}$$

The same statements apply to D_c and D_{TCP}. Negative off-diagonal entries therefore represent coupling terms rather than negative stiffness directions.

The congruence transformation preserves the symmetry and definiteness of the source gain matrices. Closed-loop robot–controller–contact behaviour with phase switching was assessed experimentally under the operating conditions reported in Chapter 4.

This formulation is a local small-motion model in which r_c is treated as fixed during one controller evaluation. For a surface-fixed lever, its components remain constant in the surface frame and therefore its base-frame vector remains constant during set-up. For a tool-fixed lever, its components remain constant in the end-effector frame, but its base-frame vector changes as the end effector rotates. The controller therefore updates the base-frame lever and the corresponding shifted gain matrices during each control cycle for the tool-fixed definition.

The local linearised CoC p_c is consequently a selected point rather than a measured physical pivot. It need not coincide with the physical contact location. The observed motion also depends on the finite rotational stiffness, the lever direction, the environmental constraint, friction, and the additional active controller terms introduced later.

2.8 Null-Space Torque

The Franka Emika Panda has seven joints, while the controlled end-effector task contains six components: three translational and three rotational. The robot is therefore kinematically redundant. The Cartesian impedance behaviour at the end-effector is the primary objective, and the remaining joint-space freedom can be used for a secondary objective.

In this thesis, the secondary joint-space action consists of a null-space damping contribution and an active singular-value conditioning contribution. Both are projected using the joint-torque null-space projector N_τ. For the full-row-rank 6×7 Panda Jacobian, one joint-space direction remains after the six task-producing directions are accounted for. The singular value decomposition (SVD) is used here for two distinct purposes: v_7 identifies this redundant direction, while σ_6 describes the weakest task-producing mapping. The controller damps motion in the projected null space and can additionally drive the robot toward the sampled null-space direction with the larger σ_6. The required null-space direction, pseudoinverse, and projector are introduced before the two torque contributions are defined.

2.8.1 SVD-Based Null-Space Characterisation

The geometric Jacobian $J(q)$ relates the current joint velocity $\dot{q}$ to the end-effector velocity $\dot{x}$:

$$\dot{x} = J(q)\dot{q}, \qquad J(q) \in \mathbb{R}^{6\times 7}, \tag{2.69}$$

where

$$\dot{x} = \begin{bmatrix} \dot{p}_{\mathrm{EE}} \\ \omega_{\mathrm{EE}} \end{bmatrix} \in \mathbb{R}^6 \tag{2.70}$$

contains the three linear and three angular velocity components of the end-effector.

A joint velocity $\dot{q}$ belongs to the kinematic null space when

$$J(q)\dot{q} = 0. \tag{2.71}$$

Such a joint velocity $\dot{q}$ changes the robot configuration while preserving the end-effector position and orientation to first order. The Jacobian $J(q)$ is the first-order term in the expansion of the forward kinematics about the current configuration, so a finite displacement along the null space leaves a pose error of second order in its magnitude. When $J(q)$ has full row rank, the null space of the 6×7 Jacobian $J(q)$ is one-dimensional.

The SVD is applied to the Jacobian $J(q)$ at the current joint configuration q [3, 6]:

$$J(q) = U_J \Sigma_J V_J^\top, \qquad U_J \in \mathbb{R}^{6\times 6}, \quad \Sigma_J \in \mathbb{R}^{6\times 7}, \quad V_J \in \mathbb{R}^{7\times 7}. \tag{2.72}$$

The Jacobian $J(q)$ is the input to the decomposition. The SVD returns the matrices U_J, Σ_J, and V_J, with the singular values contained in Σ_J.

The columns of V_J form an orthonormal basis of the seven-dimensional joint-velocity space. The joint velocity $\dot{q}$ can therefore be expressed as

$$\dot{q} = V_J c = \sum_{i=1}^{7} c_i v_i, \tag{2.73}$$

where v_i is the i-th column of V_J, and $c \in \mathbb{R}^7$ contains the components of $\dot{q}$ along these basis directions.

Similarly, the columns of U_J form an orthonormal basis of the six-dimensional end-effector-velocity space. Substituting $\dot{q} = V_J c$ into Equation (2.69) gives

$$\dot{x} = J(q) V_J c = U_J \Sigma_J c. \tag{2.74}$$

Thus, V_J provides the basis directions in which the joint velocity $\dot{q}$ is expressed, while U_J provides the corresponding basis directions in which the end-effector velocity $\dot{x}$ is expressed. The matrix Σ_J scales the mapping between these directions.

For the 6×7 Jacobian $J(q)$, the ordered Jacobian singular values σ_i occupy the diagonal of the singular-value matrix:

$$\Sigma_J = \begin{bmatrix} \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_6) & 0_{6\times 1} \end{bmatrix}, \tag{2.75}$$

where

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_6 \geq 0. \tag{2.76}$$

For the first six basis directions,

$$J v_i = \sigma_i u_i, \qquad i = 1, \ldots, 6, \tag{2.77}$$

where u_i is the i-th column of U_J. A joint velocity $\dot{q}$ in the direction v_i,

$$\dot{q} = c_i v_i, \tag{2.78}$$

therefore produces the end-effector velocity

$$\dot{x} = c_i \sigma_i u_i. \tag{2.79}$$

The singular value σ_i scales the mapping from the joint-velocity basis direction v_i to the corresponding end-effector-velocity basis direction u_i.

The seventh column of V_J corresponds to the zero column of Σ_J and therefore satisfies

$$J v_7 = 0. \tag{2.80}$$

A null-space joint velocity $\dot{q}$ may be scaled by the scalar amplitude c_7:

$$\dot{q} = c_7 v_7 \tag{2.81}$$

therefore changes the joint configuration without producing an end-effector velocity. When the Jacobian $J(q)$ has full row rank, v_7 spans its one-dimensional kinematic null space.

The minimum task-producing singular value $\sigma_{\min}$ is

$$\sigma_{\min} = \sigma_6. \tag{2.82}$$

The quantities v_7 and σ_6 have different purposes. The vector v_7 represents the redundant joint-motion direction, whereas σ_6 describes the weakest of the six mappings from joint velocity $\dot{q}$ to end-effector velocity. As σ_6 approaches zero, one end-effector velocity direction becomes increasingly difficult to produce, indicating an approach to a loss of Jacobian $J(q)$ row rank. Because the geometric Jacobian combines translational and rotational rows with different physical units, the numerical value of $\sigma_{\min}$ depends on the chosen Jacobian representation and length unit. It is therefore used in this thesis as a relative conditioning indicator for comparisons made with the same Jacobian convention, rather than as a unit-invariant manipulability measure. Within that fixed convention, $\sigma_{\min}$ is used to compare the conditioning of nearby joint configurations and to select the direction of the conditioning torque.

2.8.2 SVD-Based Pseudoinverse

The Jacobian pseudoinverse J^+ maps end-effector-velocity components back to the corresponding joint-velocity directions. Its calculation uses the singular directions obtained from the SVD.

Small singular values lead to large reciprocal values and can amplify numerical errors. A relative singular-value threshold is therefore applied. Let $\rho \geq 0$ denote the configured relative SVD tolerance. The corresponding absolute singular-value threshold σ_{thr} is calculated from the largest singular value σ_1:

$$\sigma_{\text{thr}} = \rho\sigma_1. \tag{2.83}$$

A singular direction is retained when

$$\sigma_i > \sigma_{\text{thr}}. \tag{2.84}$$

The Jacobian pseudoinverse J^+ is defined as

$$J^+ = \sum_{\substack{i=1 \\ \sigma_i > \sigma_{\text{thr}}}}^{6} \frac{1}{\sigma_i} v_i u_i^\top \in \mathbb{R}^{7\times 6}. \tag{2.85}$$

For each retained singular direction, J^+ maps an end-effector-velocity component along u_i to the corresponding joint-velocity direction v_i and scales it by $1/\sigma_i$. Directions at or below σ_{thr} receive a reciprocal value of zero.

2.8.3 Null-Space Projectors

The pseudoinverse is used to separate a joint-space vector into a task-related component and a null-space component. For an arbitrary joint velocity $\dot{q}$, this decomposition is

$$\dot{q} = \underbrace{J^{+}J\dot{q}}_{\text{task-related component}} + \underbrace{\left(I_7 - J^{+}J\right)\dot{q}}_{\text{null-space component}} . \tag{2.86}$$

The product $J^{+}J$ projects $\dot{q}$ onto the joint-space directions that contribute to the end-effector velocity. Since J includes all three linear and all three angular velocity components, this task-related component accounts for the complete controlled end-effector pose at the velocity level.

The complementary joint-velocity projector N_q is

$$N_q = I_7 - J^{+}J \in \mathbb{R}^{7\times 7}. \tag{2.87}$$

Applying N_q to an arbitrary joint velocity $\dot{q}$ gives the projected null-space velocity $\dot{q}_{\text{null}}$:

$$\dot{q}_{\text{null}} = N_q\dot{q}. \tag{2.88}$$

When J has full row rank and all six task-producing singular directions are retained,

$$J\dot{q}_{\text{null}} = JN_q\dot{q} = 0. \tag{2.89}$$

The projected joint velocity $\dot{q}_{\text{null}}$ can therefore change the robot configuration while preserving the end-effector position and orientation to first order.

A corresponding decomposition is used for a secondary joint torque τ_0. The joint-torque projector N_τ is the transpose of the velocity projector:

$$N_\tau = N_q^{\top} = I_7 - J^{\top}J^{+\top} \in \mathbb{R}^{7\times 7}. \tag{2.90}$$

The secondary torque can then be written as

$$\tau_0 = \underbrace{J^{\top}J^{+\top}\tau_0}_{\text{task-related component}} + \underbrace{N_\tau\tau_0}_{\text{null-space component}} . \tag{2.91}$$

The first component lies in the joint-torque subspace associated with Cartesian wrenches F through $J^{\top}$. The second component is retained as the secondary null-space torque τ_{null}:

$$\tau_{\text{null}} = N_\tau \tau_0. \tag{2.92}$$

The transpose relation follows from the mechanical power $\tau^\top \dot{q}$ between joint torques τ and joint velocities $\dot{q}$. Thus, N_q is applied to joint velocities $\dot{q}$, while N_τ is applied to secondary joint torques τ_0.

The projector provides an orthogonal kinematic separation with respect to the Jacobian and the Moore–Penrose pseudoinverse. It is not the inertia-weighted dynamically consistent null-space projector used in some operational-space formulations. Cartesian task retention under the implemented secondary torques is therefore also evaluated experimentally.

For the SVD-based pseudoinverse, J^+J is an orthogonal and symmetric projector. Hence,

$$N_q = N_\tau \tag{2.93}$$

holds analytically. The separate symbols distinguish the quantity on which the projector acts. In the implementation, N_τ is additionally symmetrised to reduce floating-point asymmetry.

When J has full row rank and all six task-producing singular directions are retained,

$$N_q = N_\tau = v_7 v_7^\top. \tag{2.94}$$

In this case, projection retains only the component along the redundant direction v_7. When a singular value lies at or below the selected threshold, its associated right-singular direction is also assigned to the numerical null space.

2.8.4 Null-Space Damping Torque

The current joint velocity $\dot{q}$ is separated into task-related and null-space components according to Equation (2.86). The null-space damping torque τ_d acts on the component $N_q\dot{q}$:

$$\tau_d = -d_{\text{null}} N_\tau \dot{q} = -d_{\text{null}} N_q \dot{q}, \tag{2.95}$$

where $d_{\text{null}} \geq 0$ is the null-space damping coefficient. The negative sign makes the torque oppose motion along the redundant joint-space direction and dissipate the associated motion.

2.8.5 Singular-Value Conditioning Torque

The second contribution applies an active torque along the redundant direction v_7. Its sign is selected by comparing $\sigma_{\min}$ at two nearby joint configurations.

The probe angle α_{probe} specifies the joint-space step from the current configuration q. The sampled configurations in the positive and negative null directions are q_+ and q_-:

$$q_+ = q + \alpha_{\text{probe}} v_7, \qquad q_- = q - \alpha_{\text{probe}} v_7. \tag{2.96}$$

The Jacobian $J(q)$ is recalculated at both configurations. The minimum singular values σ_+ and σ_- correspond to q_+ and q_-, respectively:

$$\sigma_+ = \sigma_{\min}(J(q_+)), \qquad \sigma_- = \sigma_{\min}(J(q_-)). \tag{2.97}$$

The sampled singular-value difference Δ_σ is

$$\Delta_\sigma = \sigma_+ - \sigma_-. \tag{2.98}$$

The singular-value-conditioning direction selector s_σ is defined as

$$s_\sigma = \begin{cases} +1, & \Delta_\sigma > \delta_\sigma, \\ -1, & \Delta_\sigma < -\delta_\sigma, \\ 0, & |\Delta_\sigma| \leq \delta_\sigma, \end{cases} \tag{2.99}$$

where $\delta_\sigma \geq 0$ is the comparison deadband. Within this deadband, the selector s_σ is set to zero and the conditioning contribution is suppressed.

The selected direction is invariant to the sign convention of v_7 returned by the SVD. Reversing v_7 exchanges q_+ and q_- and reverses s_σ, so the product $s_\sigma v_7$ remains unchanged.

The conditioning-torque magnitude k_σ is specified independently of the probe angle.

The vector v_7 represents the instantaneous null-space direction at the current configuration. For a finite probe angle, q_+ and q_- provide local approximations of the nonlinear constant-pose self-motion. The corresponding preservation of the end-effector pose is therefore a first-order property of the sampled direction.

The selector s_σ uses the sign of Δ_σ to choose the sampled direction with the larger value of $\sigma_{\min}$. This produces a two-point directional conditioning strategy based on local comparison rather than on the magnitude of the sampled variation.

The vector $s_\sigma v_7$ defines the selected secondary joint-space direction. Projecting it with N_τ gives the singular-value conditioning torque τ_σ:

$$\tau_\sigma = k_\sigma N_\tau \left(s_\sigma v_7\right), \tag{2.100}$$

where $k_\sigma \geq 0$ specifies the commanded torque magnitude.

When the Jacobian has full row rank and all six task-producing singular directions are retained, v_7 lies in the one-dimensional null space and

$$N_\tau v_7 = v_7. \tag{2.101}$$

The conditioning contribution therefore acts along the selected redundant joint direction, while the damping contribution opposes motion within the projected subspace.

2.8.6 Complete Null-Space Torque

Combining the damping and conditioning contributions gives

$$\tau_{\text{null}} = \tau_d + \tau_\sigma = -d_{\text{null}} N_\tau \dot{q} + k_\sigma N_\tau \left(s_\sigma v_7\right) \in \mathbb{R}^7. \tag{2.102}$$

When the Jacobian $J(q)$ has full row rank, its null space is one-dimensional and is spanned by v_7. If the Jacobian loses row rank, the null space becomes multidimensional and the projector includes the additional numerical-null-space directions. The damping contribution acts on motion in all these directions through $N_\tau \dot{q}$. The implemented conditioning term retains its two probes along $+v_7$ and $-v_7$.

The two contributions therefore have different roles: τ_d opposes redundant joint motion that is already present, whereas τ_σ applies an active torque that moves the redundant configuration along the selected direction toward the nearby sampled configuration with the larger $\sigma_{\min}$.

The activation of the damping and conditioning contributions is a controller configuration choice. The available combinations and the configuration used in the reported experiments are described in Section 3.6.

2.9 Complete Joint-Torque Command

The complete joint-torque command τ_{cmd} combines the Cartesian impedance torque τ_{cart}, the null-space torque τ_{null}, and the Coriolis compensation τ_c:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_{\text{null}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_{\text{null}} + \tau_c(q, \dot{q}). \tag{2.103}$$

Here, $\tau_{\text{cart}} = J^\top(q)F$ is the Cartesian impedance torque, τ_{null} is the secondary null-space torque, and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal compensation obtained from the robot model and added in the control callback. A separate gravity term is not added because gravity compensation is handled internally by the Franka Control Interface, as described in Section 2.4.3. The next chapter implements these torque contributions in the real-time control loop and combines them with the surface-relative geometry, phase-dependent references, and contact sequence.

Chapter 3

Controller Design and Real-Time Implementation

This chapter describes how the Cartesian impedance formulation of Chapter 2 is implemented as a real-time controller for the Franka Emika Panda. The controller is written in C++ using libfranka and Eigen and is executed through the Franka Control Interface (FCI) with a nominal control period of 1 ms.

The implementation is organised around the control flow required for the surface-alignment task. A surface-relative task representation first defines the surface frame, the tool orientation, and the point of the rectangular face used by the contact sequence. The real-time callback then updates the Cartesian reference, evaluates the impedance wrench, and maps it to joint torque. During set-up, the directional impedance can be shifted to a virtual CoC p_c, so that the commanded normal press also produces a moment. A secondary SVD-based null-space controller acts on the redundant joint motion without changing the Cartesian task to first order.

The architecture is introduced first, followed by the task geometry, real-time impedance calculation, phase-dependent gains and virtual CoC, contact sequence, null-space controller, and supporting real-time, safety, and recording functions. Numerical values used in the reported experiments are collected in Chapter 4 and Appendix C.

3.1 Controller Architecture

The software separates configuration and operator interaction from the real-time control calculation and from post-run data processing. The complete parameter set is read and validated before a controller session starts. The real-time callback then receives only the prepared configuration, preallocated state, robot-model interface, and non-blocking operator requests. File output and run evaluation are performed after torque control has stopped.

Table 3.1 summarises the main functional subsystems.

Table 3.1: Functional subsystems of the surface-grinding controller.

Subsystem	Responsibility
Configuration	Validates the parameter set and constructs the typed controller configuration.
Task geometry	Constructs the surface frame, tool orientation target, tool geometry, and geometric clearance quantities.
Real-time controller	Updates the active phase and Cartesian reference and evaluates the 1 kHz joint-torque callback.
Impedance and null-space control	Constructs the active Cartesian gains, optional compliance-centre shift, Cartesian wrench, and secondary null-space torque.
Operator and robot interface	Manages mode selection, manual guidance, robot preparation, gripper actions, and tool handling.
Recording and evaluation	Buffers controller signals during torque control and performs file output and run-summary calculations afterwards.

The main operating path is a phase sequence for tool orientation, surface approach, contact set-up, and grinding. Additional modes provide standard Cartesian pose hold, pose hold with the set-up impedance, and manual guidance. The same configuration system supplies the surface and tool geometry, phase-dependent Cartesian gains, damping settings, virtual CoC, null-space parameters, safety thresholds, and data-recording settings.

At program start, the robot connection is established and the robot model is loaded after error recovery and collision configuration. A separate keyboard thread receives operator requests. These requests are transferred through atomic variables, keeping terminal input outside the real-time callback. Before each new controller session, the configuration is read and validated again, which permits settings to be changed between runs without recompiling the program.

The torque path used during one impedance-controlled cycle is shown in Figure 3.1. The figure also indicates where the phase state, active Cartesian gains, robot model, and null-space contribution enter the command.

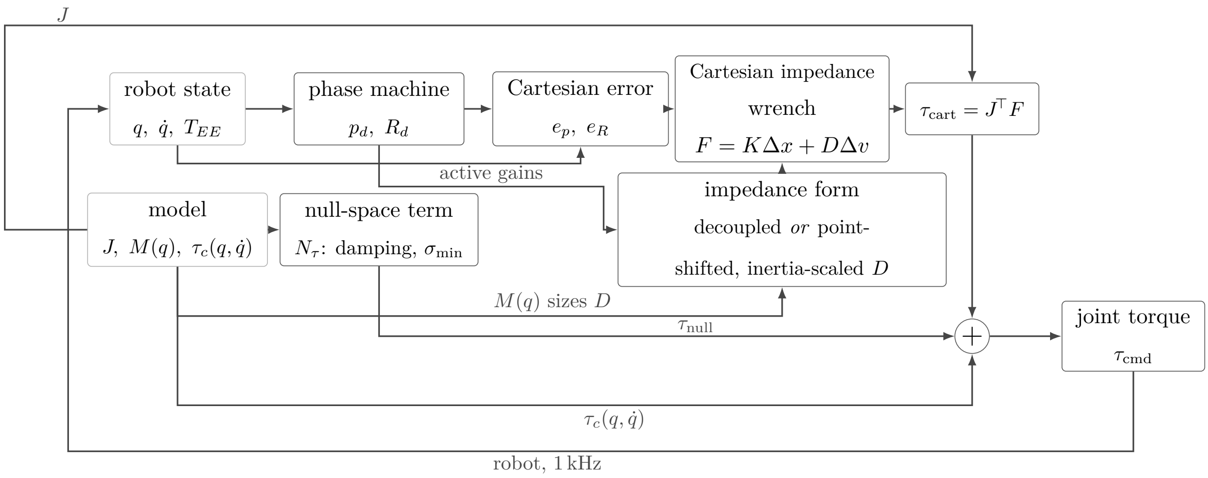


Figure 3.1: Torque contributions combined in one control cycle.

The following sections describe the task geometry, impedance calculation, phase logic, null-space controller, and real-time support functions in sequence.

3.2 Surface-Relative Task Representation

The contact sequence is formulated relative to the calibrated surface and to the known geometry of the tool face. The controller therefore constructs a surface task frame, a surface-relative tool orientation target, and a geometric reference point on the rectangular face before the set-up impedance is applied.

3.2.1 Surface Task Frame

The mathematical surface-frame construction is introduced in Section 2.5. In the implementation, the parameter set contains a point p_s on the surface, the surface orientation, and a surface-frame reference direction a_s used to define the first tangent. The resulting rotation has the fixed column order

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}.$$

The reference direction a_s is projected onto the plane orthogonal to n_s and normalised to obtain t_1. The second tangent is constructed as

$$t_2 = n_s \times t_1,$$

which completes the right-handed frame in robot-base coordinates. A fallback reference direction is used when the configured direction is approximately parallel to n_s.

This surface frame is used consistently for the approach direction, the surface-relative orientation command, directional Cartesian gains, the compliance-centre definition when a surface-fixed lever is selected, and the surface-resolved quantities used in the experimental evaluation. The descent direction is

$$-n_s,$$

from the free-space starting region towards the configured plane. The calibrated surface values used for the reported experiments are given in Chapter 4 and Appendix C.

3.2.2 Surface-Relative Tool Orientation

The tool normal is calibrated once in the end-effector frame. Its current direction in the robot base frame is calculated from the measured end-effector orientation:

$$n_{\text{Tool},0} = R_{\text{EE}} n_{\text{Tool,EE}}. \tag{3.1}$$

The calibrated surface normal n_s points outward from the surface towards the free-space region and therefore defines the flat tool-normal target. The commanded angular offsets about the two surface tangents are first combined into the rotation vector

$$\theta_{\text{tilt}} = \theta_{t_1} t_1 + \theta_{t_2} t_2 \in \mathbb{R}^3. \tag{3.2}$$

Its magnitude gives the total commanded tilt angle,

$$\phi_{\text{tilt}} = \|\theta_{\text{tilt}}\| = \sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}. \tag{3.3}$$

For $\phi_{\text{tilt}} > 0$, the corresponding unit rotation axis is

$$u_{\text{tilt}} = \frac{\theta_{\text{tilt}}}{\phi_{\text{tilt}}} = \frac{\theta_{t_1} t_1 + \theta_{t_2} t_2}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}}. \tag{3.4}$$

Using Rodrigues' relation of Equation (2.15), the pair $(u_{\text{tilt}}, \phi_{\text{tilt}})$ defines the rotation matrix R_{tilt}. Applying this rotation to the outward surface normal gives the desired tool-normal direction,

$$n_d = R_{\text{tilt}} n_s. \tag{3.5}$$

For zero commanded tilt, $\theta_{t_1} = \theta_{t_2} = 0$, the rotation is the identity and $n_d = n_s$.

The shortest rotation that aligns the captured tool direction with n_d is applied to the complete start orientation. The implementation can additionally prescribe a twist about the desired tool axis. The long axis of the rectangular face and the selected surface tangent are projected onto the plane normal to n_d, and the requested in-plane rotation is then applied about that axis. Because a centred rectangle is unchanged by a half-turn, the implementation selects the equivalent orientation closest to the captured start pose.

The construction separates the two orientation roles used later in the contact sequence: θ_{t_1} and θ_{t_2} define the commanded inclination relative to the surface, whereas the normal-axis twist defines the in-plane orientation of the rectangular face. The exact settings used for the measurements are stated in Chapter 4.

3.2.3 Tool Geometry and Geometric Clearance

The tool face is represented as a rectangle fixed in the end-effector frame. Its centre offset, half-width vector, and half-length vector define the four vertices

$$\{r_{\text{face,EE}} \pm r_{\text{width,EE}} \pm r_{\text{length,EE}}\}\,. \tag{3.6}$$

The tool geometry determines the point expected to reach the surface first. This point is denoted by p_T and is used as the contact reference during set-up; here the subscript T denotes the selected tool point. Depending on the tool orientation it corresponds to a leading corner, the midpoint of a leading edge, or the tool-face centre. It is distinct from the TCP and from the virtual CoC. Its position follows from the end-effector-frame offset $r_{T,\text{EE}}$,

$$p_T = p_{\text{TCP}} + R_{\text{EE}} r_{T,\text{EE}}, \tag{3.7}$$

where the tool centre point coincides with the reported end-effector origin in this implementation, $p_{\text{TCP}} = p_{\text{EE}}$. The offset $r_{T,\text{EE}}$ is fixed relative to the tool once the contact reference has been selected. Its base-frame representation $r_T = R_{\text{EE}} r_{T,\text{EE}} = p_T - p_{\text{TCP}}$ changes with the tool orientation. The compliance-centre shift $r_c = p_c - p_{\text{TCP}}$ of Section 2.7 is a different quantity: it is the virtual displacement used by the Cartesian impedance transformation, and it carries no tool geometry at all.

During tool orientation and surface approach, all four vertices are transformed to the base frame and compared along the descent direction. The vertex with the largest projection is geometrically closest to the surface in that direction. Vertices whose projected positions agree within the configured tie tolerance are grouped and averaged. A general orientation therefore selects a leading corner, two nearly equal neighbouring corners define a leading edge, and four nearly equal corners define the face centre.

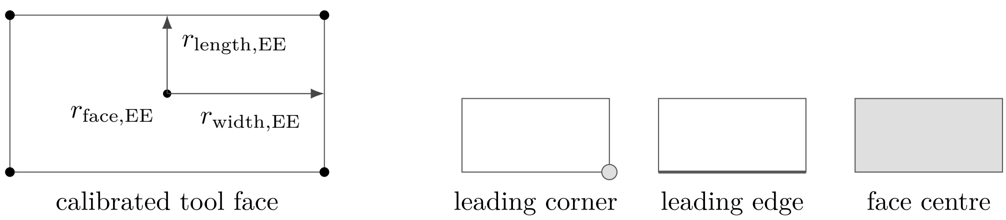


Figure 3.2: Tool face geometry and the selected leading feature.

The controller distinguishes the clearance of the foremost face point from the height of the averaged controlled point. The foremost-face clearance is used only to determine the transition from approach to set-up. The averaged point is the reference point used for the subsequent set-up motion.

At the geometric clearance transition, the implementation stores

- the selected tool-point offset in the end-effector frame;
- its projection onto the calibrated surface;
- the measured TCP pose and the orientation that will be held during set-up; and
- the current model-estimated external wrench as a reference for later changes.

The selected face offset is then kept fixed during set-up. The desired contact-point position is generated relative to its projected surface point and converted back to the corresponding TCP reference. This allows the phase logic to command the geometrically relevant point on the tool face while the Cartesian impedance law remains formulated at the TCP.

3.3 Real-Time Cartesian Impedance Controller

One controller session prepares the task geometry and phase-dependent gains and then registers a joint-torque callback with libfranka. Before torque control begins, the current robot state is read once to initialise the desired pose, phase state, external-wrench reference, and preallocated recording buffer.

The callback receives the current `RobotState` and the elapsed cycle duration and returns the seven commanded joint torques τ_{cmd}:

Listing 3.1: Registration of the joint-torque callback through `libfranka`.

```
robot.control(
    [&](const franka::RobotState& state,
        franka::Duration period) -> franka::Torques {

      // Real-time state and torque calculations

      return franka::Torques(tau_array);
    });
```

Within each callback, the measured joint state, end-effector pose, end-effector Jacobian, robot-model quantities, and model-estimated external wrench are read first. The phase logic then updates the desired Cartesian reference. The controller evaluates the Cartesian error, selects the active impedance branch, maps the resulting wrench through $J^\top$, adds the secondary torque terms, records the required signals, and returns the final joint-torque vector.

3.3.1 State and Reference Update

The robot state provides the measured joint configuration q, joint velocity $\dot{q}$, and end-effector transformation. The libfranka model returns the 6×7 geometric Jacobian $J(q)$ of the configured end-effector frame together with the joint-space mass matrix and Coriolis terms. Multiplication of the Jacobian by $\dot{q}$ gives the measured Cartesian linear and angular velocities in the base frame.

The end-effector transformation supplied by the robot state is used directly for p_{EE} and R_{EE}; the real-time controller does not reconstruct this pose from a separate Denavit–Hartenberg chain.

The phase state machine supplies the desired TCP position p_d, desired linear velocity $\dot{p}_d$, and desired orientation R_d. Their meaning depends on the active phase: tool orientation changes the orientation reference while holding position, approach advances the position reference along $-n_s$, set-up advances the selected tool point into the surface, grinding maintains the normal press and can add a tangential sweep, and pose hold regulates a captured Cartesian pose. The detailed phase logic is described in Section 3.5.

3.3.2 Cartesian Error and Impedance Wrench

The translational error is the desired-minus-measured position error. The orientation error follows the convention introduced in Section 2.2.4. The current-to-desired relative rotation is converted to an axis–angle vector and expressed in the robot base frame:

$$\Delta R = R_{\mathrm{EE}}^{\top} R_d, \qquad e_R = R_{\mathrm{EE}} u \phi. \tag{3.8}$$

A rotational mask can be applied after this transformation by resolving e_R in the surface frame, selecting the enabled components, and transforming the result back to the base frame. The mask acts on the rotational spring error; the damping term continues to use the measured angular velocity.

The six-dimensional displacement and velocity errors are

$$\Delta x = \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \qquad \Delta v = \begin{bmatrix} \dot{p}_d - \dot{p}_{\mathrm{EE}} \\ -\omega_{\mathrm{EE}} \end{bmatrix}. \tag{3.9}$$

For the decoupled impedance branch, the Cartesian wrench is evaluated from the translational and rotational blocks separately:

$$F = \begin{bmatrix} K_p e_p + D_p(\dot{p}_d - \dot{p}_{\mathrm{EE}}) \\ K_R e_R - D_R \omega_{\mathrm{EE}} \end{bmatrix}. \tag{3.10}$$

The same force–moment ordering is used for the coupled branch, but the 6×6 stiffness and damping matrices are first shifted from the selected CoC to the TCP as described in Section 3.4.3. The returned wrench is denoted

$$F = \begin{bmatrix} f \\ m \end{bmatrix} \in \mathbb{R}^6. \tag{3.11}$$

3.3.3 Joint-Torque Command

The Cartesian wrench is mapped to joint torque through

$$\tau_{\mathrm{cart}} = J^\top(q) F. \tag{3.12}$$

The SVD-based secondary controller supplies τ_{null}. During the dedicated Cartesian pose-hold disturbance study of Section 4.6, the callback also adds the internally commanded joint-torque-equivalent disturbance τ_{dist}. Outside this experiment, $\tau_{\mathrm{dist}} = 0$.

The Coriolis and centrifugal contribution $\tau_c(q, \dot{q})$ is obtained from the libfranka model. The complete impedance-controlled command is therefore

$$\tau_{\mathrm{cmd}} = \tau_{\mathrm{cart}} + \tau_{\mathrm{null}} + \tau_{\mathrm{dist}} + \tau_c(q, \dot{q}). \tag{3.13}$$

No separate gravity term is added because the Franka Control Interface handles gravity compensation for the external joint-torque command, as described in Chapter 2. The final torque vector is returned directly to libfranka; robot-side monitoring and the configured collision behaviour remain active.

3.4 Phase-Dependent Impedance and Centre of Compliance

The active Cartesian impedance is determined by three choices: the directional stiffness, the damping associated with those directions, and whether the impedance reference point is the TCP or a shifted virtual CoC. The controller constructs the directional gains first and applies the compliance-centre transformation afterwards.

3.4.1 Directional Cartesian Gains

Directional gains are defined in the frame that gives them a physical meaning for the active task. A diagonal gain G_{surface} expressed along $[t_1, t_2, n_s]$ is transformed to the base frame by

$$G_0 = R_{\text{surface}} G_{\text{surface}} R_{\text{surface}}^{\top}. \tag{3.14}$$

The base-frame matrix can then be applied directly to the base-frame Cartesian errors used by the control law. The implemented frame policy is summarised in Table 3.2.

Table 3.2: Coordinate frames used for the phase-dependent Cartesian gains.

Controller mode	**Translational gains**	**Rotational gains**
Tool orientation and surface approach	Surface frame $[t_1, t_2, n_s]$	Surface frame $[t_1, t_2, n_s]$
Set-up and grinding	Selectable between base frame and surface frame	Surface frame $[t_1, t_2, n_s]$
Cartesian pose hold	Base frame $[x, y, z]$	Base frame $[x, y, z]$
Set-up-impedance hold	Same construction as set-up	Same construction as set-up
Phase-gate hold	Configurable hold frame	Configurable hold frame

The set-up phase is the central case for the contact experiments. Its translational and rotational gains can be aligned with the surface frame so that normal and tangential stiffnesses can be selected independently. The specific matrices used in each experimental case are intentionally left to Chapter 4, where the varied and fixed entries are defined.

Set-up and grinding use the same source stiffness and damping blocks. The phase transition changes the generated Cartesian reference and the choice between the coupled and decoupled impedance branch. A phase gate can temporarily replace the active matrices with dedicated hold gains while the current reference is maintained.

3.4.2 Inertia-Scaled Damping

For phase groups configured to use inertia-scaled damping, the damping coefficients are calculated from the active directional stiffness and the Cartesian inertia associated with the current robot configuration. The joint-space inertia matrix

$$M(q) \in \mathbb{R}^{7\times 7} \tag{3.15}$$

is obtained from the libfranka model. The operational-space inertia in the robot base frame is evaluated from

$$\Lambda_0(q) = \Big(J(q)M(q)^{-1}J^\top(q) + \varepsilon I_6\Big)^{-1}, \qquad \varepsilon = 10^{-9}. \tag{3.16}$$

The inverse $M(q)^{-1}$ is not formed explicitly in the implementation. Instead, a lower–diagonal–lower-transpose (LDLT) factorisation of $M(q)$ is used to solve

$$M(q)Y(q) = J^\top(q), \qquad Y(q) = M(q)^{-1}J^\top(q) \in \mathbb{R}^{7\times 6}. \tag{3.17}$$

Here, $Y(q)$ is only the solution matrix of the linear system. The same operational-space inertia can therefore be evaluated numerically as

$$\Lambda_0(q) = \left(J(q)Y(q) + \varepsilon I_6\right)^{-1}. \tag{3.18}$$

The base-frame matrix is then transformed into the coordinate frame of the active gain block,

$$\Lambda_{\text{task}}(q) = T_R^\top \Lambda_0(q) T_R, \qquad \lambda_i(q) = \left[\Lambda_{\text{task}}(q)\right]_{ii}. \tag{3.19}$$

Thus, each directional stiffness entry k_i is paired with the corresponding Cartesian inertia $\lambda_i(q)$ of the same translational or rotational direction. The directional damping coefficient is then

$$d_i(q) = 2\zeta\sqrt{\lambda_i(q)k_i}, \tag{3.20}$$

where ζ is the configured directional damping factor. For a single decoupled direction with $\zeta = 1$, this expression corresponds to the critical-damping coefficient of that local second-order model.

The calculation is performed independently for the three translational and three rotational directions and is subject to the configured coefficient bounds and numerical validity checks. The resulting damping is cached by phase group and recalculated when that group is entered again, so the current robot configuration is reflected in $M(q)$, $J(q)$, and therefore in $\Lambda_{\text{task}}(q)$. If the calculation cannot be used, the configured damping matrix remains available as a fallback.

The reported surface-contact sequence used inertia-scaled damping in the phase groups stated in Chapter 4. Standard Cartesian pose hold is a separate case: its reported null-space trials used fixed hold damping rather than this inertia-scaled calculation.

3.4.3 Virtual Centre of Compliance

The theoretical point-shift formulation is derived in Section 2.7. In the implementation, the source translational and rotational gains are first expressed in the required task frame and assembled at a selected virtual CoC p_c. They are then shifted to the TCP.

The implemented lever convention is

$$r_c = p_c - p_{\text{TCP}}.$$

Two definitions are supported because the meaning of a fixed lever depends on the frame in which it is specified.

For a tool-fixed definition, the configured centre displacement is constant in end-effector coordinates. The corresponding base-frame lever is therefore updated from the measured end-effector orientation during every callback. It rotates in the base frame as the end effector rotates.

For a surface-fixed definition, the lever components are constant in $[t_1, t_2, n_s]$. Because the surface frame is fixed for one controller session, the resulting base-frame lever is constant during set-up. The configuration loader requires exactly one of the two definitions to be active when the virtual centre is enabled, and rejects a parameter set that selects both or neither.

The active base-frame lever is used to construct the point-shift adjoint. The source translational and rotational matrices are assembled into block-diagonal spatial gains at p_c, and the same congruence is applied independently to stiffness and damping:

$$K_{\mathrm{TCP}} = \mathrm{Ad}^\top(r_c) K_c \,\mathrm{Ad}(r_c), \qquad D_{\mathrm{TCP}} = \mathrm{Ad}^\top(r_c) D_c \,\mathrm{Ad}(r_c). \tag{3.21}$$

For a tool-fixed lever, r_c and therefore the shifted base-frame matrices are updated as the measured end-effector orientation changes. For a surface-fixed lever, the base-frame lever remains constant and the shifted matrices remain fixed as long as the source gains do not change.

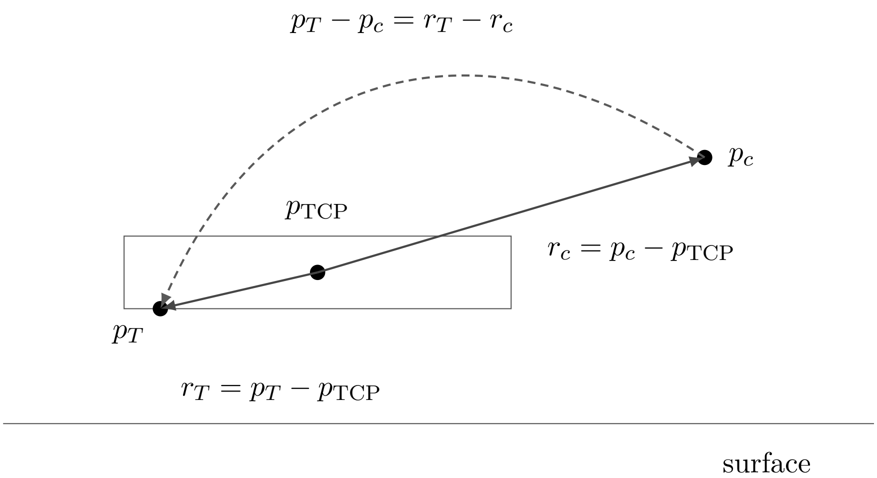


Figure 3.3: Selected tool point, tool centre point, and centre of compliance.

The off-diagonal blocks generated by the point shift introduce translation–rotation coupling. A translational position or velocity error can therefore contribute to the commanded TCP moment, and a rotational error can contribute to the commanded TCP force. When $r_c = 0$, the point shift disappears and the spatial matrices reduce to the decoupled block-diagonal form.

This coupling is the mechanism used for contact-induced alignment during set-up. The desired orientation is held at the value captured at the geometric clearance transition, so no time-varying tipping trajectory is commanded. Instead, the selected tool point is advanced towards the surface. The resulting normal press acts through the finite Cartesian impedance; when a non-zero compliance-centre lever is enabled, the shifted impedance introduces a corresponding moment that can assist or oppose the contact-induced rotation. The measured response also depends on the directional stiffness, contact geometry, robot configuration, and physical tool–surface interaction.

The coupled branch is used only in the controller modes configured for a virtual CoC. Other phases use the decoupled Cartesian impedance. The exact cases that used the point-shifted branch in the reported experiments are specified in Chapter 4 and Appendix C.

3.5 Contact Sequence and Reference Generation

The main run sequence passes through tool orientation, surface approach, set-up, and grinding, in that order. The orientation phase can be omitted by configuration, and optional operator gates can hold the current reference between phases. Standard Cartesian pose hold, set-up-impedance hold, and manual guidance are separate operator modes. Figure 3.4 summarises the implemented phase flow.

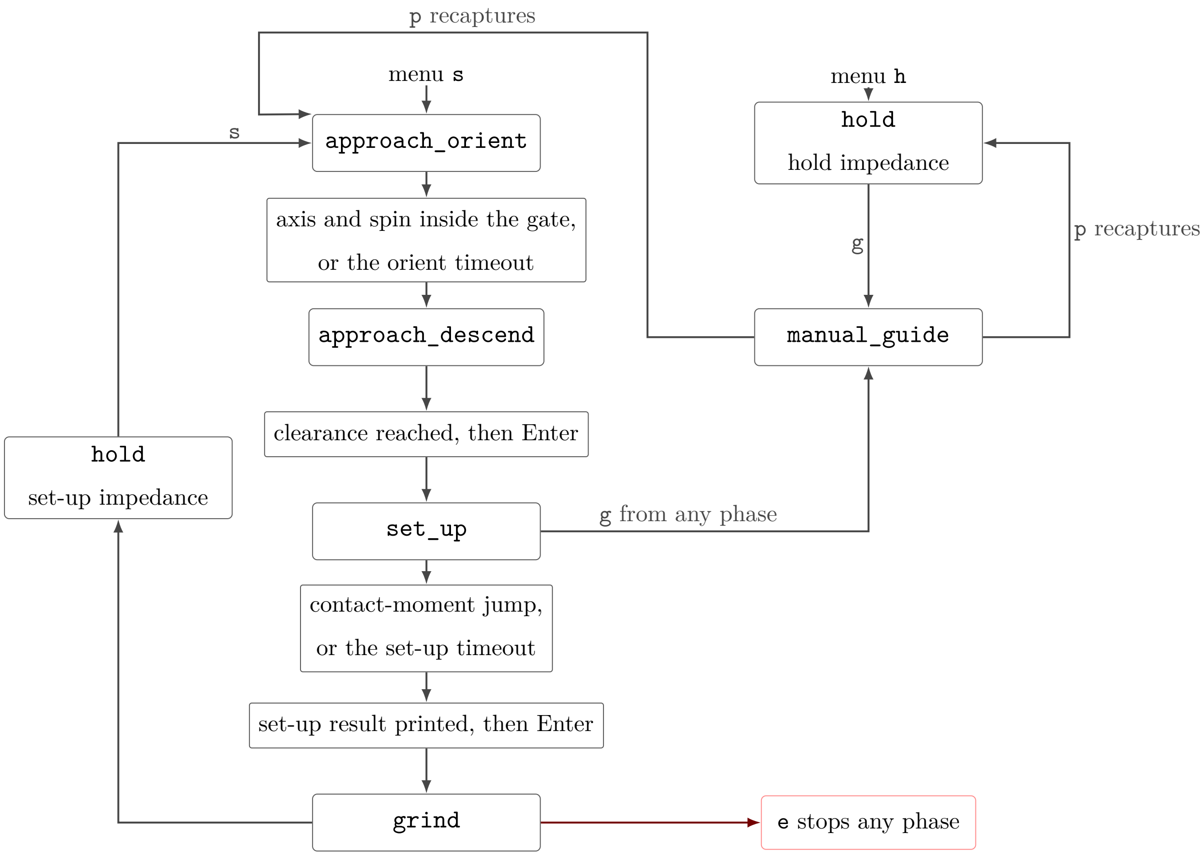


Figure 3.4: Run phase sequence and Cartesian pose hold.

3.5.1 Tool Orientation

The tool-orientation phase keeps the captured start position fixed while the desired orientation is advanced towards the surface-relative target of Section 3.2.2. The reference follows a rate-limited interpolation from the captured start orientation.

The transition is evaluated from the tool-axis error and, when the in-plane twist is enabled, the remaining spin error about the target tool axis. A configured minimum duration prevents immediate acceptance of a transient orientation, while a timeout provides an alternative transition path. The orientation reached at the end of this phase becomes the reference used during surface approach.

3.5.2 Surface Approach

During surface approach, the desired position advances from the captured start position along $-n_s$ with a rate limit and maximum travel. The rectangular tool geometry is evaluated continuously during this motion, so the controller always knows which corner or edge currently has the smallest geometric clearance to the surface.

The transition to set-up occurs when the foremost-face clearance reaches the configured geometric threshold. This is not a force-detected contact event. The model-estimated external wrench is recorded during approach, but it does not determine the phase transition.

At the clearance transition, the selected tool point is selected and held constant, and the controller stores its projected surface point, the current TCP pose, the orientation reference, and the external-wrench reference. If the optional pre-set-up gate is enabled, the controller first holds the captured pose until operator confirmation and then captures the set-up references from the current measured state. Reaching the maximum permitted approach distance before the geometric clearance is satisfied terminates the approach.

3.5.3 Set-Up

Set-up generates the contact press that is evaluated in the experiments. The calibrated normal n_s points outward from the surface, so the descent direction towards the surface is $-n_s$. At the clearance transition, the selected tool point is projected onto the calibrated plane. This projected point is denoted by $p_{T,0}$ and provides the starting reference for set-up. The scalar coordinate $s_{\mathrm{set}}(t) \geq 0$ describes the commanded displacement from this point towards the surface and is integrated at the configured speed and clamped at its configured endpoint. The desired position of the selected tool point is therefore

$$p_{T,d}(t) = p_{T,0} - s_{\mathrm{set}}(t) n_s. \tag{3.22}$$

Here, $p_{T,d}(t)$ is expressed in the robot base frame. The Cartesian impedance controller itself regulates the TCP, so this desired tool-point position must be converted into the corresponding desired TCP position.

The vector $r_{T,\mathrm{EE}}$ points from the TCP to the selected tool point and is expressed in the end-effector frame. At the clearance transition, the desired orientation for set-up is captured as R_{clr}. Expressing the stored tool-point offset in the robot base frame at this orientation gives

$$r_{T,\mathrm{clr}} = R_{\mathrm{clr}} r_{T,\mathrm{EE}}. \tag{3.23}$$

The desired tool-point position and desired TCP position are related by the same rigid-body geometry,

$$p_{T,d}(t) = p_d(t) + r_{T,\mathrm{clr}}. \tag{3.24}$$

Solving this relation for the desired TCP position gives

$$p_d(t) = p_{T,d}(t) - r_{T,\mathrm{clr}} = p_{T,d}(t) - R_{\mathrm{clr}} r_{T,\mathrm{EE}}. \tag{3.25}$$

Thus, $p_d(t)$ is the TCP reference required to place the selected tool point at $p_{T,d}(t)$ while maintaining the set-up orientation R_{clr}. The surface restricts the commanded motion of the tool point, so the resulting Cartesian position error generates the press.

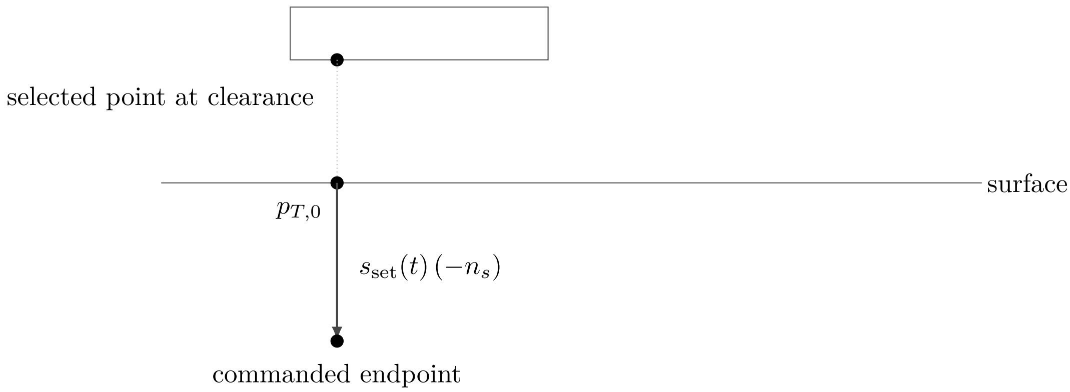


Figure 3.5: Set-up reference generation for the selected tool point.

The desired orientation remains equal to the orientation captured at the clearance transition. It therefore acts as a finite-stiffness rotational reference rather than as a rigid pose constraint. Contact-induced rotation can occur when the Cartesian wrench overcomes the corresponding rotational spring. When the virtual CoC is enabled, the point-shifted matrices of Section 3.4.3 additionally couple the translational press to the commanded moment.

The model-estimated external wrench is referenced to the value captured at the clearance transition. For the force component, $\Delta \hat{f}_{\mathrm{ext}}$ is resolved in the surface frame as

$$R_{\text{surface}}^{\top}\Delta\hat{f}_{\text{ext}} = \begin{bmatrix} t_1^{\top}\Delta\hat{f}_{\text{ext}} & t_2^{\top}\Delta\hat{f}_{\text{ext}} & n_s^{\top}\Delta\hat{f}_{\text{ext}} \end{bmatrix}^{\top}. \tag{3.26}$$

The last component is the signed model-estimated normal-force change. The commanded wrench $F = [f^{\top}, m^{\top}]^{\top}$ and the model-estimated external wrench remain separate signals. The former is generated by the impedance controller; the latter is supplied by the robot state and is used as a supporting contact quantity.

After the configured minimum duration, set-up can terminate through its moment-change condition or through the independent timeout. If the pre-grinding gate is active, the controller remains in the set-up branch while waiting for operator confirmation. The coupled impedance therefore remains active when it was selected for that run. The exact transition settings and the termination behaviour observed in the reported experiments are documented in Chapter 4.

3.5.4 Grinding

The grinding phase maintains the normal set-up reference and can superimpose a tangential sweep along a selected surface tangent. The sweep uses a smooth fifth-order trajectory with zero velocity at its reversal points. When the sweep is disabled, only the normal contact reference is maintained.

Grinding uses the same source stiffness and damping blocks as set-up, but the implemented wrench calculation returns to the decoupled impedance branch after the phase change. The reported contact measurements in this thesis ended at the pre-grinding gate; no reported run entered the grinding sweep. The grinding phase is therefore included here as implemented controller functionality rather than as part of the evaluated contact data.

3.5.5 Pose Hold, Manual Guidance, and Run Termination

Standard Cartesian pose hold regulates a captured pose with the dedicated hold impedance. Set-up-impedance hold uses the set-up gain construction and can retain the virtual CoC. These hold modes are also used for dedicated controller checks outside the main contact sequence.

Manual guidance can be entered from an impedance-controlled state. The controller then returns the Coriolis contribution together with joint-space damping,

$$\tau_{\text{guide}} = \tau_c(q, \dot{q}) - d_{\text{guide}}\dot{q}, \tag{3.27}$$

where d_{guide} is the configured guidance damping. After operator confirmation, the measured pose, joint configuration, and required references are recaptured before the selected sequence or hold mode resumes.

A controller session ends when the configured duration expires, an operator stop is requested, the maximum approach distance is exceeded, or libfranka terminates the motion because of a robot-side event. File writing and post-run evaluation occur only after the real-time callback has finished.

3.6 SVD-Based Null-Space Control

The Cartesian impedance defines the primary task. The Panda has one additional joint-space degree of freedom when the 6×7 task Jacobian has full row rank, so the implementation includes a secondary null-space controller for redundant motion. The theoretical projector, damping term, and two-point singular-value-conditioning strategy are derived in Section 2.8; this section describes their execution in the real-time callback.

For each impedance-controlled cycle, the full singular value decomposition of $J(q)$ is evaluated. The relative SVD tolerance is multiplied by the largest current singular value to obtain the threshold used in the Moore–Penrose pseudoinverse. Singular directions above the threshold are inverted; the remaining reciprocal entries are set to zero. The resulting joint-torque null-space projector N_τ is symmetrised numerically.

The projected joint velocity and damping torque are

$$\dot{q}_{\text{null}} = N_\tau \dot{q}, \qquad \tau_d = -d_{\text{null}}\dot{q}_{\text{null}}. \tag{3.28}$$

For the Moore–Penrose SVD construction used here the orthogonal projector is symmetric, so $N_\tau = N_q$ analytically. The implementation uses the numerically symmetrised N_τ both for the secondary torque and for the recorded projected joint velocity. The two symbols are kept apart in Section 2.8 only to distinguish the quantity the projector acts on.

The controller provides the four secondary-torque modes listed in Table 3.3.

Table 3.3: Selectable null-space torque contributions.

Mode	**Returned secondary torque**
Disabled	0
Damping only	τ_d
Sigma only	τ_σ
Combined	$\tau_d + \tau_\sigma$

For singular-value conditioning, the current redundant right-singular direction v_7 is sampled in both signs. Two nearby joint configurations are formed at $q \pm \alpha_{\text{probe}} v_7$ and their Jacobians are evaluated using the current robot transformations. The smallest singular values at the two sampled configurations are compared. If their difference exceeds the configured deadband, the sign associated with the larger $\sigma_{\min}$ is selected; within the deadband, no conditioning torque is requested.

The selected direction is projected again through N_τ before the conditioning torque is returned. This keeps the commanded contribution in the numerical null space defined by the current pseudoinverse. The combined mode adds this contribution to the projected joint-velocity damping.

The surface-contact experiments used one fixed null-space configuration, while the dedicated pose-hold study varied the secondary terms to examine them separately. The corresponding numerical settings belong to the experimental method and are given in Chapter 4, Section 4.6, and Appendix C.

3.7 Real-Time Operation, Safety, and Data Recording

The functions around the control law are designed so that robot preparation, operator interaction, and file handling do not add blocking operations to the joint-torque callback. Safety monitoring remains active on the robot side, and the signals required for the experimental evaluation are written to preallocated memory during control.

3.7.1 Robot Preparation and Safety

Before a torque-controlled session begins, the software establishes the FCI connection, requests error recovery, applies the selected collision behaviour, and loads the robot model, in that order.

The robot can then be moved to a configured initial joint posture using a libfranka joint-motion generator, or the operator can start from a pose captured after manual guidance. The initial posture used in the reported experiments is documented in Appendix C rather than embedded in the controller description.

The complete joint-torque command of Equation (3.13) is returned directly to libfranka.

No additional application-side saturation or torque-rate limiter is applied after the Cartesian, null-space, disturbance, and Coriolis terms have been assembled. The configured robot-side collision/reflex monitoring therefore remains an independent robot-side protection and reflex mechanism. A robot communication, motion-generation, or torque-control error exits the callback through `franka::Exception`. Configuration, file, and numerical errors are handled separately through standard exceptions.

Gripper verification and tool handling. The start-up interface also supports opening, grasping, and recalibrating the Franka Hand. Gripper state is checked before and after an action so that an incorrectly calibrated finger stroke or an unsuccessful grasp can be detected before torque control begins. An optional tool-transfer routine moves between the configured initial posture, a calculated stand-off posture, and the stored holder posture. The stand-off pose is obtained from damped least-squares inverse kinematics and is accepted only when the resulting joint configuration satisfies the configured joint-limit margins. The tool-transfer sequence is shown in Figure 3.6.

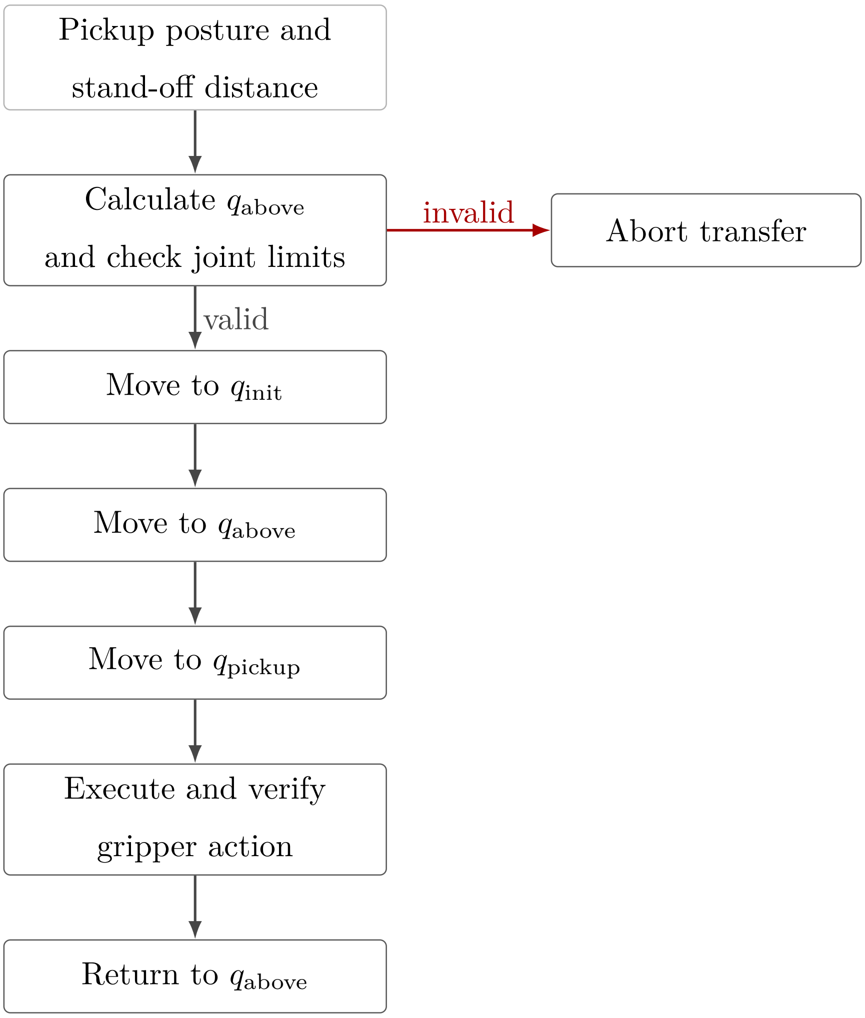


Figure 3.6: Tool collection and return sequence.

These preparation functions support robot operation while leaving the Cartesian control law and experimental parameter definitions unchanged.

3.7.2 Real-Time-Safe Data Recording

The controller stores the signals required for post-run analysis in a bounded in-memory ring buffer. The buffer memory is allocated before `robot.control` starts, so the callback does not grow the log dynamically. Each sampled row contains the measured state, active reference, calculated error and wrench, controller phase, secondary-controller state, and returned joint torque.

Table 3.4 groups the principal recorded signals. The complete column-level description used for the experimental evaluation is given in Appendix B.

Table 3.4: Signal groups stored during the real-time control loop.

Signal group	**Recorded quantities**
Time and controller state	Run time, active control phase, and null-space mode.
TCP state and reference	Measured and desired TCP pose quantities, Cartesian errors, and linear and angular velocities.
Tool and surface geometry	Controlled tool point, stored clearance geometry, desired contact target, and set-up coordinate.
Cartesian wrench	Commanded wrench and model-estimated external wrench, including the clearance-transition reference.
Null-space controller	Active null-space parameters, projected motion quantities, singular-value data, and secondary torque.
Disturbance study	Disturbance waveform, application point, equivalent force, and joint-torque contribution.
Joint-torque command	The seven final commanded joint torques returned to libfranka.

The values captured at the clearance transition refer to the geometric event described in Section 3.5.2; they do not denote a force-detected first-contact event. Each sampled row holds the measured state, the active reference, the Cartesian error, the commanded wrench, the controller phase, and the returned joint torque.

When the allocated capacity is reached, the write index wraps and new samples replace the oldest rows. This keeps the memory use bounded during real-time control.

3.7.3 Post-Run Output and Evaluation Support

After the torque callback ends, the ring-buffer contents are reconstructed in chronological order. The ordered samples are then written to the CSV file and reduced to the run summary, both outside the real-time control path.

The output preserves the commanded Cartesian wrench and the model-estimated external wrench as separate quantities. It also stores the geometric clearance-transition references needed to calculate subsequent force and moment changes. The implementation produces a set-up summary containing the final pose, selected tool point, external-wrench change, phase termination condition, and active controller settings. Detailed experimental metrics are defined in Chapter 4 and Appendix B. This chapter therefore remains focused on the generation and recording of the controller signals.

The controller structure described in this chapter is used throughout the reported experiments. Chapter 4 next specifies the physical system, calibration procedure, experimental parameter matrix, contact procedure, and evaluation quantities used to assess the implemented controller.

Chapter 4

Experimental Methodology

This chapter describes how the controller developed in Chapter 3 was evaluated experimentally. The methodology is organised in the order in which the experiments can be reproduced: the physical system is introduced first, followed by the surface and tool calibration, the common contact-test configuration and procedure, the parameter matrix for Cases A to H, the quantities used for evaluation, and the separate Cartesian pose-hold study for the null-space controller.

The main data set consists of surface-contact measurements in which the commanded tool orientation offset, directional Cartesian stiffness, and virtual CoC p_c were varied under otherwise matched conditions. The primary controller-response quantity is the signed end-effector rotation during set-up. The alignment angle calculated from the end-effector pose, TCP-height flatness classification, displacement, commanded wrench, and model-estimated external load are retained as supporting quantities. A second, independent experiment evaluates the null-space terms under a repeatable internally commanded disturbance.

4.1 Experimental System

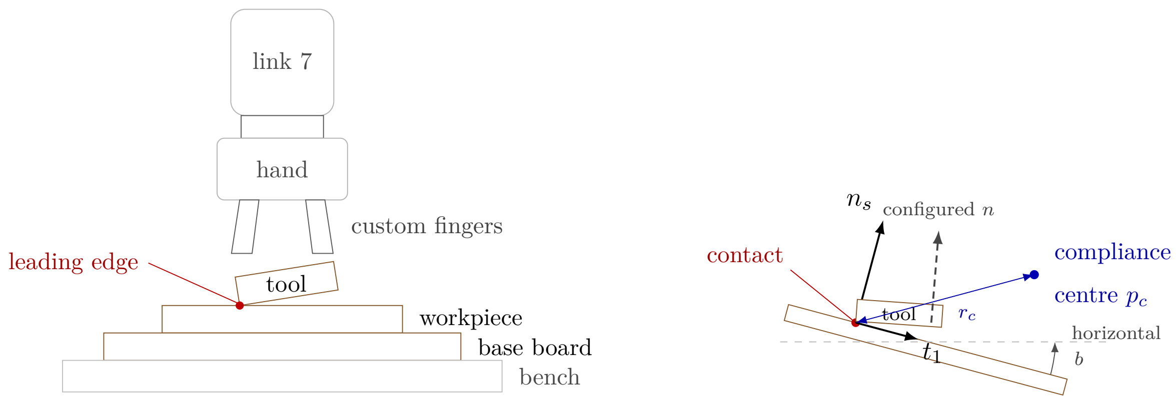
4.1.1 Robot, Tool, and Surface

The experiments were performed with a seven-degree-of-freedom Franka Emika Panda operated through the Franka Control Interface. Joint-torque commands were transmitted through libfranka at the nominal control rate of 1 kHz, corresponding to a sampling period of 1 ms. The real-time controller, Cartesian impedance law, contact geometry, and phase sequence are described in Chapter 3.

The rectangular grinding tool was held by the Franka Hand using custom gripper fingers. The tool face measured 40 × 120 mm, with the width along X_{EE} and the length along Y_{EE}. Its centre was located 20 mm along $+Z_{\mathrm{EE}}$ from the reported end-effector origin. The tool was clamped by custom pads screwed to the gripper fingers. Mechanical clearance in that clamping allowed the tool to rotate relative to the gripper by approximately ±2° about y_{EE}, which corresponds approximately to the t_2 direction in the flat experimental configuration. The robot does not measure this relative rotation: it is absent from the joint and end-effector state. The ±2° was measured in the unloaded condition, and the instantaneous relative tool–gripper rotation was not tracked during the contact experiments. It is therefore a mechanical property of the mounting rather than a calibrated measurement uncertainty.

Orientation quantities reconstructed from R_{EE} and the calibrated tool normal consequently describe the tool orientation under the assumption of a fixed tool-to-end-effector transformation, and any motion of the tool within the gripper is not included in them.

The surface was a plane plate mounted on a raised base board. All reported measurements used dry contact. The physical surface orientation and the tool normal were calibrated separately before the measurements, as described in Section 4.2.



(a) side elevation

(b) contact geometry

Figure 4.1: Experimental system and surface-frame contact geometry.

4.1.2 Real-Time Computer and Software

The controller computer ran Ubuntu 20.04.3 with the `5.9.1-rt20` Linux kernel and the `PREEMPT_RT` patch. The controller was compiled using `g++ 9.3`, C++14, and optimisation level `-O2`. Eigen 3.3.7 was used for matrix calculations and libfranka 0.7.1 for communication with the robot.

Before the measurements, the computer–robot connection was checked with the `communication_test` program supplied with libfranka 0.7.1 [8]. The program reports the fraction of control commands received by the robot and issues its lower communication-quality warning below an average command-success rate of 0.90. The experimental system passed this criterion. This check verifies communication quality for operation through the nominal 1 kHz interface; it is not a worst-case execution-time or scheduling-jitter measurement.

4.1.3 Operating and Safety Configuration

Before torque control started, the robot-side collision behaviour was configured with equal acceleration and nominal joint-torque thresholds of 140 N m. The six Cartesian force and moment channels used thresholds of 240 N or 240 N m, according to the component type. These thresholds supplied the Franka reflex mechanism and were kept unchanged throughout the reported measurements.

The complete run-specific controller configuration is given in Appendix C.

4.2 Surface and Tool Calibration

Two geometric quantities were required before the contact experiments could be performed: a surface reference expressed in the robot base frame, and the normal direction of the physical tool face expressed in the end-effector frame. Each came from its own calibration procedure.

The surface-plane calibration of Section 4.2.1 supplies a point p_s on the plate and its outward normal n_s, both in the base frame. The two tangent directions t_1 and t_2 are constructed from n_s, which completes the surface frame $R_{\text{surface}} = [\, t_1 \quad t_2 \quad n_s \,]$. The tool-normal calibration of Section 4.2.2 supplies $n_{\mathrm{Tool,EE}}$, the direction normal to the physical tool face in end-effector coordinates. That second calibration is needed because the physical face is not exactly perpendicular to the nominal $+Z_{\mathrm{EE}}$ axis.

The subscripts name the frame each direction is expressed in: $n_{\mathrm{Tool,EE}}$ is the tool-face normal in end-effector coordinates, and $n_{\mathrm{Tool,0}}$ is the same physical direction in the base frame. Only the first is calibrated. The second follows from it during operation, as Section 3.2.2 sets out, and Section 4.5 uses it for the alignment angle θ_{align}.

Neither calibration entered the other. The surface reference was obtained first, from one seated pose, using the nominal $+Z_{\mathrm{EE}}$ direction; the tool normal calibrated afterwards was not used to revise it. Determining $n_{\mathrm{Tool,EE}}$ in turn used only the relative orientations of four seated yaw poses, and no value of n_s. That separation matters, because the calibrated $n_{\mathrm{Tool,EE}}$ later differed from nominal $+Z_{\mathrm{EE}}$ by approximately 1.56°.

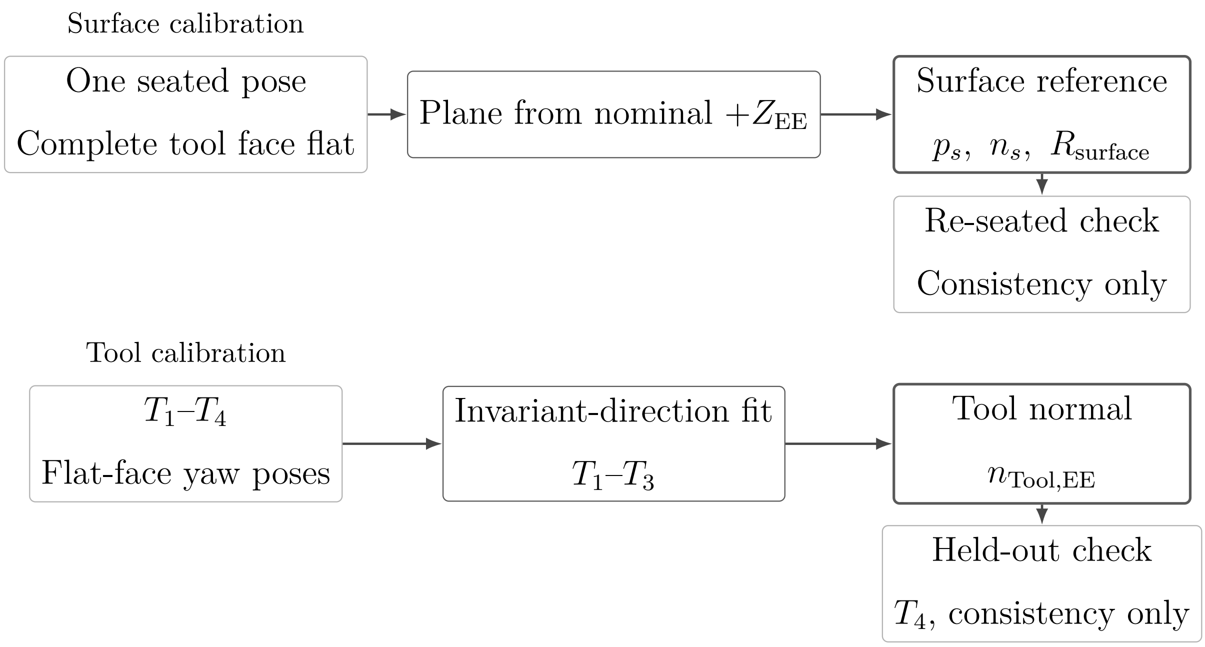


Figure 4.2: The two independent calibrations and the result each produces.

4.2.1 Surface-Plane Calibration

This calibration determines a fixed plane reference (p_s, n_s) in the robot base frame. The surface tangents and the surface frame used throughout the experiments are constructed from that pair.

The complete tool face was seated flat on the plate, and one end-effector pose $(p_{\mathrm{EE}}, R_{\mathrm{EE}})$

was recorded. No motion was commanded while that pose was taken. At this stage the physical tool normal $n_{\text{Tool,EE}}$ had not yet been calibrated, so the nominal $+Z_{\text{EE}}$ axis was used as the seated direction. The recorded orientation then carries the orientation of the plane,

$$n_s = R_{\text{EE}}[\,0,\ 0,\ 1\,]^\top, \tag{4.1}$$

with the sign taken to give a positive base-frame z component, so that n_s is the outward normal.

The same seated pose also supplies a point on the plane. The centre of the tool face sits at the fixed end-effector-frame offset $r_{\text{face,EE}}$ of Equation (4.17), so its base-frame position is

$$p_{\text{cal}} = p_{\text{EE}} + R_{\text{EE}} r_{\text{face,EE}}. \tag{4.2}$$

Because the complete face is seated on the plate, that centre lies on the plane, and $p_s = p_{\text{cal}}$. The calibrated plane is consequently

$$n_s^\top (p - p_s) = 0, \tag{4.3}$$

with n_s the unit outward surface normal. The plane point stored in the controller was

$$p_s = \begin{bmatrix} 0.5153 \\ -0.1072 \\ 0.0031 \end{bmatrix} \text{m}. \tag{4.4}$$

The controller stores the plane orientation through rotations α_s about the base x-axis and β_s about the base y-axis. With e_z the base-frame z-axis,

$$n_s = R_y(\beta_s) R_x(\alpha_s) e_z = \begin{bmatrix} \sin(\beta_s)\cos(\alpha_s) \\ -\sin(\alpha_s) \\ \cos(\beta_s)\cos(\alpha_s) \end{bmatrix}. \tag{4.5}$$

Inverting that relation returns the stored angles from the measured normal,

$$\alpha_s = \arcsin(-n_{s,y}), \qquad \beta_s = \operatorname{atan2}(n_{s,x}, n_{s,z}). \tag{4.6}$$

The calibrated angles were

$$\alpha_s = -1.585^\circ, \qquad \beta_s = +0.988^\circ, \tag{4.7}$$

which give

$$n_s \approx \begin{bmatrix} 0.017245 \\ 0.027663 \\ 0.999469 \end{bmatrix}, \qquad \|n_s\| = 1. \tag{4.8}$$

The surface calibration was completed before the physical tool normal was identified, so n_s was calculated with the nominal $+Z_{\mathrm{EE}}$ axis and retained unchanged for all reported experiments. The subsequent tool calibration showed that the physical tool-face normal differs from that axis by approximately 1.56°. Substituting $n_{\mathrm{Tool,EE}}$ for the nominal axis in Equation (4.1) would therefore rotate the stored surface normal. Since the surface was not recalibrated after $n_{\mathrm{Tool,EE}}$ had been determined, the originally stored p_s, n_s and R_{surface} were used consistently in both the controller and the evaluation, and the resulting difference is common to the complete data set.

Repeating the measurement while the tool face remains seated gives the consistency check of the stored plane. The repeat pose supplies a normal n_{cal} through Equation (4.1) and a face centre p_{cal} through Equation (4.2). Comparing them with the stored values gives

$$\psi_{\mathrm{plane}} = \arccos\left(n_{\mathrm{cal}}^\top n_s\right), \qquad \varepsilon_{\mathrm{plane}} = \left|n_s^\top (p_{\mathrm{cal}} - p_s)\right|. \tag{4.9}$$

Reading the orientation per axis adds $\Delta\alpha_s$ and $\Delta\beta_s$. Both are differences between the tilt angles of n_{cal}, returned by Equation (4.6), and the stored angles of Equation (4.7). ψ_{plane}, $\Delta\alpha_s$ and $\Delta\beta_s$ describe the orientation, and $\varepsilon_{\mathrm{plane}}$ the position of the plane along its own normal. A single seated point constrains no orientation, so neither part describes the agreement on its own. ψ_{plane} additionally carries any difference between the axis configured for the repeat measurement and the nominal axis of Equation (4.1). An orientation of the plate measured independently of the tool was not recorded.

The in-plane orientation was fixed by projecting the base-frame reference direction $a_s = e_x$ onto the calibrated plane:

$$t_1 = \frac{(I - n_s n_s^\top) a_s}{\|(I - n_s n_s^\top) a_s\|}. \tag{4.10}$$

The second tangent completes the right-handed frame,

$$t_2 = n_s \times t_1. \tag{4.11}$$

For the calibrated plane,

$$t_1 \approx \begin{bmatrix} 0.999851 \\ -0.000477 \\ -0.017238 \end{bmatrix}, \qquad t_2 \approx \begin{bmatrix} 0.000000 \\ 0.999618 \\ -0.027667 \end{bmatrix}. \tag{4.12}$$

The surface frame used throughout the contact measurements is therefore

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix} \approx \begin{bmatrix} 0.999851 & 0.000000 & 0.017245 \\ -0.000477 & 0.999618 & 0.027663 \\ -0.017238 & -0.027667 & 0.999469 \end{bmatrix}. \tag{4.13}$$

All surface-resolved quantities use the coordinate order $[t_1, t_2, n_s]$.

4.2.2 Tool Normal Calibration

This calibration determines $n_{\text{Tool,EE}}$, the direction normal to the physical tool face in end-effector coordinates. It is not assumed to coincide with nominal $+Z_{\text{EE}}$, because the manufactured tool and its mounting introduce a small geometric offset.

To identify $n_{\text{Tool,EE}}$, the complete tool face was seated flat on the same plate at four different yaw orientations. Rotating a flat tool about the surface normal changes the end-effector orientation, and leaves the physical direction normal to the seated face unchanged in the base frame. The calibration therefore searches for the one direction fixed in the end-effector frame that becomes the same base-frame direction in every flat yaw pose.

Only the recorded orientations enter that search, so no value of the calibrated plane of Section 4.2.1 reaches $n_{\text{Tool,EE}}$.

The first three captured rotations were used for the estimate, and the fourth was retained for validation. With $R_{\text{cal},i}$ the rotational part of the captured pose, perfect seating would carry the tool normal onto one base-frame direction $\bar{n}_B$ in all three,

$$R_{\text{cal},1} n_{\text{Tool,EE}} \approx R_{\text{cal},2} n_{\text{Tool,EE}} \approx R_{\text{cal},3} n_{\text{Tool,EE}} = \bar{n}_B. \tag{4.14}$$

Seating scatter leaves no direction that satisfies this exactly. The tool normal was therefore taken as the unit direction that comes closest across every pair of captures,

$$n_{\text{Tool,EE}} = \underset{\|n\|=1}{\arg\min} \sum_{i<j} \left\| (R_{\text{cal},i} - R_{\text{cal},j}) n \right\|^2, \tag{4.15}$$

whose minimiser is the eigenvector of $\sum_{i<j} (R_{\text{cal},i} - R_{\text{cal},j})^\top (R_{\text{cal},i} - R_{\text{cal},j})$ belonging to the smallest eigenvalue. The sign was chosen toward nominal $+Z_{\text{EE}}$. The final value used in every reported run was

$$n_{\text{Tool,EE}} = \begin{bmatrix} 0.026387057 \\ -0.006678514 \\ 0.999629492 \end{bmatrix}, \qquad \|n_{\text{Tool,EE}}\| = 1. \tag{4.16}$$

It differs from nominal $+Z_{\text{EE}}$ by approximately 1.56°. The calibration identifies the effective geometric direction used by the controller and evaluation; it does not determine the physical source of this offset.

$n_{\text{Tool,EE}}$ is the output of this calibration, and Section 3.2.2 carries it into the base frame during operation. Flat contact requires that base-frame direction to coincide with the inward surface normal, so the target sign is $s_a = -1$. The commanded spin about the desired tool axis, measured from t_1, was 90° for all reported runs. In the flat target orientation the projected long face axis is aligned with the t_2 line.

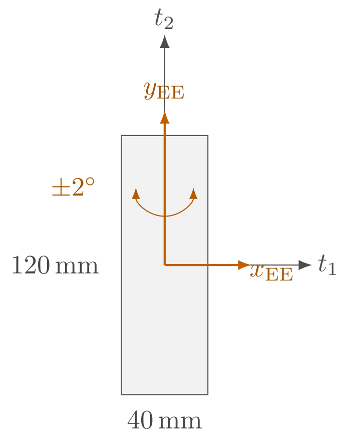


Figure 4.3: Calibrated tool face and mounting axes in the surface plane.

4.2.3 Calibrated Experimental Geometry

The calibrated tool face is represented in end-effector coordinates by its centre offset

$$r_{\mathrm{face,EE}} = \begin{bmatrix} 0 \\ 0 \\ 0.020 \end{bmatrix} \mathrm{m}, \tag{4.17}$$

and the half-width and half-length vectors

$$r_{\mathrm{width,EE}} = \begin{bmatrix} 0.020 \\ 0 \\ 0 \end{bmatrix} \mathrm{m}, \qquad r_{\mathrm{length,EE}} = \begin{bmatrix} 0 \\ 0.060 \\ 0 \end{bmatrix} \mathrm{m}. \tag{4.18}$$

These calibrated offsets, together with a feature-selection tolerance of $\varepsilon_g = 0.1\,\mathrm{mm}$, were used by the tool-point selection described in Section 3.2.3. Corners whose heights differ by no more than that tolerance are treated as one edge or face and averaged. The present chapter records the calibrated geometry; the selection algorithm that consumes it belongs to Chapter 3.

4.3 Common Contact-Test Configuration and Procedure

All surface-contact cases used the same calibrated geometry, phase sequence, normal press trajectory, fixed gain entries, and null-space configuration except where a parameter was explicitly varied. This section defines that common experimental basis before the individual cases are introduced in Section 4.4.

4.3.1 Common Cartesian Gains and Damping

The phase-dependent gains are diagonal in the calibrated surface frame $[t_1, t_2, n_s]$ before any compliance-centre shift. Table 4.1 gives the entries used by every reported contact run, together with the damping ratio ζ of the corresponding phase. The configured grinding phase carried the set-up entries unchanged. The pre-set-up gate was configured with the same stiffness as set-up and was disabled in the reported runs, so it did not act as a separate phase.

Table 4.1: Common Cartesian gains and damping ratio by phase.

	K_p [N/m]			K_R [N m/rad]			
Phase	t_1	t_2	n_s	t_1	t_2	n_s	ζ
Orientation and approach	150	150	150	180	180	90	1.9
Set-up and grinding	2000	2000	350	5	5	50	1.0

During orientation and approach the translational stiffness was isotropic and the rotational stiffness was higher about the surface tangents. In set-up the lower normal entry limits the elastic normal-force increase for a given normal position error, while the higher tangential entries restrict motion in the surface plane. The low rotational stiffnesses about t_1 and t_2 permit contact-induced alignment rotation, and the higher entry about n_s mainly resists in-plane spin of the tool face. Because the set-up translational stiffness $K_{p,\text{set}}$ is diagonal in the surface frame, the effective normal stiffness $K_{p,n} = n_s^\top K_{p,\text{set}} n_s$ equals the tabulated normal entry of 350 N/m.

The directional damping coefficients were calculated from the Cartesian inertia and the active stiffness at the tabulated damping ratio, as described in Section 3.4.2. In contrast, the fallback damping values, used only if the inertia-based calculation could not be applied, are provided in Appendix C.

4.3.2 Contact Sequence and Set-Up Motion

The common trajectory and transition parameters are collected in Table 4.2. The table is the single reference for the fixed contact-test sequence used by Cases A to H.

Table 4.2: Common phase-transition and trajectory parameters for Cases A to H.

Quantity	**Value**	**Role**
Minimum orientation time	1.0 s	Prevents an immediate phase transition.
Tool-axis transition tolerance	0.016 rad	Maximum remaining tool-axis error before descent.
Spin transition tolerance	0.009 rad	Maximum remaining normal-axis spin error.
Orientation timeout / maximum rate	8.0 s / 20°/s	Bounds the orientation phase.
Descent speed	0.1 m/s	Reference speed along the descent direction.
Maximum descent distance	0.640 m	Terminates an unsuccessful approach.
Surface clearance	0.020 m	Geometric transition from approach to set-up.
Minimum set-up time	2.0 s	Earliest activation of the moment-based exit condition.
Set-up timeout	5.0 s	Time-based set-up termination.
Moment-change threshold	150 N m	Alternative set-up termination based on the model-estimated external moment.

Table 4.2: Common phase-transition and trajectory parameters for Cases A to H (continued).

Quantity	**Value**	**Role**
Set-up push speed	0.080 m/s	Rate of the signed press coordinate.
Set-up push endpoint	+0.240 m	Final signed Cartesian spring reference.
Pre-set-up gate	disabled	Approach transfers directly to set-up.
Pre-grind gate	enabled	Holds the reached pose before the optional grinding transition.
Configured grinding sweep	t_2, 0.030 m, 0.2 Hz	Implemented but not entered in the reported runs.

Before each run, the active parameter set was loaded and the robot was moved to the configured initial joint posture. The calibrated geometry, the phase gains and damping, and the compliance-centre definition were prepared before the torque callback started.

Approach ended when the selected tool point reached 20 mm above the calibrated plane. That transition was defined by geometry rather than by a force threshold. At the same instant the controller stored the TCP pose, the orientation reference, the selected tool point and its projection onto the plane, and the model-estimated external wrench, which serve as the references for the evaluation quantities of Section 4.5.

The set-up press endpoint of +0.240 m is a Cartesian spring reference beyond the calibrated plane and is not interpreted as a physical penetration of the same magnitude. The reference reached it after approximately 3.25 s and was then held. Throughout set-up the rotational reference remained the orientation captured at the clearance transition, so the measured alignment rotation is a response to contact rather than the tracking of a time-varying orientation command. In Cases A, B, and C the decoupled Cartesian impedance was active during set-up, and in Cases D to H the point-shifted 6×6 impedance associated with the selected centre of compliance.

Every reported contact run ended at the 5 s set-up timeout, and the moment-change threshold was reached in none of them. The pre-grind gate then held the reached pose.

No run entered the configured grinding sweep, so the recorded data cover orientation, approach, and set-up only.

The set-up stiffness and commanded press range also define a virtual elastic command scale. For a completely blocked 0.260 m reference change along the calibrated normal,

$$F_{n,\text{block}} = 0.260\, n_s^\top K_{p,\text{set}} n_s = 91.0\,\text{N}. \tag{4.19}$$

This value is a command-scale bound for a fully blocked virtual spring and not a measured contact load.

4.3.3 Common Null-Space Configuration

Projected null-space damping and singular-value conditioning were active together throughout the surface-contact measurements. Keeping this secondary controller fixed prevents changes in the null-space policy from becoming an additional variable across Cases A to H. The gains were

$$d_{\text{null}} = 2\,\text{N m s/rad}, \qquad k_\sigma = 1\,\text{N m}. \tag{4.20}$$

This is the combined secondary-torque mode of the implementation described in Section 3.6. The contact measurements therefore evaluate the surface-alignment controller with both null-space terms active. Their individual effects are examined separately in the pose-hold experiment of Section 4.6.

4.3.4 Repetitions and Data Recording

The real-time logging buffer recorded the measured robot state, Cartesian reference and error, selected tool point, active control phase, commanded Cartesian wrench, final commanded joint torque, null-space mode, and libfranka model-estimated external wrench. The complete signal list and CSV structure are given in Section 3.7.2 and Appendix B.

Three repetitions were recorded for each parameter setting, giving 165 surface-contact runs across 55 settings. No run was discarded, and the quantitative evaluation uses all 165. Within each controlled comparison, the calibrated geometry, common phase parameters, fixed gains, and null-space settings remained unchanged. Only the parameter assigned to the corresponding case was varied.

4.4 Surface-Contact Experimental Matrix

The surface-contact experiments compare the TCP-centred condition with stiffness variations and displaced centres of compliance across a controlled set of initial angular misalignments. Case A establishes the TCP-centred reference, where $r_c = 0$ and the point shift contributes no coupling moment. Cases B and C quantify the influence of rotational and cross-axis translational stiffness. Case D evaluates the sign and direction dependence of a non-zero tangential centre across both principal surface tangents. Cases E to G examine the lever-definition frame, the displacement direction, and the commanded orientation-offset magnitude. Case H extends the comparison to intermediate tangent-plane rotation directions and contrasts a direction-selected lever with one fixed lever.

Table 4.3 lists the complete parameter variations and the shared reference conditions. The case letters are retained throughout Chapter 5 so that each experimental setting maps directly to its reported result. Two cases need a geometric explanation that the table cannot carry, and both are treated after it.

Table 4.3: Experimental cases and parameter variations.

Case	**Varied quantity**	**Commanded tool orientation offset**	**Tested values**	**Runs**	**Evaluated effect**
A	Zero lever; CoC at the TCP	None and $\pm 10°$ about t_1 or t_2	No lever.	15	Contact response without lever coupling; the reference for the other cases.
B	Rotational stiffness	$+10°$ about t_1 or t_2	Command-axis $K_R = 15$ and 50 N m/rad, orthogonal tangent at 5 N m/rad.	12	Command-axis rotational stiffness on the direction-dependent response.

Table 4.3: Experimental cases and parameter variations (continued).

Case	**Varied quantity**	**Commanded tool orientation offset**	**Tested values**	**Runs**	**Evaluated effect**
C	Translational stiffness	$+10°$ about t_1 or t_2	Cross-axis $K_p = 300$ and $800\,\mathrm{N/m}$.	12	Cross-axis translational stiffness on the direction-dependent response.
D	Tangential compliance-centre lever, direction and sign	$\pm10°$ about t_1 or t_2	-40, -20, -10, $+10$, $+20$, $+40$ mm along the tangent perpendicular to the commanded rotation.	72	Direction and sign dependence across both principal tangents.
E	Compliance-centre lever-definition frame	None and $+10°$ about t_1	The same nominal 40 mm lever: end-effector frame at no offset, surface frame at both offsets.	9	Dependence on the frame holding the displacement, not on its magnitude alone.
F	Compliance-centre position along the tool axis	$+10°$ about t_1	-40, -20, -10, $+10$, $+20$, $+40$ mm along the tool axis.	18	Tangential against tool-axis displacement; the axial moment contribution is small at the tested inclination.
G	Commanded orientation-offset magnitude	$+5°$ about t_1 or t_2	0 and $+40$ mm along the assisting tangent.	12	Response against the commanded orientation-offset magnitude at a fixed centre.
H	Commanded rotation direction and lever selection	$+10°$ in four tangent-plane directions	A 40 mm lever, direction-selected and fixed at the t_1 selection.	15	Agreement between a fixed lever and the direction-selected rule.

The run counts in Table 4.3 give the three scheduled repetitions of each setting. A condition that a case shares with an earlier one is counted with that earlier case.

In Case F the CoC was displaced along the tool axis. An axial displacement produces no tangential lever component when the tool axis is exactly parallel to the normal press direction. At finite inclination a small tangential component remains. Case F therefore quantifies the residual influence of this nominally weak displacement direction.

4.4.1 Case-H Direction and Lever Definition

The compliance-centre lever follows the convention

$$r_c = p_c - p_{\mathrm{TCP}}. \tag{4.21}$$

This quantity defines the shift of the impedance reference point. The selected tool-point offset $p_T - p_{\mathrm{TCP}}$ follows from the physical tool geometry and is treated separately, as defined in Section 4.2.3 and Section 2.7.

A tool-fixed lever is constant in end-effector coordinates and therefore rotates in the base frame with the end effector. A surface-fixed lever is constant in the surface frame $[t_1, t_2, n_s]$. In both cases, the active lever is transformed to base coordinates during each control cycle before the stiffness and damping matrices are shifted to the TCP.

For a commanded rotation represented by surface components $(\theta_{t_1}, \theta_{t_2})$, the direction-selected tangential lever is

$$r_{c,t} = \frac{\rho_c}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}} \left(\theta_{t_1} t_2 - \theta_{t_2} t_1\right), \tag{4.22}$$

where $\rho_c > 0$ is the selected lever magnitude. Case H used $\rho_c = 40\,\mathrm{mm}$. The selected lever lies perpendicular to the commanded rotation direction in the tangent plane, and its sign is chosen so that the press-induced coupling moment acts in the required alignment direction. It therefore changes with the commanded rotation direction, whereas the fixed comparison retains the lever selected for the t_1 command.

Table 4.4: Case-H orientation commands and lever vectors.

Commanded direction	Lever setting	$[\theta_{t_1}, \theta_{t_2}]$ (deg)	$[r_{c,t_1}, r_{c,t_2}, r_{c,n}]$ (mm)
About t_1	selected	$[+10.0, 0.0]$	$[0, +40.0, 0]$
$-45°$ from t_1	selected	$[+7.1, -7.1]$	$[+28.3, +28.3, 0]$
$+45°$ from t_1	selected	$[+7.1, +7.1]$	$[-28.3, +28.3, 0]$
About t_2	selected	$[0.0, +10.0]$	$[-40.0, 0, 0]$
$-45°$ from t_1	fixed at the t_1 selection	$[+7.1, -7.1]$	$[0, +40.0, 0]$
$+45°$ from t_1	fixed at the t_1 selection	$[+7.1, +7.1]$	$[0, +40.0, 0]$
About t_2	fixed at the t_1 selection	$[0.0, +10.0]$	$[0, +40.0, 0]$

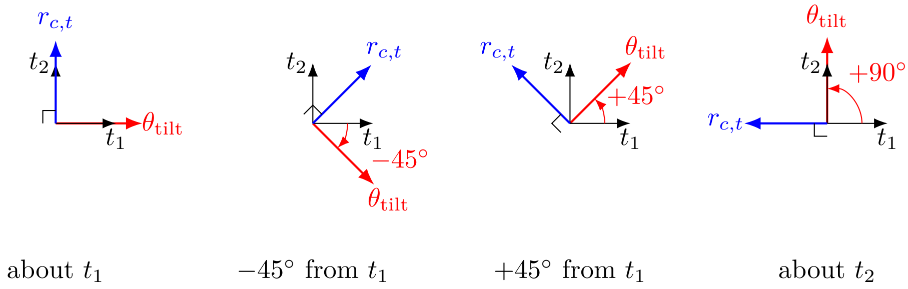


Figure 4.4: Selected tangential lever for each commanded rotation direction.

The principal-axis selected conditions are shared with Case D. The five Case-H settings are the two diagonal directions with both lever definitions and the t_2 direction with the fixed lever; each was repeated three times.

4.5 Evaluation Quantities

The evaluation separates the controller response from geometric quantities that depend on the calibrated tool model. The primary comparison quantity for Cases A to H is the signed end-effector rotation during set-up, $\Delta\theta_{\mathrm{set}}$. Supporting quantities are the alignment angle θ_{align}, the TCP-height flatness classification, the selected tool-point displacement Δp_T and its magnitude s_T, the TCP displacement Δp_{TCP} and its magnitude s_{TCP}, the commanded Cartesian wrench $F = [f^\top, m^\top]^\top$, and the model-estimated external normal-load magnitude F_n. These quantities describe different aspects of the response and are not treated as interchangeable measurements of the same physical quantity.

4.5.1 Signed Set-Up Rotation

The primary response quantity measures how the end effector actually rotated between the clearance transition and the end of set-up. The relative end-effector rotation is $R_{\mathrm{after}} R_{\mathrm{before}}^\top$. Using the finite-rotation relations introduced in Equations (2.18) and (2.20), its angle ϕ_{set} and unit axis u_{set} define the rotation vector

$$\Delta\theta_{\mathrm{set}} = \phi_{\mathrm{set}} u_{\mathrm{set}}. \tag{4.23}$$

The vector $\Delta\theta_{\mathrm{set}}$ is expressed in the robot base frame and contains both the magnitude and direction of the finite end-effector rotation during set-up. For comparison with the commanded surface directions, it is resolved in the surface frame:

$$\Delta\theta^{S}_{\mathrm{set}} = R^{\top}_{\mathrm{surface}}\Delta\theta_{\mathrm{set}}. \tag{4.24}$$

The signed component along the commanded rotation direction is the primary controller-response metric. A positive value denotes end-effector rotation in the intended alignment direction. This quantity is calculated directly from the measured end-effector orientations and does not require the calibrated tool normal $n_{\mathrm{Tool,EE}}$.

4.5.2 Alignment Angle and Flatness Classification

The tool normal used for the alignment calculation is obtained from the measured end-effector orientation and the calibrated direction in the end-effector frame:

$$n_{\mathrm{Tool},0} = R_{\mathrm{EE}} n_{\mathrm{Tool,EE}}. \tag{4.25}$$

The alignment angle is

$$\theta_{\mathrm{align}}(t) = \cos^{-1}\Big(-n_{\mathrm{Tool},0}(t)^{\top} n_s\Big). \tag{4.26}$$

The negative sign follows the experimental flat target $R_d n_{\mathrm{Tool,EE}} = -n_s$. The quantity $\theta_{\mathrm{align}}(t) \geq 0$ is an unsigned instantaneous geometric error: it states how far the reconstructed tool normal is from the flat target at time t. It is calculated from the end-effector pose and the calibrated tool normal and therefore assumes that the calibrated relation between the tool and end effector remains fixed. Relative motion in the tool mounting is not captured by this calculation.

The reduction of this unsigned alignment error over set-up is reported as

$$\Delta\theta_{\mathrm{align}} = \theta_{\mathrm{align,before}} - \theta_{\mathrm{align,after}}, \tag{4.27}$$

where the before value is taken at the clearance transition and the after value at the end of set-up. A positive $\Delta\theta_{\mathrm{align}}$ therefore means that the reconstructed angular difference from the flat target decreased. This quantity is a change of two unsigned alignment-error magnitudes; it is not the finite end-effector rotation itself. The latter is measured separately by $\Delta\theta_{\text{set}}$ in Equation (4.23).

The alignment angle carries no direction, and the case comparisons need one. The same geometric relation therefore also supplies a signed decomposition. Let θ_{dev} be the rotation vector of the shortest rotation taking the pose-based tool axis $n_{\text{Tool},0}$ onto the flat target direction $-n_s$. Resolved on the surface axes it gives

$$\theta_{\text{dev}}^{S} = R_{\text{surface}}^{\top}\theta_{\text{dev}} = [\,\theta_{\text{dev},t_1},\ \theta_{\text{dev},t_2},\ \theta_{\text{dev},n}\,]^{\top}, \tag{4.28}$$

whose components are reported in degrees and whose norm is the alignment angle itself, $\|\theta_{\text{dev}}^{S}\| = \theta_{\text{align}}$. The two quantities are the magnitude and the signed components of one vector, not independent measurements.

Two properties of the sign convention matter when the values are read. θ_{dev}^{S} is the rotation still required to reach the flat target, so its sign is opposite to that of the commanded tool orientation offset that produced the deviation: a commanded $+10^\circ$ about t_1 appears as a negative t_1 component. Its magnitude is also somewhat below the commanded value, because the orientation phase exits on an angular tolerance rather than driving the offset to completion. The value at set-up entry is written $\theta_{\text{dev},t_i,\text{before}}$, and it is distinct from $\theta_{\text{align,before}}$, which is unsigned.

A separate geometry-based flatness classification is obtained from the final TCP height relative to the calibrated plane and the height expected from the calibrated rectangular tool geometry when the tool face is flat. This classification uses the end-effector position and calibrated geometry; it is not a direct angular measurement of the physical tool face. The alignment angle and the TCP-height classification are therefore reported separately from the primary end-effector set-up rotation. Appendix D.3 compares $\Delta\theta_{\text{align}}$ and $\Delta\theta_{\text{set}}$ against each other, as a check on the choice of primary quantity.

4.5.3 Displacement and Wrench Quantities

Let p_T denote the tool point selected at the clearance transition. Its displacement and magnitude are

$$\Delta p_T = p_{T,\text{after}} - p_{T,\text{before}}, \qquad s_T = \|\Delta p_T\|. \tag{4.29}$$

The TCP displacement is evaluated in the same form:

$$\Delta p_{\text{TCP}} = p_{\text{TCP,after}} - p_{\text{TCP,before}}, \qquad s_{\text{TCP}} = \|\Delta p_{\text{TCP}}\|. \tag{4.30}$$

Both displacement vectors were additionally resolved in the surface frame to separate tangential and normal motion.

The supporting normal-load quantity is derived from the change in the libfranka model-estimated external force relative to the value stored at the clearance transition:

$$F_n = \left| n_s^\top \left(\hat{f}_{\text{ext}} - \hat{f}_{\text{ref}} \right) \right|. \tag{4.31}$$

This is a model-estimated force change, not an independently measured contact force. The commanded normal force and commanded TCP moment used in the controller-response plots are instead obtained from $F = [f^\top, m^\top]^\top$:

$$F_{n,\text{cmd}} = n_s^\top f, \qquad M_{t_i,\text{cmd}} = t_i^\top m. \tag{4.32}$$

Thus $F_{n,\text{cmd}}$ is a signed command, while F_n is the positive magnitude of a model-estimated change. During a quasi-static press the two are expected to have approximately opposite magnitude under the chosen sign convention, but they remain distinct controller signals.

4.5.4 Treatment of Repeated Runs

Repeated settings are reported using the arithmetic mean and the sample standard deviation over their usable repetitions.

The within-setting standard deviations are reported with the corresponding results in Chapter 5. Comparisons are restricted to settings with the same calibrated geometry, common phase timing, fixed gain entries, and null-space configuration.

4.6 Secondary Null-Space Pose-Hold Experiment

The surface-contact measurements keep projected damping and singular-value conditioning active together. A separate Cartesian pose-hold experiment was therefore used to compare their individual effects under a repeatable internally commanded disturbance.

4.6.1 Experimental Configuration

The pose-hold controller maintained the initial TCP pose with the gains of Table 4.5. Every entry was isotropic, so each matrix is the stated value times the identity. Unlike the contact-sequence phases, the twelve pose-hold trials used fixed Cartesian damping rather than inertia-scaled damping.

Table 4.5: Isotropic Cartesian gains of the pose-hold trials.

Quantity		**Value**
Translational stiffness	$K_{p,\text{hold}}$	$2000\,\mathrm{N/m}$
Rotational stiffness	$K_{R,\text{hold}}$	$50\,\mathrm{N\,m/rad}$
Translational damping	$D_{p,\text{hold}}$	$50\,\mathrm{N\,s/m}$
Rotational damping	$D_{R,\text{hold}}$	$8\,\mathrm{N\,m\,s/rad}$

Four null-space settings were tested with three repetitions each: no null-space torque, projected damping with $d_{\text{null}} = 2\,\mathrm{N\,m\,s/rad}$, sigma conditioning with $k_\sigma = 1.5\,\mathrm{N\,m}$, and sigma conditioning with $k_\sigma = 2.0\,\mathrm{N\,m}$. The probe step, sigma deadband, and relative SVD threshold remained at $0.08\,\mathrm{rad}$, 10^{-6}, and 10^{-4}, respectively.

4.6.2 Commanded Disturbance

The disturbance represents the joint-torque equivalent of a commanded point force acting at a specified location on link 3. The point was defined by the link-frame offset

$$r_p = \begin{bmatrix} 0 \\ 0 \\ 0.200 \end{bmatrix} \mathrm{m}. \tag{4.33}$$

At the initial posture q_{init}, the structural null direction $v_7(q_{\text{init}})$ and the translational point Jacobian $J_p(q_{\text{init}})$ define the fixed base-frame force direction

$$u_f = \frac{J_p(q_{\text{init}})v_7(q_{\text{init}})}{\|J_p(q_{\text{init}})v_7(q_{\text{init}})\|_2}. \tag{4.34}$$

The commanded force magnitude was $F_d = 20\,\mathrm{N}$. The virtual force and its equivalent joint torque were

$$f_d(t) = F_d s_d(t) u_f, \qquad \tau_{\text{dist}}(t) = J_p(q(t))^\top f_d(t). \tag{4.35}$$

Here u_f is the fixed unit force direction defined in Equation (4.34), and $s_d(t) \in [0,1]$ is a dimensionless time-scaling function that switches the commanded virtual disturbance smoothly on and off. It was defined as

$$s_d(t) = \begin{cases} 0, & t < 5\,\mathrm{s}, \\ \frac{1}{2}\left[1 - \cos\!\left(\pi \frac{t - 5\,\mathrm{s}}{2\,\mathrm{s}}\right)\right], & 5\,\mathrm{s} \le t < 7\,\mathrm{s}, \\ 1, & 7\,\mathrm{s} \le t < 8\,\mathrm{s}, \\ \frac{1}{2}\left[1 + \cos\!\left(\pi \frac{t - 8\,\mathrm{s}}{1\,\mathrm{s}}\right)\right], & 8\,\mathrm{s} \le t < 9\,\mathrm{s}, \\ 0, & t \ge 9\,\mathrm{s}. \end{cases} \tag{4.36}$$

Thus the disturbance increased smoothly from zero to the full commanded magnitude between 5 and 7 s, remained at full magnitude until 8 s, and decreased smoothly to zero by 9 s. The equivalent disturbance torque was limited to

$$\|\tau_{\text{dist}}\|_2 \le 3\,\mathrm{N\,m}.$$

The disturbance is internal to the controller and therefore provides a repeatable comparison input; it is not a separately applied physical force.

In every trial the selected null-space law was active from the start of the pose-hold phase. The first 5 s therefore formed a pre-disturbance settling interval, during which the conditioning torque of the sigma-only settings could already move the redundant configuration.

The disturbance then acted from 5 to 9 s, followed by an observation interval until the end of the trial. No secondary term was enabled or disabled within a trial: the sigma-conditioning torque remained active throughout the complete sigma-only trial and was not switched on after the disturbance had been removed.

4.6.3 Evaluation Quantities

All quantities in this subsection are evaluated over the interval in which the internally commanded virtual disturbance acts, $t \in [5, 9]\,\mathrm{s}$. The first criterion checks whether the primary Cartesian pose-hold task remains approximately unchanged while the disturbance and the selected null-space law act. The predefined task-retention criterion is

$$\max_{5\,\mathrm{s} \le t \le 9\,\mathrm{s}} \|e_p(t)\|_2 < 2\,\mathrm{mm}, \tag{4.37}$$

where $e_p(t)$ is the Cartesian position-error vector. A trial therefore satisfies the criterion when the TCP position-error norm remains below 2 mm throughout the commanded disturbance interval.

The redundant joint motion is evaluated from the projected joint velocity

$$\dot{q}_{\mathrm{null}}(t) = N_q(q(t))\dot{q}(t), \tag{4.38}$$

where $N_q(q)$ is the kinematic null-space projector. The vector $\dot{q}_{\mathrm{null}} \in \mathbb{R}^7$ has units of rad/s and represents the component of the measured joint velocity that lies in the instantaneous kinematic null space.

Two complementary quantities are formed from this projected velocity. The first is the cumulative projected null-space motion

$$E_N = \int_{5\,\mathrm{s}}^{9\,\mathrm{s}} \|\dot{q}_{\mathrm{null}}(t)\|_2 \,\mathrm{d}t. \tag{4.39}$$

Because the norm is taken before integration, motion in opposite directions contributes positively in both cases. E_N therefore measures the total amount of redundant joint motion that occurred during the commanded disturbance interval. Its unit is radians and the reported values are converted to degrees. For example, motion of $+5°$ along a redundant direction followed by $-5°$ back along the same direction contributes approximately $10°$ to E_N, even though the final redundant configuration can be close to its starting value. In this sense E_N is a joint-space path-length quantity, not a torque or energy measure.

The second quantity retains the direction of the motion. Integrating the projected velocity without taking its norm gives the accumulated projected joint displacement

$$\Delta q_{\text{null}} = \int_{5\,\text{s}}^{9\,\text{s}} \dot{q}_{\text{null}}(t)\,\mathrm{d}t = \int_{5\,\text{s}}^{9\,\text{s}} N_q(q(t))\dot{q}(t)\,\mathrm{d}t. \tag{4.40}$$

The vector $\Delta q_{\text{null}} \in \mathbb{R}^7$ has units of radians. It describes the net accumulated projected joint motion during the disturbance interval, so motion in opposite directions can cancel. It is not an integrated torque: the integrated quantity is joint velocity.

A common direction is introduced to express this seven-dimensional net displacement by one signed scalar. Let $\Delta q_{\text{null},0}$ denote the reference net projected displacement obtained from the condition without null-space torque. The corresponding unit direction is

$$v_{\text{ref}} = \frac{\Delta q_{\text{null},0}}{\|\Delta q_{\text{null},0}\|_2}. \tag{4.41}$$

Thus v_{ref} represents the redundant joint-space direction in which the commanded virtual disturbance moves the robot in the reference condition without a secondary null-space torque. The same direction is used for all tested null-space settings. Their signed net displacement along this common direction is

$$\Delta\eta_{\text{dist}} = v_{\text{ref}}^{\top}\Delta q_{\text{null}}. \tag{4.42}$$

A positive $\Delta\eta_{\text{dist}}$ denotes net motion in the same direction as the reference disturbance response, a negative value denotes net motion in the opposite direction, and a value near zero indicates little remaining net displacement along that direction. The quantity is a signed projection of the accumulated projected joint motion rather than an exact nonlinear redundant-coordinate displacement, because $N_q(q)$ changes with the robot configuration while v_{ref} is held fixed. At full row rank the Panda has a one-dimensional instantaneous null space, so this common direction provides a meaningful signed comparison of the four tested settings.

The two redundant-motion metrics therefore answer different questions: E_N measures how much redundant joint motion occurred in total, whereas $\Delta\eta_{\mathrm{dist}}$ measures how much net displacement remained along the common reference disturbance direction. A trajectory can consequently have a comparatively large E_N while ending close to its starting redundant configuration and therefore having a small $\Delta\eta_{\mathrm{dist}}$.

Singular-value conditioning was evaluated from the change in $\sigma_{\min}$ across the same disturbance interval, and Cartesian task retention from the peak position-error norm. The corresponding results are presented in Chapter 5.

Chapter 5

Results and Discussion

The surface-contact results are presented according to the experimental cases defined in Chapter 4, and each result is interpreted where it appears. The signed set-up rotation $\Delta\theta_{\text{set}}$ about the commanded rotation direction is the primary controller-response quantity, as defined in Section 4.5. The TCP-height flatness criterion and the alignment angle θ_{align} calculated from the end-effector pose provide supporting information, together with the displacements, the commanded wrench, and the model-estimated normal load.

Repeated settings are reported by their mean and standard deviation. The median within-setting standard deviation of the signed set-up rotation was approximately $0.04°$ about t_1 and $0.14°$ about t_2, so differences below approximately $0.1°$ are treated as being of the same order as the observed scatter. This value is a descriptive comparison scale and not a statistical significance limit. Larger scatter occurred at individual transition conditions in Case D, which are identified where they arise.

The final response was close to steady state for most tested conditions. During the final second of set-up, the median changes in the model-estimated normal load and the alignment angle were $0.10\,\text{N}$ and $0.00°$; the largest were $1.10\,\text{N}$ and $0.60°$. Selected time histories are therefore shown only where they help explain the underlying controller behaviour.

The alignment angle at the beginning of set-up was not identical to the commanded orientation offset, because the preceding orientation phase ended with finite tracking error. Matched conditions nevertheless started reproducibly: a commanded $+10°$ offset reached $9.34 \pm 0.02°$ about t_1 and $10.04 \pm 0.03°$ about t_2. The commanded offset identifies the experimental condition, and the measured initial value is retained in the tables where it is required.

5.1 Surface-Contact Results

The cases are reported in the order in which they progressively constrain whether one CoC can be fixed before the sign and the tangent-plane direction of the initial angular misalignment are known. Case A provides the zero-lever reference. Cases B and C quantify stiffness sensitivity. Case D then tests the tangential compliance-centre lever over both surface tangents and both signs of the commanded orientation offset, which is where a fixed non-zero displacement has to hold if it is to hold at all. Cases E, F, and G examine the lever-definition frame, the tool-axis centre position, and the commanded orientation-offset magnitude. Case H is presented last because it tests whether the direction rule obtained from the principal surface tangents extends to intermediate tangent-plane directions.

5.1.1 Zero-Lever Reference: Case A

The first experiment establishes what compliant contact produces on its own, before any virtual lever is introduced. In Case A the CoC is placed on the TCP, so that $r_c = 0$. The TCP-centred condition removes the coupling introduced by the virtual point shift, while leaving the ordinary rotational impedance and the tool-geometry lever r_T unchanged. The commanded rotational moment m_0 remains and the rotational compliance stays finite, so contact-induced rotational motion remains possible. The measured response is the zero-coupling reference for the remaining contact cases under the common Cartesian impedance and null-space settings, and not yet an answer about where the CoC should be placed.

Table 5.1: Case-A zero-lever response.

Commanded offset [°]	$\theta_{\text{align,before}}$ [°]	$\Delta\theta_{\text{set}}$ [°]	$\Delta\theta_{\text{align}}$ [°]	Flat by TCP height
None	0.74	−0.97	-1.56 ± 0.01	yes
+10° about t_1	9.32	+7.57	$+5.94 \pm 0.03$	yes
−10° about t_1	9.43	−10.83	$+7.19 \pm 0.24$	yes
+10° about t_2	10.07	−1.63	-1.74 ± 0.17	yes
−10° about t_2	9.59	−1.25	$+1.20 \pm 0.19$	yes

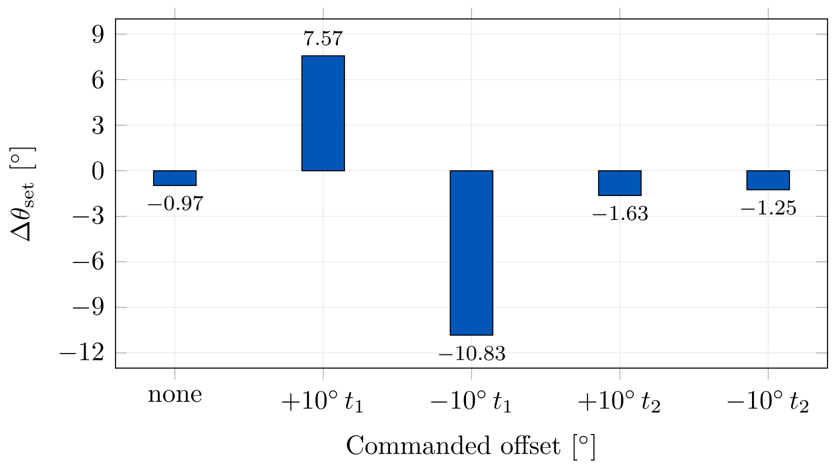


Figure 5.1: Set-up rotation with the centre of compliance at the TCP.

With the CoC at the TCP, the response is strongly direction-dependent. As Table 5.1 records and Figure 5.1 plots against the commanded tangent, the positive 10° command rotates the end effector by +7.57° about t_1 but by −1.63° about t_2. Reversing the commanded orientation gives −10.83° about t_1 and −1.25° about t_2. The t_1 direction therefore already exhibits substantial alignment-directed motion without a compliance-centre lever. About t_2 the TCP-centred condition produces no alignment-directed end-effector rotation under this metric, although the final configuration satisfies the TCP-height flatness criterion, as the following paragraphs set out.

This baseline conditions the reading of the following cases, because the effect of a parameter change has to be interpreted relative to the response already present without lever-induced coupling. The two principal surface tangents cannot be treated as dynamically equivalent. The asymmetry is consistent with the rectangular contact geometry and with the direction-dependent compliance of the tool mounting: the tool face has different half-dimensions across the two surface tangents, and unloaded clearance was observed near the mounting direction associated with t_2. The experiments do not isolate the individual contributions of contact geometry and tool-mount motion.

All five Case-A settings satisfy the TCP-height criterion. That criterion and the alignment angle answer different questions and are not equivalent measurements of physical tool orientation. TCP height rests on the calibrated plane and tool geometry, whereas the alignment angle is calculated from the measured end-effector orientation and the calibrated tool normal. A difference between them is consistent with relative tool–gripper motion or with the assumptions of the geometric reconstruction. The mounting play was measured unloaded, and the instantaneous relative tool–gripper rotation was not tracked separately during the contact runs.

The zero-orientation-offset condition is the reason the alignment angle is kept separate from the controller-response metric. It changes from 0.74° at set-up entry to approximately 2.30° after set-up although no initial orientation offset is commanded, while the TCP-height criterion still classifies the tool as flat. The signed set-up rotation is therefore used for the controller comparisons that follow.

Case A establishes the response available without virtual coupling, and shows that this response differs between the two surface tangents. Where the centre of compliance should be placed remains open at this point, since no alternative placement has yet been measured. The cases that follow supply that comparison.

5.1.2 Influence of Cartesian Stiffness

Case B: Rotational Stiffness

The following trials analyse the rotational stiffness about the commanded surface tangent, which is the first of the two Cartesian gains available before the CoC is moved. Case B varies that entry alone, and its $5\,\mathrm{N\,m/rad}$ condition is the corresponding zero-lever reference of Case A.

Table 5.2: Case-B response by rotational stiffness.

Commanded offset [°]	Varied entry	Value [N m/rad]	$\Delta\theta_{set}$ [°]	Flat by TCP height
+10° about t_1	K_{R,t_1}	5	+7.57	yes
		15	+7.29	yes
		50	+2.73	no
+10° about t_2	K_{R,t_2}	5	−1.63	yes
		15	−1.52	yes
		50	−0.48	yes

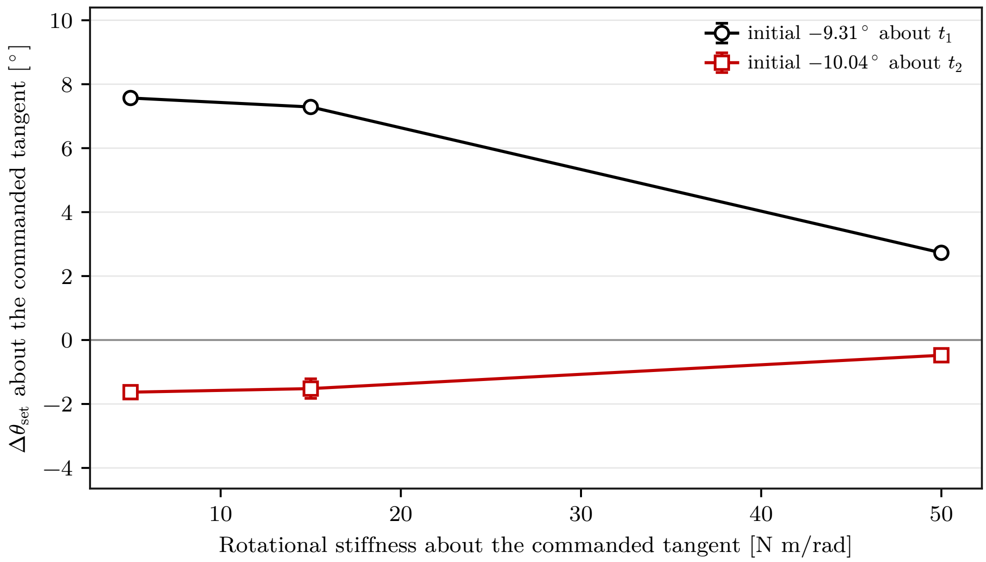


Figure 5.2: Set-up rotation about the commanded tangent by rotational stiffness.

Increasing the rotational stiffness about the commanded tangent suppresses the contact-induced rotation. Table 5.2 lists the response at the three tested stiffnesses and Figure 5.2 plots it. About t_1, raising K_{R,t_1} from 5 to 50 N m/rad reduces the signed set-up rotation from +7.57 to +2.73°. The final TCP-height classification changes from flat at the two lower stiffnesses to tilted at the highest value, where the TCP remains 3.6 mm above the flat-tool height. In contrast, about t_2 the response stays small across the whole tested range, changing from −1.63 to −0.48°, and all three settings are classified as flat.

Rotational stiffness therefore controls the resistance to alignment rotation directly, and most strongly about t_1. However, changing it alone does not produce the missing alignment-directed response about t_2: every tested value leaves that response negative.

The axis dependence of the zero-lever behaviour survives the rotational-stiffness variation.

Case C: Cross-Axis Translational Stiffness

The second Cartesian gain is examined next. Case C varies the translational stiffness along the surface tangent perpendicular to the commanded rotation direction, the direction in which translational compliance can influence how the contacting tool moves and rotates.

Table 5.3: Case-C response by cross-axis translational stiffness.

Commanded offset [°]	Varied entries	Setting	Cross-axis [N/m]	Command-axis [N m/rad]	$\Delta\theta_{\mathrm{set}}$ [°]	Flat by TCP height
+10° about t_1	K_{p,t_2}	translation only	300	5	+7.59	yes
		translation only	800	5	+7.56	yes
		reference	2000	5	+7.57	yes
+10° about t_2	K_{p,t_1}	translation only	300	5	−2.24	yes
		translation only	800	5	−1.78	yes
		reference	2000	5	−1.63	yes

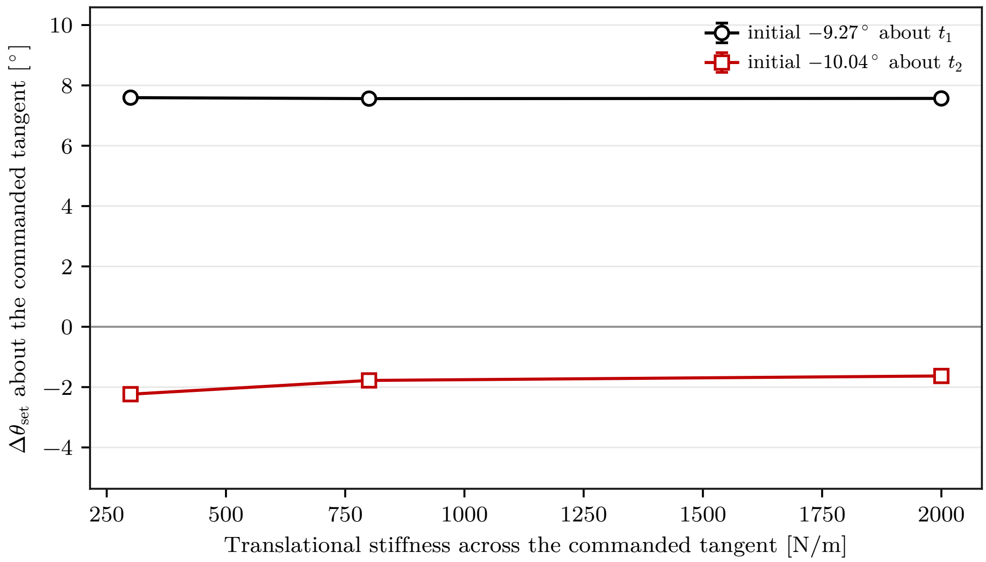


Figure 5.3: Set-up rotation by cross-axis translational stiffness.

The effect of cross-axis translational stiffness is considerably smaller. About t_1, changing the cross-axis stiffness from 300 to 2000 N/m changes the signed set-up rotation by only 0.02°. About t_2, the same variation changes it by 0.61°, from −2.24 to −1.63°. All tested settings are classified as flat.

Taken together, the two stiffness cases span 4.84° about t_1 for the rotational entry and 0.61° about t_2 for the translational entry. Within the tested ranges, the Cartesian gains change the magnitude of the response without removing the axis dependence of the zero-lever behaviour: no tested stiffness setting produced alignment-directed rotation about t_2. Changing the virtual centre position produces the stronger change and is examined next.

5.1.3 Tangential Compliance-Centre Lever: Case D

The present trials analyse the variation of the tangential centre of compliance, and they carry the central measurement of the chapter. Case D establishes whether one fixed non-zero tangential centre can assist different signs and different principal directions of the initial angular misalignment. The CoC is moved along the surface tangent perpendicular to the commanded rotation direction. Seven positions from −40 to +40 mm are tested for both surface tangents and for both signs of the commanded orientation offset, so that a displacement suitable for all four conditions would appear as one column of the table.

Table 5.4: Case-D signed set-up rotation $\Delta\theta_{\text{set}}$ [deg] versus the signed tangential tangential lever $r_c = p_c - p_{\text{TCP}}$.

Commanded offset [°]	$\theta_{\text{dev,before}}$ [°]	−40	−20	−10	0	+10	+20	+40 mm
+10° about t_1	−9.32	+0.18	+0.99	+7.45	+7.57	+7.67	+7.73	+7.87
−10° about t_1	+9.36	−11.30	−11.18	−11.14	−10.83	−10.36	−5.01	−0.09
+10° about t_2	−10.03	−4.51	−1.95	−1.85	−1.63	−1.31	−1.12	+4.43
−10° about t_2	+9.55	−10.51	−1.51	−1.29	−1.25	−1.29	−1.28	+4.68

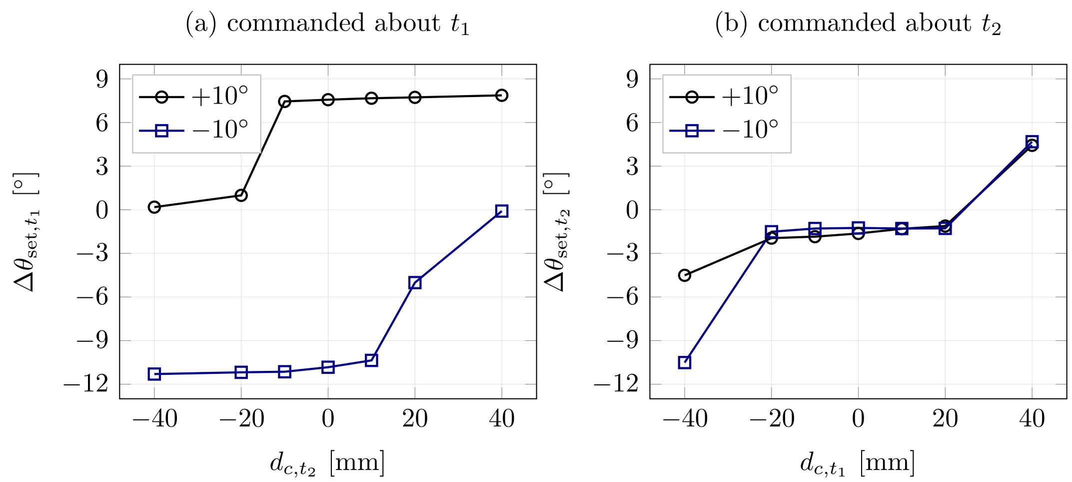


Figure 5.4: Set-up rotation by tangential centre position, by commanded axis.

The tangential compliance-centre position produced the largest response variation over the tested parameter ranges. Table 5.4 gives the seven tested positions, and Figure 5.4 separates the two commanded axes so that the sign reversal is readable within each panel. Reversing the commanded orientation direction reverses the side of the TCP that assists the motion. About t_1, the positive command produces $+7.87°$ at $+40\,\text{mm}$ but only $+0.18°$ at $-40\,\text{mm}$; the reversed command exchanges the assisting and opposing sides, giving $-11.30°$ at $-40\,\text{mm}$ and $-0.09°$ at $+40\,\text{mm}$.

About t_2, the lever matters most relative to the zero-lever reference. For the positive command, the signed response changes from $-1.63°$ at the TCP to $+4.43°$ at the assisting $+40\,\text{mm}$ position, and the final configuration satisfies the TCP-height flatness criterion. For the reversed command, the assisting end is exchanged and gives $-10.51°$ at $-40\,\text{mm}$. The alignment angle calculated from the end-effector pose does not by itself establish a residual physical tool orientation, because the instantaneous relative tool–gripper rotation was not tracked separately during the contact runs.

The sign dependence follows directly from the point-shifted impedance. With the elastic commanded force directed approximately along the negative surface normal, $f_K \approx -F_{K,n} n_s$, its coupling moment is

$$m_{c,K} = r_c \times f_K. \tag{5.1}$$

This is the elastic sign argument. The commanded moment plotted below carries the complete point-shifted spring-damper response, of which $m_{c,K}$ is the part that persists once the reference stops advancing.

For the two principal surface tangents, the selected tangential lever is

$$r_{c,t} = -\rho_c t_2 \quad \text{about } t_1, \qquad r_{c,t} = +\rho_c t_1 \quad \text{about } t_2, \qquad \rho_c > 0, \tag{5.2}$$

which gives, under the same normal-force approximation,

$$m_{c,K} \approx -F_{K,n} \rho_c t_1 \quad \text{about } t_1, \qquad m_{c,K} \approx -F_{K,n} \rho_c t_2 \quad \text{about } t_2. \tag{5.3}$$

The lever must therefore lie along the tangent perpendicular to the commanded rotation direction, with a sign chosen so that the induced moment acts in the alignment direction.

Reversing the commanded orientation offset reverses the required lever sign, as the four curves show.

A fixed non-zero tangential centre would have to remain suitable when the sign or the axis of the initial misalignment changes. The measurements show the opposite. For the positive command about t_1 the assisting side lies at positive coordinates, giving $+7.87°$ against $+0.18°$ on the opposing side; for the reversed command the two sides are exchanged. About t_2 the selected positive side gives $+4.43°$ where the zero-lever condition gives $-1.63°$, and the reversed command again exchanges the assisting side. The lever that assists a command about t_1 lies along t_2, and the lever that assists a command about t_2 lies along t_1, so the two axes do not share one displacement either. The direction and sign that make a non-zero tangential centre assist alignment therefore follow from the initial misalignment. That misalignment is not known before contact. No tested non-zero tangential centre consequently satisfies the definition of a direction-independent fixed CoC used in this thesis.

The two tangents also differ in their sensitivity on the assisting side. About t_1, positions from -10 to $+40$ mm all give set-up rotations between $+7.45$ and $+7.87°$, close to the $+7.57°$ zero-lever value, because substantial alignment-directed rotation is already present there. About t_2, the lever is required to obtain any positive alignment-directed rotation from the positive-command zero-lever response.

The transition between assisting and opposing behaviour is not equally smooth at all positions. The positive t_1 condition changes sharply between -10 and -20 mm, and the -20 mm setting has a standard deviation of $0.42°$. The positive t_2 condition at -40 mm has a standard deviation of $2.35°$. These points are treated separately from the approximately $0.1°$ descriptive comparison scale. Figure D.1 plots all four command conditions with the spread of each setting, so the two can be read against the rest.

The preceding lever positions establish the final dependence on lever position. To examine the mechanism responsible for that dependence, one representative command direction is considered in more detail. The positive t_1 condition is selected because its two outer lever positions produce a large difference in set-up rotation while the TCP-centred condition provides a common reference.

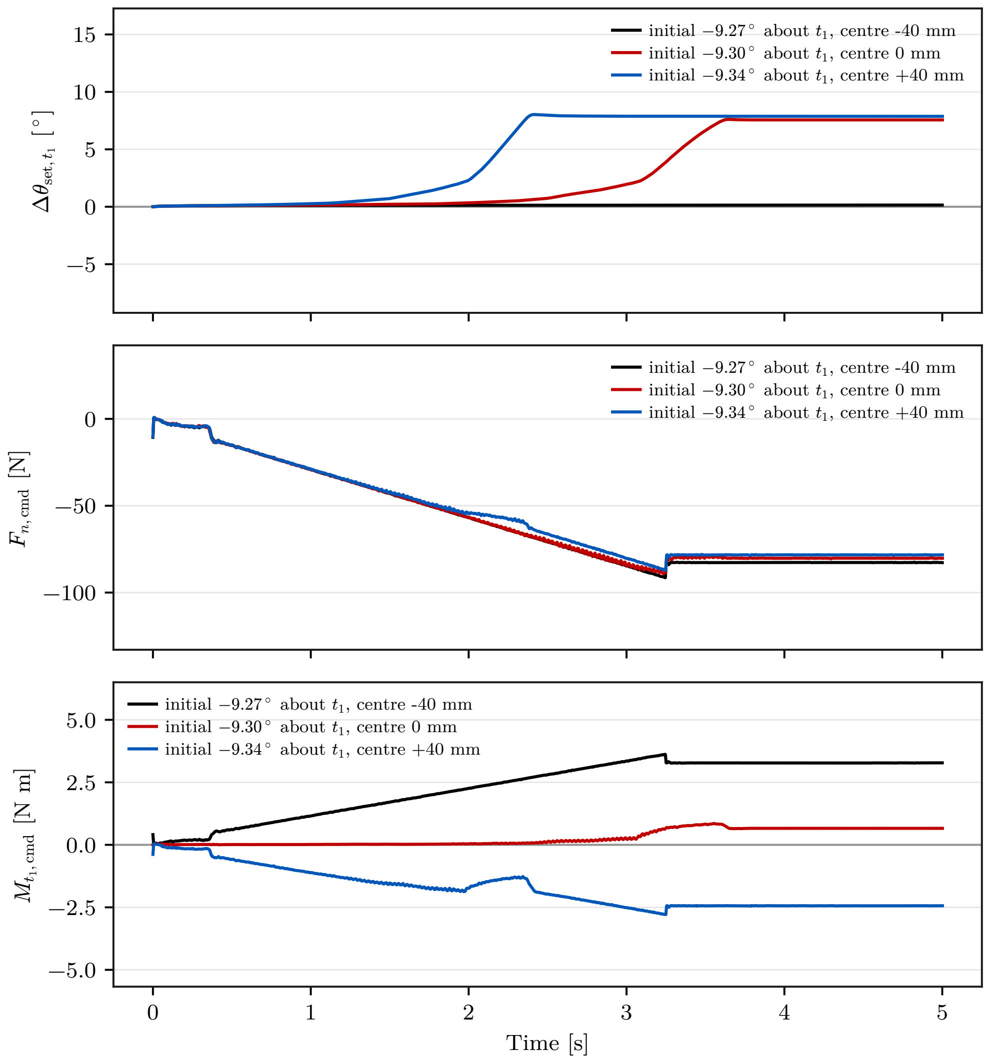


Figure 5.5: Set-up rotation, commanded force and moment at three centre positions.

Figure 5.5 places the set-up rotation $\Delta\theta_{\text{set},t_1}$, the commanded normal force $F_{n,\text{cmd}}$, and the commanded TCP moment $M_{t_1,\text{cmd}}$ on a common time axis for three centre positions. Every condition shown carries a commanded orientation offset of +10° about t_1, and the legends state the measured orientation at the beginning of set-up. At −40, 0, and +40 mm, the commanded normal force settles in a similar range of approximately −77 to −82 N, while the commanded TCP moment about t_1 settles at approximately +3.28, +0.66, and −2.44 N m. The outer positions therefore produce moments of similar magnitude and opposite sign while their set-up rotations differ by 7.73°. The difference in set-up rotation is therefore associated primarily with the moment generated by the displaced impedance reference rather than with a substantial change in the normal pressing force.

The mechanism is a displacement of the reference point, not of the force. A tangential displacement of the CoC moves the impedance reference point away from the TCP, so the translational press no longer acts independently of the rotational part of the impedance. The physical surface force continues to act where the tool touches; the change is that the elastic contribution now generates the additional commanded moment $m_{c,K} = r_c \times f_K$ with $r_c = p_c - p_{\mathrm{TCP}}$, converting part of the normal press into rotation.

That conversion carries a preferred direction. For as long as the normal press is present, a fixed tangential $r_c \neq 0$ keeps generating a moment with one rotational sense, so the same displacement assists an angular deviation of one sign while opposing a deviation of the opposite sign. Figure 5.4 shows this directly: reversing the commanded orientation offset reverses the side of the TCP on which the centre of compliance must lie to assist alignment. The asymmetry follows from the lateral displacement of the impedance reference while the press remains active, not from any change in the magnitude of the press itself.

With the CoC at the TCP, $r_c = 0$ and therefore $m_{c,K} = 0$. The normal press then produces no additional translation–rotation coupling through the virtual centre, so that condition is rotationally neutral with respect to this point-shift mechanism: it selects no preferred tangential direction and adds no one-sided moment. The neutrality is confined to that mechanism, since the tool geometry and the tool mounting remain direction dependent, as Case A shows. Within that scope it supplies the property a fixed centre requires. The TCP-centred condition can be set before the misalignment is known, because it selects no tangential direction at all, whereas every tested non-zero tangential displacement has to be chosen from the misalignment it is meant to correct.

The two roles of the displacement are therefore distinct. During set-up a non-zero tangential displacement can deliberately generate a corrective moment where the natural contact response is weak or acts in the undesired direction, as the t_2 results show. That role is condition dependent by construction. Once the tool has reached the surface-contact configuration, retaining the same displacement while the press continues also retains its preferred moment direction, so the TCP-centred condition is also the appropriate virtual reference for the sustained contact that follows. The tangentially shifted centre is accordingly read as a means of adding alignment authority during set-up rather than as a candidate fixed setting.

The implemented phase structure follows the same division. During set-up, Cases D to H use the point-shifted 6×6 impedance, so that the normal press can generate the additional alignment moment, while the implemented grinding phase returns to the decoupled impedance branch described in Section 3.5.4, removing this coupling for the subsequent sustained-contact task. That arrangement is a consistency of the implemented architecture and not an experimental result: the reported runs ended at the pre-grinding gate, so no displacement was measured during sustained contact.

5.1.4 Compliance-Centre Dependencies

Case E: Lever-Definition Frame

The following experiment investigates whether a lever is fully specified by its magnitude. Case E commands the same nominal 40 mm displacement in the tool frame and in the surface frame. The tool-fixed lever follows the end-effector rotation, whereas the surface-fixed lever remains fixed relative to the calibrated surface.

Table 5.5: Case-E response by lever-definition frame.

Commanded offset [°]	$\theta_{\text{align,before}}$ [°]	$\Delta\theta_{\text{set}}$, tool-fixed lever [°]	$\Delta\theta_{\text{set}}$, surface-fixed lever [°]
None	0.76	-0.06 ± 0.01	-1.05 ± 0.02
$+10°$ about t_1	9.33	$+7.87 \pm 0.01$	$+0.13 \pm 0.01$

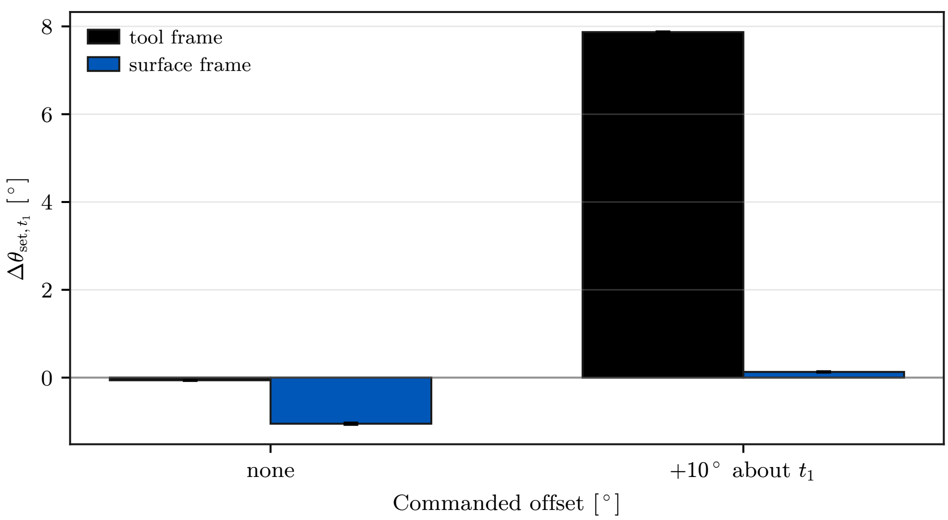


Figure 5.6: Case-E set-up rotation for tool- and surface-frame lever definitions.

The two lever definitions produced markedly different responses. Table 5.5 pairs the two definitions at each command, its tool-fixed $+10°$ condition being the Case-D reference, and Figure 5.6 shows how far apart they fall. For the $+10°$ command about t_1, the tool-fixed definition gives $+7.87 \pm 0.01°$ of set-up rotation, whereas the surface-fixed definition gives only $+0.13 \pm 0.01°$. The surface-fixed condition is also classified as tilted by TCP height. At zero commanded orientation offset, the two definitions produce $-0.06 \pm 0.01°$ and $-1.05 \pm 0.02°$. The coordinate frame in which the compliance-centre displacement is defined is therefore part of the controller parameter itself, because a tool-fixed lever rotates with the tool while a surface-fixed lever stays constant in the surface frame.

A non-zero CoC is therefore not characterised by its magnitude. The same 40 mm carries a direction as well, and that direction moves relative to the contact geometry as the tool rotates unless the frame holding it is selected deliberately. Specifying a displacement of 40 mm is consequently not a complete controller setting, which is a further reason for treating a displaced centre as condition dependent rather than fixed. Returning the centre to the TCP removes the direction and the definition frame along with the coupling.

Case F: Tool-Axis Centre Position

The present trials separate the distance between the CoC and the TCP from the direction of that displacement. Case F contrasts the tangential displacement component with a displacement along the tool axis. The centre is moved along the tool axis at a fixed $+10°$ command about t_1, so that the results are directly comparable with the tangential displacement of Case D at the same command.

Table 5.6: Case-F response by tool-axis centre position.

Position	−40	−20	−10	0	+10	+20	+40 mm
$\Delta\theta_{\mathrm{set},t_1}$	+7.26	+7.49	+7.58	+7.57	+7.58	+7.55	+7.60
Standard deviation	0.13	0.01	0.01	0.01	0.01	0.03	0.04

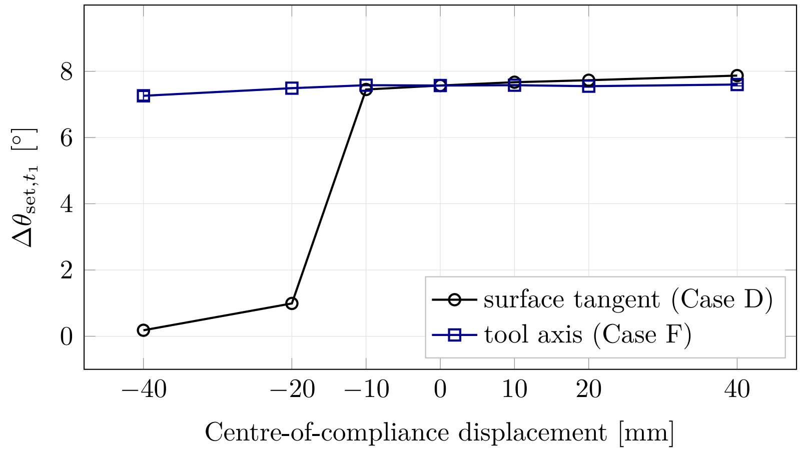


Figure 5.7: Tangential against tool-axis centre displacement at the same command.

The tangential displacement produced the substantially larger response variation under the tested condition. Table 5.6 lists the seven axial positions with their standard deviations, and Figure 5.7 places the two displacement directions on one pair of axes for the same command. The tool-axis response is flat across the whole tested range: the signed set-up rotation changes from +7.26 to +7.60° between −40 and +40 mm, a span of 0.34° against 7.73° for the tangential displacement over the same interval. Every Case-F position is classified as flat, and the response is not strictly monotonic.

This supports the cross-product interpretation of Equation (5.3): a lever component parallel to the normal press contributes no alignment moment. The parameter that acts is not the distance between the CoC and the TCP, but the lever component perpendicular to the press that enters $m_{c,K} = r_c \times f_K$, which is a directional quantity. For a tool inclined by approximately 10°, an axial 40 mm displacement has only $40 \sin 10° \approx 6.9$ mm of tangential projection, which explains the weak residual dependence. The measured variation can also arise from tangential force components, finite rotation, evolving contact geometry, or tool-mount compliance; these contributions are not isolated by the experiment.

Case G: Commanded Orientation-Offset Magnitude

The following experiment investigates whether one displaced centre gives a proportionally scaled response as the commanded misalignment magnitude changes. Case G evaluates that dependence. The zero-lever and assisting-lever conditions are repeated at +5° and +10° about both surface tangents.

Table 5.7: Case-G response by commanded orientation-offset magnitude.

Centre position	$+5°$ about t_1	$+10°$ about t_1	$+5°$ about t_2	$+10°$ about t_2
$\theta_{\text{align,before}}$	4.29	9.34	5.19	10.04
0 mm	+2.55	+7.57	−1.40	−1.63
+40 mm	+2.70	+7.87	+5.91	+4.43

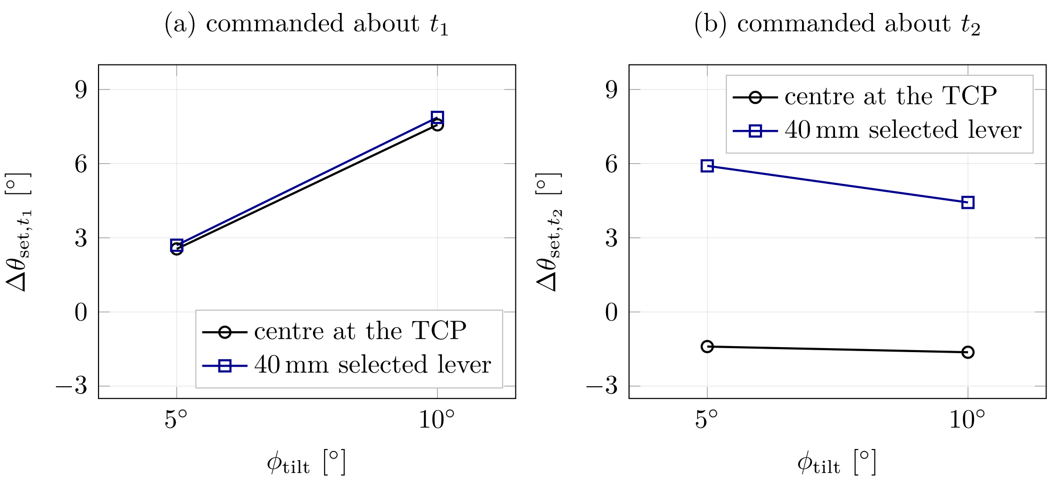


Figure 5.8: Set-up rotation by commanded orientation offset, with and without the lever.

The lever effect depended on the commanded orientation-offset magnitude. Table 5.7 sets the two commanded magnitudes side by side, and Figure 5.8 plots the response against the commanded offset with the two centre positions as the compared series. About t_1, the assisting lever adds only 0.15° at the 5° command and 0.30° at 10°, because substantial zero-lever alignment is already present. About t_2, the lever reverses the response at both tested magnitudes: from −1.40 to +5.91° at 5° and from −1.63 to +4.43° at 10°.

The set-up rotation is not proportional to the commanded orientation offset. The zero-lever t_1 response changes from +2.55° at the 5° command to +7.57° at the 10° command, and the assisting lever adds a different amount at each command about t_2. The commanded orientation magnitude is therefore treated as an independent operating parameter rather than a scale factor on the measured rotation. No proportional scaling was established over the two tested magnitudes, so no single displacement magnitude identified at one initial deviation can be carried to another without measurement.

5.1.5 Generalisation of the Lever-Direction Rule: Case H

Cases A and D establish the required lever direction on the principal tangents. The final contact experiment investigates whether that direction rule also holds between them. Case H determines whether one fixed non-zero tangential lever can serve as a common centre across different tangent-plane misalignment directions, and of whether the direction rule extends to directions between the principal tangents. Each direction is compared with a direction-selected 40 mm lever and with the fixed lever that assists the t_1 command.

Table 5.8: Case-H direction-rule comparison at the common 40 mm lever magnitude.

Commanded rotation direction [°]	Selected $[r_{c,t_1}, r_{c,t_2}]$ [mm]	$\Delta\theta_{\mathrm{set}}$ selected [°]	$\Delta\theta_{\mathrm{set}}$ fixed at t_1 [°]
About t_1	$[0, +40.0]$	+7.87	+7.87
−45° from t_1	$[+28.3, +28.3]$	+4.46	+4.39
+45° from t_1	$[-28.3, +28.3]$	+4.85	+4.80
About t_2	$[-40.0, 0]$	+4.43	−1.61

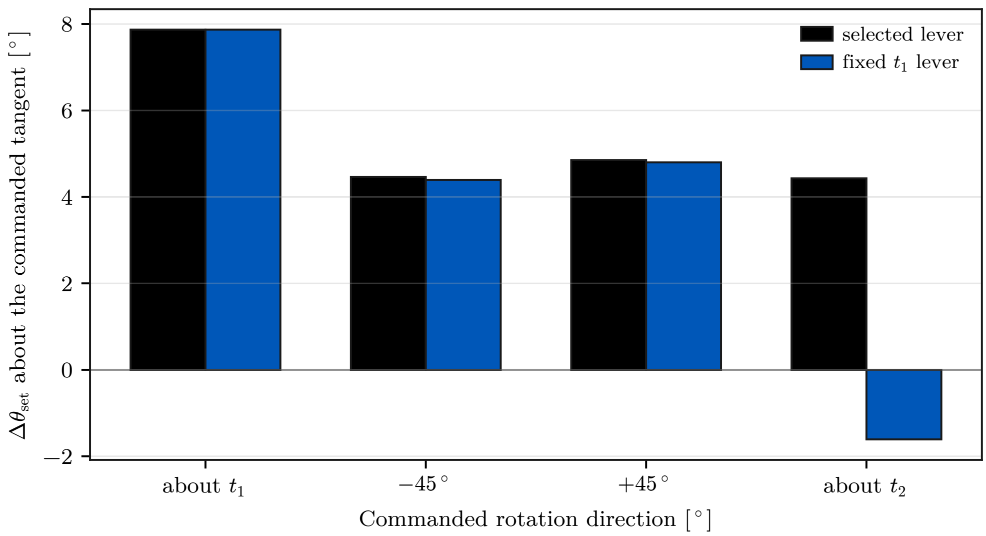


Figure 5.9: Case-H set-up rotation with the direction-selected and fixed levers.

Table 5.8 gives the lever vectors with both responses, and Figure 5.9 shows the two levers against each other at every tested direction. At the two diagonal directions the selected and fixed levers differ by only 0.07° and 0.05°, which is within the observed scatter, because the fixed lever still has a substantial projection along the selected direction. The clearest separation occurs about t_2, where the direction-selected lever gives +4.43° while the fixed t_1 lever gives −1.61°.

The measurements therefore support the geometric lever-direction rule of Equation (4.22) over the four tested directions. They do not establish one lever magnitude that suits every tested direction. The two diagonal results are also not evidence that the fixed lever is general: at $\pm 45°$ the lever selected for the t_1 command still projects onto the direction-selected lever, so the two agree there. Where that projection vanishes, at the t_2 direction, the two differ by $6.04°$. One fixed non-zero tangential lever consequently does not reproduce the direction-selected effect across the tested directions.

Cases D and H together settle the point the contact study was arranged to establish. No tested non-zero tangential CoC can be selected independently of the initial misalignment: its assisting side reverses with the sign of the commanded orientation offset, and the direction it must take follows the required alignment rotation. A displaced centre has to be chosen from that rotation. The TCP-centred condition, $r_c = 0$, is the only condition among those tested that requires no such selection. Within the investigated task and parameter range it is therefore identified as the direction-independent fixed CoC in the sense defined in Chapter 1.

Direction independence is not the same property as alignment authority. The centre that requires no prior knowledge of the misalignment is not the centre that produces the largest alignment-directed rotation in every condition. About t_2 the direction-selected 40 mm lever changed the response from -1.63 to $+4.43°$, where the TCP-centred condition produced none. The TCP provides the fixed default condition. A displaced centre provides greater condition-specific alignment authority where the required rotation is known.

Within a similar contact geometry, the measurements suggest treating the compliance-centre displacement as the first parameter to select where additional rotational authority is needed during set-up. The direction follows from the commanded rotation, the sign is chosen so that the press-induced moment assists alignment, and the definition frame must be stated with the coordinates, since Case E shows that the same nominal coordinates in two frames give very different responses. The rotational stiffness about the commanded axis can then be refined, and the perpendicular translational stiffness treated as a secondary variable for this geometry. No magnitude carries forward as a setting for sustained contact. While the press continues, a fixed tangential r_c keeps commanding rotation in one direction, so the TCP-centred condition is the one to return to once alignment no longer needs assistance. This ordering is a practical interpretation of the present measurements rather than a general design rule, and the scheduling it implies was not implemented or measured.

Across the 55 tested settings, 49 satisfied the TCP-height flatness criterion; the six that did not are listed in Appendix D.

5.2 Null-Space Pose-Hold Results

The remaining experiment is separate from the contact cases and analyses the redundant joint-space direction while the Cartesian pose is held. Matched trials compare the selectable null-space terms under the same internally commanded point-force-equivalent disturbance. The commanded disturbance reached 20.00 N in all twelve trials, with peak equivalent joint-torque norms between 2.17 and 2.45 N m. The torque-limit scale remained equal to one, so the waveform was applied without clipping.

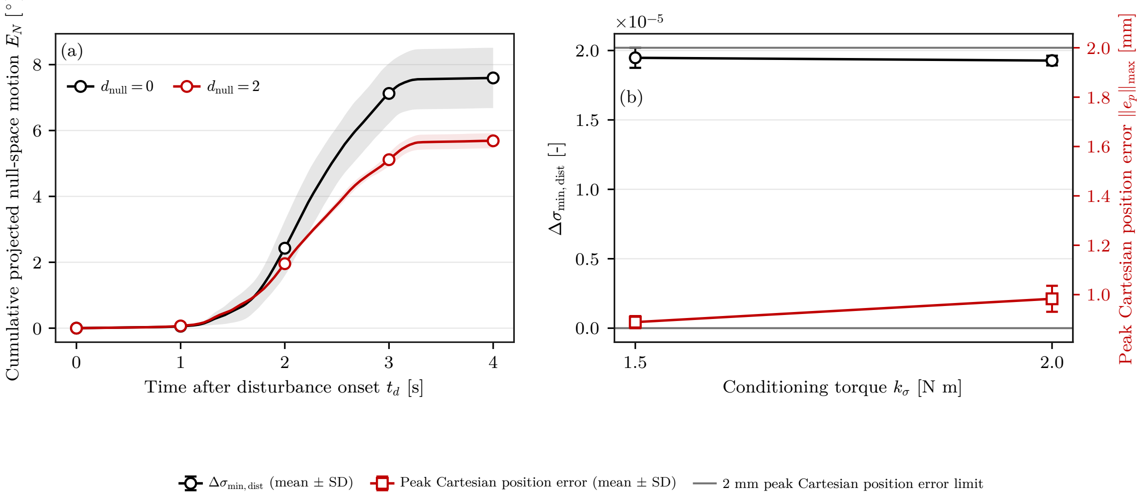


Figure 5.10: Null-space pose-hold response.

Figure 5.10 reports the response during the Cartesian pose-hold disturbance test in two panels. Panel (a) gives the mean cumulative projected null-space motion E_N without null-space torque and with projected damping, the shaded bands showing ± one standard deviation across the three repetitions. Panel (b) gives the change in the minimum Jacobian singular value over the disturbance interval, $\Delta\sigma_{\mathrm{min,dist}}$, together with the peak Cartesian position error, for the two singular-value conditioning magnitudes; its error bars show ± one standard deviation and the dashed line marks the 2 mm task-retention limit. The values for all four settings are given below. Projected null-space damping reduced E_N from

$(7.596 \pm 0.916)^\circ$ without null-space torque to $(5.687 \pm 0.228)^\circ$ at $d_{\text{null}} = 2\,\text{N m s/rad}$, a reduction of approximately 25 %. The net redundant displacement fell in the same proportion, from 0.131 ± 0.016 rad to 0.098 ± 0.004 rad, and the change in $\sigma_{\min}$ across the disturbance interval became less negative, from $(-2.13 \pm 0.47) \times 10^{-3}$ to $(-1.26 \pm 0.10) \times 10^{-3}$. Damping acts on the projected joint velocity through $\tau_d = -d_{\text{null}} N_\tau \dot{q}$, so it vanishes as the redundant motion stops and requests no return towards the initial configuration.

Singular-value conditioning behaved differently from both the uncontrolled and the damping-only settings. It was already active during the 5 s preceding the disturbance and remained active throughout it, so it did not act as a term applied after the redundant configuration had been displaced. Both sigma-only settings ended the disturbance interval with a net redundant displacement whose magnitude was smaller than the scatter across the three repetitions: $(0.3 \pm 0.3) \times 10^{-3}$ rad at $k_\sigma = 1.5\,\text{N m}$ and $(-0.1 \pm 0.2) \times 10^{-3}$ rad at 2.0 N m. Both are smaller than the 0.131 rad measured without null-space torque by more than two orders of magnitude. Over the same interval $\sigma_{\min}$ changed by $(1.95 \pm 0.07) \times 10^{-5}$ and $(1.93 \pm 0.03) \times 10^{-5}$, where it fell by $(-2.13 \pm 0.47) \times 10^{-3}$ without null-space torque. Under the tested disturbance direction, the conditioning term therefore acted by preventing substantial redundant displacement rather than by restoring the configuration after the disturbance had been removed.

The two sigma magnitudes differed in motion activity rather than in net displacement. At $k_\sigma = 2.0\,\text{N m}$ the cumulative projected null-space motion was $(1.619 \pm 0.130)^\circ$, against $(0.283 \pm 0.007)^\circ$ at 1.5 N m, a factor of about six, while both net displacements stayed near zero. The larger torque magnitude was associated with greater switching activity near the locally preferred region. This is consistent with the form of the law: the conditioning term commands a fixed torque magnitude along the selected direction, and the selector s_σ is set to zero only inside the comparison deadband. On that reading the larger cumulative motion is switching activity around the same region rather than poorer disturbance rejection. At 1.5 N m the mean peak Cartesian position error was also lower, 0.889 ± 0.024 mm against 0.983 ± 0.053 mm. The largest position error measured in the complete pose-hold study was 1.304 mm, below the predefined 2 mm acceptance limit. Of the two tested magnitudes, 1.5 N m achieved the same suppression of net disturbance-induced displacement with substantially less redundant motion.

The pose-hold trials compare the four selectable null-space modes under the repeatable commanded disturbance. The two terms act on different quantities. Projected damping dissipates redundant joint motion that is already present, whereas the singular-value term drives the redundant configuration towards the sampled neighbour with the larger $\sigma_{\min}$ whether or not such motion is present. The conditioning law is not dissipative: a disturbance acting towards larger $\sigma_{\min}$ would move the configuration in the same direction as τ_σ rather than against it, so the rejection observed here is specific to the tested disturbance direction. Both terms act through the null-space projector, so the Cartesian task stays primary.

Several bounds apply to these results. Because the conditioning torque was active before the disturbance, the sigma-only trials entered it from a redundant configuration displaced by approximately 0.013 rad from the one at which the trials without conditioning began. The four settings therefore compare the closed-loop behaviour of the complete modes. They do not isolate an instantaneous opposing torque at identical joint configurations. The surface-contact experiments use both contributions together in the combined secondary-torque mode and therefore do not separate their individual effects during contact. The geometric Jacobian combines translational and rotational rows without a common length scale, so the reported $\sigma_{\min}$ is used only as a relative indicator under the fixed Jacobian convention.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This thesis developed and experimentally evaluated a real-time Cartesian impedance controller for contact-induced alignment of a grinding tool with a plane surface. The controller runs in the Franka Control Interface torque loop and combines surface-relative Cartesian impedance control, a virtual centre of compliance, and null-space control. During set-up, stiffness and damping defined at the selected CoC p_c are shifted to the TCP. The resulting translation–rotation coupling allows the commanded normal pressing action to generate an additional alignment moment without a time-varying rotational trajectory.

The experiments focused on the measured end-effector response during contact set-up and on the selection of a fixed CoC when the sign and tangent-plane direction of the initial angular misalignment are unknown before contact. The TCP-centred condition, Cartesian stiffness, and displaced centres were evaluated under matched conditions.

Firstly, the robot with the CoC located at the TCP was tested. The natural contact response was strongly direction dependent: the end effector already rotated considerably toward alignment about t_1, whereas the corresponding response about t_2 was weak and even had the opposite sign under the primary metric. Changing rotational stiffness changed the amount of rotation, and cross-axis translational stiffness had a smaller effect, but neither removed this directional asymmetry.

Moving the CoC tangentially had the strongest effect. Because the normal pressing force acts through an offset virtual reference, the shift creates an additional moment according to $r_c \times f$. The lever therefore has to lie perpendicular to the required alignment rotation and its sign must reverse when the initial angular error reverses. This was especially clear about t_2, where a 40 mm selected lever changed the response from $-1.63°$ at the TCP to $+4.43°$.

The experiments also showed that the displacement direction and its coordinate frame matter, not simply its magnitude. A tool-axis displacement produced almost no change compared with a tangential displacement, and the same nominal 40 mm lever behaved very differently when defined in the tool frame and surface frame.

Testing intermediate rotation directions confirmed the geometric lever-direction rule, but also showed that one fixed non-zero tangential lever cannot provide the same effect for every unknown misalignment direction. Therefore the TCP is the direction-independent fixed CoC among the tested positions, while a displaced CoC can be used when the required alignment direction is already known and additional alignment authority is desired.

Finally, the null-space tests showed that projected damping reduced cumulative redundant joint motion by about 25 %, while singular-value conditioning nearly eliminated the net disturbance-induced redundant displacement for the tested disturbance direction without violating the Cartesian pose-error limit.

6.2 Limitations

6.2.1 Experimental Range

The contact study used one Franka Panda configuration, one rectangular tool and mounting arrangement, one plane surface, one press trajectory, and a bounded range of stiffness and compliance-centre values. The direction-and-sign relation was evaluated about both principal surface tangents and at two diagonal directions. The measured response magnitudes therefore apply to this experimental geometry and parameter range.

No single compliance-centre displacement magnitude was established for all rotation directions or contact geometries. The usable magnitude is also limited by the resulting joint torques. An exploratory tool-axis displacement beyond the reported Case-F range reached a joint-torque limit, which links the centre position directly to the available torque margin.

The TCP-centred condition is identified as the direction-independent fixed centre within the definition and experimental scope of Chapter 1. The measurements establish the direction dependence of every tested non-zero tangential lever for the investigated robot configuration, tool, surface, press trajectory, and orientation-offset range. Other robots, tools, contact geometries, surface orientations, and grinding processes require separate evaluation. Sustained grinding was also outside the measured scope.

6.2.2 Measurement and Tool-Mount Uncertainty

The primary controller-response quantity, the signed set-up rotation, was obtained from the measured end-effector motion. The supporting alignment angle θ_{align} was calculated from the end-effector pose and the calibrated tool normal and therefore uses a fixed tool-to-end-effector transformation.

The tool mounting exhibited approximately $\pm 2^{\circ}$ of relative rotational play about y_{EE} in the unloaded condition. The instantaneous tool–gripper rotation under contact load was not measured, so the value is a mounting characteristic rather than a quantified angular uncertainty during the experiment. The TCP-height quantity provides a geometry-based flatness classification and is not a direct measurement of the physical tool orientation.

The supporting external wrench was obtained from the libfranka model estimate relative to the value stored at the clearance transition. No independent force/torque sensor was used. The reported load quantities therefore represent model-estimated values on one consistent evaluation basis.

6.2.3 Null-Space Configuration and Real-Time Operation

Projected damping and singular-value conditioning were active together during the surface-contact experiments. Their separate assessment used the Cartesian pose-hold study with an internally commanded disturbance. The conditioning torque was already active before the disturbance, so the sigma-only trials entered the disturbance from a redundant configuration displaced by approximately 0.013 rad from the reference configuration used by the trials without conditioning. The comparison therefore describes the closed-loop behaviour of the complete modes.

The disturbance acted along one redundant direction derived from the initial posture. Singular-value conditioning is directional and non-dissipative, so the measured response is specific to this disturbance direction. Real-time operation was demonstrated through the nominal 1 kHz Franka control loop. A separate worst-case execution-time and scheduling-jitter study was outside the experimental scope.

6.3 Future Work

A first extension is to evaluate the fixed-centre result over a broader contact geometry. Relevant variations include other tool shapes, surface orientations, robot configurations, initial angular directions, commanded offsets, and press trajectories. A denser set of tangent-plane rotation directions combined with independent variation of the lever magnitude would further characterise the relation between required displacement magnitude, initial orientation, and contact geometry.

A second extension is an online direction-selected compliance-centre strategy. The experimental results motivate the following sequence:

1. Begin the run with the CoC at the TCP.
2. Estimate the current tool–surface angular deviation at the clearance transition.
3. Evaluate the alignment response available with the TCP-centred impedance.
4. Introduce the direction-selected tangential lever when additional alignment authority is required.
5. Schedule its magnitude ρ_c within the geometric and joint-torque limits.
6. Return the centre smoothly to the TCP after the alignment criterion is reached and before sustained grinding.

This extension would replace a lever selected before the run with a geometry-based, run-specific displacement. Its implementation requires a validated deviation estimate, a criterion for activating the displaced centre, and a criterion for returning to the TCP. These elements remain to be designed and evaluated experimentally.

Independent measurement of the physical tool response would strengthen the contact analysis. An external orientation measurement of the tool face would separate end-effector rotation from relative motion within the tool mount. A direct force/torque sensor would provide an independent reference for the model-estimated external wrench.

The null-space study can also be extended to physical disturbances, additional redundant directions, and contact experiments that vary projected damping and singular-value conditioning separately. The increased switching activity observed at the larger conditioning magnitude suggests two further controller variants: a wider comparison deadband with hysteresis, or a conditioning torque scaled continuously by the sampled singular-value difference. Both variants would require a new pose-hold evaluation before use in the surface-contact task.

Bibliography

[1] Alin Albu-Schäffer, Christian Ott, Udo Frese, and Gerd Hirzinger. Cartesian impedance control of redundant robots: Recent results with the DLR-light-weight-arms. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, volume 3, pages 3704–3709, 2003. doi: 10.1109/ROBOT.2003.1242165.

[2] Benjamin Alt, Florian Stöckl, Silvan Müller, Christopher Braun, Julian Raible, Saad Alhasan, Oliver Rettig, Lukas Ringle, Darko Katic, Rainer Jäkel, Michael Beetz, Marcus Strand, and Marco F. Huber. RoboGrind: Intuitive and interactive surface treatment with industrial robots. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2140–2146, 2024. doi: 10.1109/ICRA57147.2024.10611143.

[3] Stefano Chiaverini, Bruno Siciliano, and Olav Egeland. Review of the damped least-squares inverse kinematics with experiments on an industrial robot manipulator. *IEEE Transactions on Control Systems Technology*, 2(2):123–134, 1994. doi: 10.1109/87.294335.

[4] John J. Craig. *Introduction to Robotics: Mechanics and Control.* Pearson Prentice Hall, Upper Saddle River, NJ, 3 edition, 2005.

[5] Jacques Denavit and Richard S. Hartenberg. A kinematic notation for lower-pair mechanisms based on matrices. *Journal of Applied Mechanics*, 22(2):215–221, 1955.

[6] Alexander Dietrich, Christian Ott, and Alin Albu-Schäffer. An overview of null space projections for redundant, torque-controlled robots. *The International Journal of Robotics Research*, 34(11):1385–1400, 2015. doi: 10.1177/0278364914566516.

[7] Ernest D. Fasse and Jan F. Broenink. A spatial impedance controller for robotic

manipulation. *IEEE Transactions on Robotics and Automation*, 13(4):546–556, 1997. doi: 10.1109/70.611320.

[8] Franka Emika GmbH. libfranka: Communication Test Example. https://github.com/frankarobotics/libfranka/blob/0.7.1/examples/communication_test.cpp, 2019. Official libfranka 0.7.1 example, file `examples/communication_test.cpp`.

[9] Franka Emika GmbH. Franka control interface documentation. https://frankaemika.github.io/docs/, 2023. Accessed for robot interface, libfranka model access, and torque control documentation.

[10] Franka Emika GmbH. libfranka: Cartesian Impedance Control Example. https://github.com/frankaemika/libfranka/blob/master/examples/cartesian_impedance_control.cpp, 2023. Official libfranka example, file `examples/cartesian_impedance_control.cpp`.

[11] Christian Friedrich, Patrick Frank, Marco Santin, and Matthias Haag. Highly dynamic physical interaction for robotics: Design and control of an active remote center of compliance. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. doi: 10.1109/ICRA55743.2025.11128451.

[12] Richard Hartisch and Kevin Haninger. Flexure-based environmental compliance for high-speed robotic contact tasks. In *2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, pages 1608–1613, 2022. doi: 10.1109/AIM52237.2022.9863334.

[13] Neville Hogan. Impedance control: An approach to manipulation—parts i–iii. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):1–24, 1985.

[14] International Federation of Robotics. World robotics 2023 report. Technical report, IFR Statistical Department, Frankfurt am Main, 2023.

[15] Oussama Khatib. A unified approach for motion and force control of robot manipulators. *IEEE Journal on Robotics and Automation*, 3(1):43–53, 1987.

[16] Kevin M. Lynch and Frank C. Park. *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017. URL https://hades.mech.northwestern.edu/images/7/7f/MR.pdf. Free preprint.

[17] Matthias Mayr and Julian M. Salt-Ducaju. A C++ implementation of a cartesian impedance controller for robotic manipulators. *Journal of Open Source Software*, 9 (93):5194, 2024. doi: 10.21105/joss.05194.

[18] Kang Min, Fenglei Ni, Zhaoyang Chen, and Hong Liu. A force control method integrating human skills for complex surface finishing. *Machines*, 12(11):756, 2024. doi: 10.3390/machines12110756.

[19] Richard M. Murray, Zexiang Li, and S. Shankar Sastry. *A Mathematical Introduction to Robotic Manipulation*. CRC Press, 1994.

[20] James L. Nevins and Daniel E. Whitney. Computer-controlled assembly. *Scientific American*, 1989. Background source on robotic assembly and compliance.

[21] Christian Ott, Alin Albu-Schäffer, Andreas Kugi, and Gerd Hirzinger. On the passivity-based impedance control of flexible joint robots. *IEEE Transactions on Robotics*, 24(2):416–429, 2008.

[22] Mathew Jose Pollayil, Franco Angelini, Guiyang Xin, Michael Mistry, Sethu Vijayakumar, Antonio Bicchi, and Manolo Garabini. Choosing stiffness and damping for optimal impedance planning. *IEEE Transactions on Robotics*, 39(2):1281–1300, 2023. doi: 10.1109/TRO.2022.3216078.

[23] Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, and Giuseppe Oriolo. *Robotics: Modelling, Planning and Control*. Springer, London, 2009.

[24] Stefano Stramigioli, Claudio Melchiorri, and Stefano Andreotti. Intrinsically passive grasping and manipulation. *IEEE Transactions on Robotics and Automation*, 2000. URL http://robby.caltech.edu/~jwb/courses/CDS270/papers/passive.grasp.pdf. Open-access preprint.

[25] Markku Suomalainen, Sylvain Calinon, Emmanuel Pignat, and Ville Kyrki. Improving dual-arm assembly by master-slave compliance. In *2019 International Conference on Robotics and Automation (ICRA)*, pages 8676–8682, 2019. doi: 10.1109/ICRA.2019.8793977.

[26] Valeria Villani, Francesco Pini, Francesco Leali, and Cristian Secchi. Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and

applications. *Mechatronics*, 55:248–266, 2018. doi: 10.1016/j.mechatronics.2018.02.009.

[27] Sen Wang, Guodong Chen, Hui Xu, and Zheng Wang. A robotic peg-in-hole assembly strategy based on variable compliance center. *IEEE Access*, 2019. doi: 10.1109/ACCESS.2019.2954459.

[28] Paul C. Watson. Remote center compliance system. U.S. Patent 4,098,001, 1978. Assigned to The Charles Stark Draper Laboratory.

[29] Chong Wu, Kai Guo, and Jie Sun. Dual PID adaptive variable impedance constant force control for grinding robot. *Applied Sciences*, 13(21):11635, 2023. doi: 10.3390/app132111635.

Appendix A

Controller Source Listings

ORANGE — sufficient. The listings are abridged to the mathematical paths used by the thesis and connect each important equation to an implementation excerpt.

This appendix contains abridged source excerpts from the controller implementation described in Chapter 3. Only the paths needed to connect the equations to the implementation are shown. The excerpts use fixed-size Eigen aliases such as `Vec3`, `Vec7`, `Mat3`, `Mat6x6`, and `Mat6x7`. The identifier `R_base_surface` denotes the surface-frame orientation used in the chapters.

A.1 Surface Task Frame and Spatial Gains

The frame order is $[t_1, t_2, n_s]$. Because $t_2 = n_s \times t_1$, this column order is right-handed. A configured tangent that is close to the surface normal n_s is replaced by a fallback direction, so the frame stays defined for any configured plane.

Listing A.1: Surface frame and gain transformation used by the controller.

```
Mat3 makeSurfaceFrameFromNormalTangent(
    const Vec3& normal_input, const Vec3& tangent1_input) {
  const Vec3 normal =
      normalizedOrFallback(normal_input, Vec3(1.0, 0.0, 0.0));
  Vec3 tangent1 =
      tangent1_input - normal * normal.dot(tangent1_input);

  if (tangent1.norm() <= 1e-9) {
    Vec3 fallback(0.0, 1.0, 0.0);
    if (std::abs(normal.dot(fallback)) > 0.95) {
      fallback = Vec3(0.0, 0.0, 1.0);
    }
    tangent1 = fallback - normal * normal.dot(fallback);
  }

  tangent1.normalize();
  Vec3 tangent2 = normal.cross(tangent1);
```

```
  tangent2.normalize();

  Mat3 R_base_surface;
  R_base_surface.col(0) = tangent1;
  R_base_surface.col(1) = tangent2;
  R_base_surface.col(2) = normal;
  return R_base_surface;
}

Mat3 makeSpatialGainMatrix(
    const Vec3& diagonal_in_task_frame, const Mat3& R_task) {
  return R_task * diagonal_in_task_frame.asDiagonal()
       * R_task.transpose();
}
```

A.2 Tool Orientation and Rotation Error

The physical tool normal $n_{\mathrm{Tool,EE}}$ is configured in the EE frame. The desired orientation is the minimum rotation of the captured start orientation that maps this axis onto the signed normal; it is not obtained by assigning the complete surface frame to the EE frame. The commanded angular offsets of a case are applied to that axis before the rotation is formed. A commanded tool orientation offset about t_1 or t_2 therefore enters as a rotation of the orientation target rather than as a trajectory.

Listing A.2: Tool-axis target, commanded tool orientation offsets, and rotation-vector error.

```
Vec3 surfaceToolAxisInBase(
    const ControllerConfig& params, const Mat3& R_base_surface) {
  const double sign =
      (params.tool_axis_target_sign >= 0.0) ? 1.0 : -1.0;
  return sign * R_base_surface.col(2);  // normal = 3rd column
}

Vec3 desiredToolAxisInBase(
    const ControllerConfig& params, const Mat3& R_base_surface) {
  const Vec3 flat_axis =
      surfaceToolAxisInBase(params, R_base_surface);
  const double deg_to_rad = M_PI / 180.0;
  const Vec3 rotation_vector =
      deg_to_rad *
      (params.tool_target_offset_tangent1_deg
           * R_base_surface.col(0) +
       params.tool_target_offset_tangent2_deg
           * R_base_surface.col(1));
  const double angle = rotation_vector.norm();
  if (angle <= 1e-12) {
    return flat_axis;
  }
  return Eigen::AngleAxisd(angle, rotation_vector / angle)
       * flat_axis;
}

Vec3 orientationError(
    const Mat3& R_current, const Mat3& R_desired) {
  Mat3 R_error = R_current.transpose() * R_desired;
  Eigen::AngleAxisd angle_axis(R_error);
  if (std::abs(angle_axis.angle()) < 1e-9) {
    return Vec3::Zero();
  }
  return R_current * angle_axis.axis() * angle_axis.angle();
}
```

The shortest arc that carries the current tool axis onto the target turns about an axis perpendicular to the tool axis, so it leaves the spin about that axis at whatever the start pose held. That spin decides which edge of a rectangular face leads into the surface, so it is commanded separately through `tool_target_offset_normal_deg` rather than inherited from q_{init}.

Listing A.3: Commanded spin about the tool axis (abridged).

```
const Mat3 R_axis =
    rotationBetweenUnitVectors(tool_axis_start, tool_axis_target)
    * R_start;
if (!params.command_tool_twist) {
  return R_axis;
}

// Both directions are measured in the plane perpendicular to the
// tool axis.
const Vec3 current = perpendicular(R_axis * face_long_axis_ee);
const Vec3 zero_reference =
    perpendicular(R_base_surface.col(0));
if (current.norm() <= 1e-9 || zero_reference.norm() <= 1e-9) {
  return R_axis;  // no twist is defined
}

const Vec3 target =
    Eigen::AngleAxisd(
        (M_PI / 180.0) * params.tool_target_offset_normal_deg,
        tool_axis_target)
    * zero_reference.normalized();
const Vec3 current_unit = current.normalized();
double twist = std::atan2(
    tool_axis_target.dot(current_unit.cross(target)),
    current_unit.dot(target));

// A face centred on the tool axis maps onto itself every half turn,
// so take the representative nearest the start pose.
if (face_center_off_axis.norm() <= 0.05 * face_scale) {
  twist -= M_PI * std::round(twist / M_PI);
}
return Eigen::AngleAxisd(twist, tool_axis_target) * R_axis;
```

A.3 Geometric Clearance and Phase Targets

The leading point, edge, or face of the rectangular tool is selected along the descent direction. The transition into set-up is based on the exact most-leading corner height; the average of corners tied within 0.1 mm supplies the controlled contact point and projected reference. The transition depends on this height; the external-force estimate does not enter the decision. The following excerpt is abridged from the phase switch.

Listing A.4: Phase-dependent tool-feature targets (abridged).

```
const Vec3 descend_direction = -R_base_surface.col(2);
double leading_contact_projection = 0.0;
const Vec3 tool_contact_offset_ee =
    selectLeadingToolContact(R_EE, leading_contact_projection);
const Vec3 tool_contact_point =
    p_EE + R_EE * tool_contact_offset_ee;

case ControlPhase::kSurfaceApproach: {
  // The gate freezes this clock, so waiting does not go on ramping
  // the commanded distance.
```

```
  const double distance = std::min(
      params.descend_speed * (phase_time - gate_paused_time),
      params.descend_max_distance);
  desired.p_d = p_start + distance * descend_direction;
  desired.pdot_d = params.descend_speed * descend_direction;

  const Vec3 surface_normal = (-descend_direction).normalized();
  const double height_above_surface =
      surface_normal.dot(p_EE - surface_point_runtime)
      - leading_contact_projection;
  const double controlled_point_height =
      surface_normal.dot(tool_contact_point - surface_point_runtime);
  const Vec3 projected_surface_point =
      tool_contact_point - controlled_point_height * surface_normal;
  const bool clearance_reached =
      height_above_surface <= params.descend_surface_clearance;

  if (clearance_reached && !gate_blocking) {
    active_tool_contact_offset_ee = tool_contact_offset_ee;
    first_contact_tcp = p_EE;  // captured at clearance
    first_contact_point = projected_surface_point;
    setup_push_start = -controlled_point_height;
    R_contact_start = R_EE;
    contact_force_bias = external_force;
    contact_moment_bias = external_moment;
    phase = ControlPhase::kSetup;
  }
  break;
}

case ControlPhase::kSetup: {
  const double push =
      setUpPush(params, phase_time - gate_grind_paused_time,
                setup_push_start, setup_push_velocity);
  const Vec3 edge_target =
      first_contact_point + push * descend_direction;
  desired.p_d =
      edge_target - R_contact_start * tool_contact_offset_ee;
  desired.pdot_d = setup_push_velocity * descend_direction;
  // Exit after minimum time on moment-delta norm, or on timeout.
  break;
}

case ControlPhase::kGrinding: {
  const Vec3 n = descend_direction;  // unit, into the surface
  Vec3 edge_target;
  if (params.grind_sweep_enabled) {
    edge_target = first_contact_point + grind_push * n
                + sweep_s * grind_tangent;
    desired.pdot_d = sweep_s_dot * grind_tangent;
  } else {
    const double edge_penetration =
        n.dot(tool_contact_point - first_contact_point);
    edge_target =
        tool_contact_point + (grind_push - edge_penetration) * n;
  }
  desired.p_d =
      edge_target - R_contact_start * tool_contact_offset_ee;
  break;
}
```

The identifiers `first_contact_*` are retained source names. They are captured at the configured clearance, not at a force-detected first contact. The model-estimated wrench recorded at the same instant is subtracted from every later sample.

Both operator gates latch once armed. The gate compares a held pose against the clearance height, and re-testing that comparison every cycle released the gate for single cycles on measurement noise; on each release the descent clock advanced and commanded the tool further down, so a longer wait produced a harder press. Latching removes that dependence on waiting time.

A.4 Centre of Compliance and the Offset Adjoint

The lever $r_c = p_c - p_{\mathrm{TCP}}$ points from the TCP to the selected centre of compliance, and the implementation uses the same convention. The same congruence is applied to stiffness and damping.

Listing A.5: TCP-to-compliance-centre adjoint and congruence transform.

```
Mat3 skewMatrix(const Vec3& v) {
  Mat3 s;
  s << 0.0, -v(2),  v(1),
       v(2),  0.0, -v(0),
      -v(1),  v(0),  0.0;
  return s;
}

Mat6x6 complianceShiftMatrix(const Vec3& r_c) {
  Mat6x6 adjoint = Mat6x6::Zero();
  adjoint.block<3, 3>(0, 0) = Mat3::Identity();
  adjoint.block<3, 3>(0, 3) = -skewMatrix(r_c);
  adjoint.block<3, 3>(3, 3) = Mat3::Identity();
  return adjoint;
}

Mat6x6 shiftGainToTcp(
    const Mat6x6& gain_at_center, const Vec3& r_c) {
  const Mat6x6 adjoint = complianceShiftMatrix(r_c);
  return adjoint.transpose() * gain_at_center * adjoint;
}
```

The wrench routine selects between the decoupled and the coupled law and constructs the lever. The centre can be defined relative to the TCP in end-effector coordinates, or the lever can be given directly in surface coordinates. The surface-frame parameter keeps its original name and stores $-r_c$, which the sign in the listing corrects.

Listing A.6: Selection of the decoupled or coupled spring wrench (abridged).

```
const bool coupled =
    params.use_virtual_compliance_center &&
    (phase == ControlPhase::kSetup ||
     (phase == ControlPhase::kPoseHold &&
      params.use_setup_impedance_hold));
if (!coupled) {
  // Two independent springs. dv.tail is -omega, so the damping
  // signs match.
  Vec6 wrench;
  wrench.head<3>() = Kp * dx.head<3>() + Dp * dv.head<3>();
  wrench.tail<3>() = KR * dx.tail<3>() + DR * dv.tail<3>();
  return wrench;
}

Vec3 r_c = Vec3::Zero();
if (params.compliance_center_in_tool_frame) {
  r_c = R_EE * params.compliance_center_offset_ee;
} else {
  r_c = -(R_base_surface
      * params.r_tcp_from_compliance_center_surface);
}
const Mat6x6 K_tcp =
    shiftGainToTcp(blockDiagonal(Kp, KR), r_c);
```

```
const Mat6x6 D_tcp =
    shiftGainToTcp(blockDiagonal(Dp, DR), r_c);
return K_tcp * dx + D_tcp * dv;
```

A.5 Task-Inertia Damping

The joint-space mass matrix M is solved rather than explicitly inverted. A 10^{-9} diagonal regularisation is used only for the task-inertia matrix Λ calculation; it is not the null-space pseudoinverse cutoff. Every intermediate result is checked before use, and a single non-finite or non-positive entry marks the estimate invalid.

Listing A.7: Task-inertia estimate and per-axis damping (abridged).

```
Eigen::LDLT<Mat7x7> mass_ldlt(joint_mass);
const Mat7x6 Minv_Jt = mass_ldlt.solve(J.transpose());
Mat6x6 lambda_inv = J * Minv_Jt;
lambda_inv = 0.5 * (lambda_inv + lambda_inv.transpose());

Mat6x6 lambda_inv_damped = lambda_inv;
lambda_inv_damped.diagonal().array() += 1e-9;
Eigen::LDLT<Mat6x6> lambda_ldlt(lambda_inv_damped);
Mat6x6 Lambda_base = lambda_ldlt.solve(Mat6x6::Identity());
Lambda_base = 0.5 * (Lambda_base + Lambda_base.transpose());

const Mat6x6 T_task = blockDiagonalRotation(R_task);
Mat6x6 Lambda_task = T_task.transpose() * Lambda_base * T_task;
Lambda_task = 0.5 * (Lambda_task + Lambda_task.transpose());

for (int i = 0; i < 3; ++i) {
  const double m = Lambda_task(i, i);
  const double I = Lambda_task(i + 3, i + 3);
  if (m <= 0.0 || I <= 0.0) {
    return result;  // estimate stays invalid
  }
  result.translational(i) = m;
  result.rotational(i) = I;
}

Vec3 criticalDampingFromStiffness(const Vec3& inertia,
                                  const Vec3& stiffness,
                                  double damping_ratio,
                                  const Vec3& min_damping,
                                  double max_damping) {
  Vec3 damping = Vec3::Zero();
  const double zeta = std::max(0.0, damping_ratio);
  for (int i = 0; i < 3; ++i) {
    const double m = std::max(0.0, inertia(i));
    const double k = std::max(0.0, stiffness(i));
    const double critical = 2.0 * zeta * std::sqrt(m * k);
    damping(i) = std::min(
        max_damping,
        std::max(std::max(0.0, min_damping(i)), critical));
  }
  return damping;
}
```

Only finite, positive directional inertia values λ_i are used. Otherwise, the phase-specific fixed damping matrix D is applied. Each inertia-scaled coefficient is bounded above by `auto_damping_max`; the value configured for the reported experiments is listed in Appendix C. The lower bound `min_damping` is the configured manual matrix only when `auto_damping_min_from_manual` is set, and is zero otherwise.

A.6 Null-Space Torque

For $J \in \mathbb{R}^{6\times7}$, Eigen returns six singular values and a full 7×7 right-singular matrix. Column six in zero-based C++ indexing is therefore v_7, the structural null vector; it is distinct from v_6, the right-singular vector paired with $\sigma_{\min} = \sigma_6$. The two sampled configurations are evaluated by re-querying the robot model for the Jacobian $J(q_\pm)$ at $q \pm \alpha n$, so the comparison uses the same kinematics as the control cycle.

Listing A.8: Thresholded-SVD joint-torque projector and null-space law (abridged).

```
Eigen::JacobiSVD<Mat6x7> svd_current(
    J, Eigen::ComputeFullU | Eigen::ComputeFullV);
const auto singular_values = svd_current.singularValues();
const double svd_cutoff =
    std::max(0.0, params.nullspace_svd_relative_tolerance)
    * singular_values.maxCoeff();

Mat7x6 J_pinv = Mat7x6::Zero();
for (int i = 0; i < 6; ++i) {
  if (singular_values(i) > svd_cutoff) {
    J_pinv += (1.0 / singular_values(i))
            * svd_current.matrixV().col(i)
            * svd_current.matrixU().col(i).transpose();
  }
}

Mat7x7 N_tau =
    Mat7x7::Identity() - J.transpose() * J_pinv.transpose();
N_tau = 0.5 * (N_tau + N_tau.transpose());
const Vec7 dq_nullspace = N_tau * dq;

// Mode 0 commands nothing, but the projector above still observes
// the redundant axis for the diagnostics.
if (params.nullspace_mode == NullspaceMode::kOff) {
  return Vec7::Zero();
}

const bool damping_enabled =
    params.nullspace_mode == NullspaceMode::kDampingOnly ||
    params.nullspace_mode == NullspaceMode::kDampingAndSigma;
Vec7 tau_damping = Vec7::Zero();
if (damping_enabled) {
  tau_damping.noalias() =
      -params.nullspace_damping * dq_nullspace;
}
if (params.nullspace_mode == NullspaceMode::kDampingOnly) {
  return tau_damping;
}
const bool sigma_only =
    params.nullspace_mode == NullspaceMode::kSigmaOnly;
const Vec7 sigma_fallback = sigma_only ? Vec7::Zero() : tau_damping;

Vec7 n = svd_current.matrixV().col(6);      // v_7
if (n.norm() <= 1e-9) { return sigma_fallback; }
n.normalize();
const double alpha =
    std::abs(params.nullspace_probe_step_rad);
if (alpha <= 1e-12) { return sigma_fallback; }

// The Jacobian is re-evaluated at the two sampled configurations.
Map<const Mat6x7> J_plus(model.zeroJacobian(
    Frame::kEndEffector, vec7ToArray(Vec7(q_current + alpha * n)),
    state.F_T_EE, state.EE_T_K).data());
Map<const Mat6x7> J_minus(model.zeroJacobian(
    Frame::kEndEffector, vec7ToArray(Vec7(q_current - alpha * n)),
```

```
    state.F_T_EE, state.EE_T_K).data());

const double sigma_difference =
    smallestSingularValue(J_plus) - smallestSingularValue(J_minus);
const double deadband =
    std::max(0.0, params.nullspace_sigma_deadband);
if (std::abs(sigma_difference) <= deadband) {
  return sigma_fallback;
}

const double sigma_direction = (sigma_difference > 0.0) ? 1.0 : -1.0;
const Vec7 best_direction = sigma_direction * n;
// alpha determines only where sigma is sampled. k_sigma is the
// commanded torque magnitude along the better direction.
const Vec7 tau_sigma =
    params.nullspace_sigma_gain * N_tau * best_direction;

return sigma_only ? tau_sigma : tau_damping + tau_sigma;
```

Here the code variable `alpha` represents α_{probe} and changes only the two sampled configurations. It does not multiply the active torque. Singular values at or below the cutoff are omitted from the inverse. The routine also fills a diagnostics structure that carries the sampled singular values, the selected direction, and the projected joint velocity $\dot{q}_{\mathrm{null}}$ into the recorded data.

A.7 Core Control Callback

The following excerpt shows the state and model quantities acquired at the start of each torque callback. The remaining listing then shows how these quantities enter the torque command.

Listing A.9: Robot-state and model quantities acquired by the torque callback.

```
Map<const Vec7> q_current(state.q.data());
Map<const Vec7> dq(state.dq.data());

const auto jacobian_array =
    model.zeroJacobian(Frame::kEndEffector, state);
Map<const Mat6x7> J(jacobian_array.data());
const Vec6 xdot = J * dq;

Map<const Mat4x4> T_EE(state.O_T_EE.data());
const Vec3 p_EE = T_EE.block<3, 1>(0, 3);
const Mat3 R_EE = T_EE.block<3, 3>(0, 0);

Map<const Vec6> external_wrench(
    state.O_F_ext_hat_K.data());
```

Listing A.10: Core impedance, coupled set-up, and torque composition.

```
const Vec3 e_p = desired.p_d - p_EE;
const Vec3 e_R = applyRotationalAxisMask(
    params, orientationError(R_EE, R_d_used), R_base_surface);

Vec6 dx;
dx.head<3>() = e_p;
dx.tail<3>() = e_R;
Vec6 dv;
dv.head<3>() = desired.pdot_d - pdot;
```

```
dv.tail<3>() = -omega;

const Vec6 wrench =
    computeCartesianImpedanceWrench(
        params, phase, Kp_used, Dp_used,
        KR_used, DR_used, R_base_surface,
        dx, dv, R_EE);

const Vec7 tau_nullspace = computeNullspaceTorque(
    params, model, state, D, dq);

AutomaticDisturbance disturbance;
if (phase == ControlPhase::kPoseHold &&
    params.disturbance_auto_enabled) {
  disturbance = computeAutomaticDisturbance(
      params, model, state, disturbance_force_direction_base,
      time - phase_start_time);
}

Array7 coriolis_array = model.coriolis(state);
Map<const Vec7> coriolis(coriolis_array.data());
const Vec7 tau_task = J.transpose() * wrench;
const Vec7 tau_cmd =
    tau_task + tau_nullspace + disturbance.tau + coriolis;
```

The null-space function is called in every non-manual phase. Manual guidance uses its separate Coriolis-plus-joint-damping branch. The disturbance term is non-zero only in the Cartesian pose-hold study of Section 4.6; it is zero in every surface-contact run.

Appendix B

Recorded Experimental Data

ORANGE — sufficient. The appendix records the signals actually used in the evaluation and identifies the common calibrated plane used by the controller and analysis.

The controller stores experimental signals in a preallocated ring buffer. This avoids file access and dynamic buffer growth inside the 1 kHz control callback. After the controller stops, the retained rows are written in chronological order to a CSV file.

B.1 Control Phases

ORANGE — sufficient. The numeric phase values are mapped directly to their recorded state names.

The integer `phase` identifies the active part of the experiment:

`phase`	0	1	2	3	4	5
state	`approach_orient`	`approach_descend`	`set_up`	`grind`	`hold`	`manual_guide`.

B.2 Signals Used in the Evaluation

ORANGE — sufficient. The table is a useful reproducibility record and keeps units, frames, and model-estimated quantities explicit.

Table B.1 lists the groups used to calculate the reported quantities. A suffix `_x,_y,_z` denotes base-frame components, and `_1.._7` denotes joint components.

Table B.1: Recorded signal groups used in the evaluation.

Column group	Meaning
`time`, `phase`	Run time and active control state.
`p_EE`, `p_d`	Measured and desired TCP positions [m].
`tool_contact`, `edge_target`	Selected contact point and its reference [m].
`tool_contact_offset_ee`	Offset of the selected contact point from the TCP, in EE coordinates [m].
`first_contact_tcp`, `first_contact`	TCP position and projected surface reference captured at the geometric clearance transition; the transition is not force triggered.
`e_p`, `e_R`, `pdot`, `pdot_d`, `omega`	Cartesian errors and velocities. During set-up, `e_R` is measured relative to the clearance orientation selected at the transition and held constant during set-up.
`angular_deviation_deg`	The alignment angle of Equation (4.26), $\theta_{\text{align}} = \cos^{-1}(-n_{\text{Tool},0}^{\top} n_s)$, between the pose-based tool axis $n_{\text{Tool},0}$ and the flat target $-n_s$, with n_s the calibrated physical-plane normal stored in the controller configuration [°]. It is zero for a tool face parallel to the surface. This recorded quantity produced the deviation values reported in the result tables.
`angular_deviation_t1_deg`, `angular_deviation_t2_deg`, `angular_deviation_normal_deg`	The same angular deviation resolved on t_1, t_2, and n_s [°]. These supply the per-tangent decomposition reported in the results.
`p_CoC`, `r_eff`	CoC $p_{\text{TCP}} + r_c$ and the offset from it to the selected tool point, $p_T - p_c$, both in the base frame [m]. A zero lever leaves the centre on the TCP, so the pair is written for every run and no case has to be separated.

Table B.1 continued

Column group	Meaning
`t_align`	Time from the start of set-up at which the deviation first fell inside the configured tolerance and stayed there [s]; negative until that happens, and negative throughout for a run that never does. Observed only: reaching it does not end the phase.
`f`, `m`	Commanded Cartesian force [N] and moment [N m].
`external_force`, `external_moment`	Model-estimated external wrench from libfranka, expressed in the base orientation and referenced to stiffness frame K.
`contact_force_bias`, `contact_moment_bias`	External-wrench values captured at the clearance transition.
`force_after_contact`, `moment_after_contact`	The model-estimated wrench with those reference values already subtracted. The reported normal load is taken from this group.
`push`	Set-up penetration reference or grind preload retained after set-up [m].
`nullspace_mode`, `sigma_current`	Selected null-space law and current smallest singular value $\sigma_{\min}$.
`sigma_plus`, `sigma_minus`, `sigma_difference`, `sigma_direction`	Smallest singular value $\sigma_{\min}$ at the two sampled configurations $q \pm \alpha_{\text{probe}} v_7$, their difference, and the selected sign. Recorded in every mode, including when no torque is commanded.
`nullspace_dq_1..7`	Projected joint velocity $\dot{q}_{\text{null}}$ [rad/s].
`tau_sigma_1..7`, `tau_sigma_norm`, `tau_nullspace_norm`	Singular-value torque τ_σ and total null-space torque τ_{null} [N m].
`nullspace_damping`	Active projected damping gain [N m s/rad].

Table B.1 continued

Column group	Meaning
`disturbance_scale`, `disturbance_force_base_x..z`, `disturbance_point_base_x..z`	Commanded hold-disturbance waveform, point force [N], and the base-frame position of the link point it acts at [m].
`tau_disturbance_1..7`	Equivalent joint torque $J_p^\top f_d$ in the internally commanded hold study [N m].
`tau_cmd_1..7`	Final commanded joint torque [N m].

B.3 Evaluation Quantities

ORANGE — sufficient. The short summary routes each reported metric to its recorded source without reintroducing unused fields.

The contact-alignment angle θ_{align} was calculated from the measured end-effector orientation and the calibrated surface normal n_s. Its before value was taken at the start of set-up, and its after value from the settled set-up interval. No other plane normal was substituted in the analysis. Selected contact-point displacement was calculated from `tool_contact`. The reported normal load F_n was the model-estimated force change since the clearance transition.

Three markers of that same angle are written once into the set-up report rather than into every row. All three read θ_{align} and the clock alone. None feeds the phase exit or the command. `t_align_fraction` is the time at which θ_{align} first fell below a configured fraction of its value at first contact. Being a relative threshold, it is reached by conditions that end far from flat, which the absolute tolerance is not. `deviation_min` is the smallest θ_{align} the run reached, together with the time at which it was reached. Both are defined for every run, so a condition that never aligned still records how close it came. `align_status` reads `aligned` where θ_{align} settled inside the tolerance for the required hold time, and `not_aligned` otherwise. Runs that never settled are identified rather than discarded.

None of these determines whether the tool itself ended flat. That is calculated afterwards from the height of the TCP above the calibrated plane, $n_s^\top(p_{\text{EE}} - p_s)$, against the height a flatly seated tool face would give. Only p_{EE} and the stored plane enter it. Unlike θ_{align}, it therefore carries no assumption about the rotation of the tool within the gripper.

For the internally commanded hold study, redundant motion is calculated by integrating the norm of the recorded projected joint velocity $\dot{q}_{\mathrm{null}}$ over the disturbance interval. The conditioning response is the final change in σ_{min}, and Cartesian task retention is represented by the peak position-error norm $\|e_p\|$.

Appendix C

Controller Parameters

ORANGE — sufficient. The active geometry, phase gains, damping policy, timings, null-space values, and collision settings are collected in one reproducibility record.

This appendix lists the effective controller parameters used in the reported experiments. It is a reproducibility record for the implementation described in Chapters 3 and 4.

C.1 Surface and Tool Geometry

ORANGE — sufficient. The configured geometry and initial joint vector are stated with their exact keys and units.

The plane satisfies

$$n_s^\top (p - p_s) = 0, \qquad n_s = R_y(\beta_s) R_x(\alpha_s) e_z,$$

and the surface-frame column order is $[t_1, t_2, n_s]$. Table C.1 collects the corresponding surface and tool parameters.

Table C.1: Surface and tool parameters.

Quantity	Configuration key	Value
Surface point	`surface_point`	$[0.515267, -0.107172, 0.003108]$ m
Surface tilts	α_s, β_s	$-1.585°$, $+0.988°$
First-tangent hint	`surface_tangent1_hint_base`	$[1, 0, 0]$
Tool axis in EE	`tool_axis_ee`	$[0.026387057, -0.006678514, 0.999629492]$
Signed alignment target	`tool_axis_target_sign`	$-n_s$
Spin about the tool axis, from t_1	`tool_target_offset_normal_deg`	$90.0°$
Tool-face centre in EE	`tool_contact_face_center_ee`	$[0, 0, 0.020]$ m
Tool face size	width $\times$ length	0.040×0.120 m
Feature tie tolerance	`tool_contact_feature_tie_tolerance`	10^{-4} m

The initial joint configuration used in the reported experiments is

$$q_{\text{init}} = [0.014777,\ 0.210911,\ -0.009320,\ -2.284835,\\ -0.002725,\ 2.493410,\ 0.780694]^\top \text{ rad.}$$

C.2 Phase Sequence and Impedance

ORANGE — sufficient. The phase-specific gain frames and transition parameters reconcile the values stated across Chapters 3 and 4.

Table C.2: Phase and mode gains.

Phase / quantity	Active stiffness and frame	Configured damping
Approach translation	$[150, 150, 150]$ N/m, surface $[t_1, t_2, n_s]$	$[20, 20, 20]$ N s/m
Approach rotation	$[180, 180, 90]$ N m/rad, surface $[t_1, t_2, n_s]$	$[15, 15, 12]$ N m s/rad
Pre-set-up gate translation	$[2000, 2000, 350]$ N/m, surface $[t_1, t_2, n_s]$	$[10, 10, 175]$ N s/m
Pre-set-up gate rotation	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Set-up translation (compliance-centre source)	$[2000, 2000, 350]$ N/m in all Cases A to H, surface $[t_1, t_2, n_s]$	$[10, 10, 175]$ N s/m
Set-up rotation (compliance-centre source)	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Grind translation (decoupled)	as the set-up entry above, surface $[t_1, t_2, n_s]$	$[10, 10, 175]$ N s/m
Grind rotation (decoupled)	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Hold translation	2000 N/m, isotropic base	50 N s/m
Hold rotation	50 N m/rad, isotropic base	8 N m s/rad
Manual guidance	no Cartesian stiffness; joint coordinates	0.5 N m s/rad joint damping

The pre-set-up gate was disabled in all reported contact runs, so its defined gains were inactive. The enabled pre-grind gate remains in set-up and keeps the coupled pressed law rather than using the pre-set-up gate gains.

The reported experiments used the surface-frame translational gains listed above. The active timing and trajectory values are listed in Table C.3.

Table C.3: Phase-transition and trajectory parameters.

Quantity	Configuration key	Value
Orient minimum time	`approach_orient_min_time`	1.0 s
Tool-axis switch tolerance	`approach_orient_error_threshold`	0.016 rad
Spin switch tolerance	`approach_orient_spin_error_threshold`	0.009 rad
Orient timeout / maximum rate	`approach_orient_timeout`, `approach_orient_max_rate_deg`	8.0 s, 20°/s
Descent speed / maximum distance	`descend_speed`, `descend_max_distance`	0.1 m/s, 0.640 m
Surface clearance	`descend_surface_clearance`	0.020 m
Set-up minimum time / timeout	`setup_min_time`, `setup_timeout`	2.0 s, 5.0 s
Set-up moment-change threshold	`setup_moment_threshold`	150 N m
Set-up push speed / endpoint	`setup_push_speed`, `setup_push_end`	0.080 m/s, 0.240 m in all Cases A to H
Configured grinding sweep	`grind_sweep_enabled`	enabled along t_2, 0.030 m, 0.2 Hz; not entered in the reported runs
Pre-set-up / pre-grind gates	`pause_before_set_up`, `pause_before_grind`	disabled, enabled

C.3 Inertia-Scaled Damping and Centre of Compliance

Inertia-scaled damping follows Equations (3.16) and (3.20). It is enabled for approach, set-up, the configured grinding phase, and the pause gate. The approach factor is $\zeta = 1.9$, while set-up and grinding use $\zeta = 1.0$. The pause gate fits separate translational and rotational factors to its two configured damping targets. The configured maximum coefficient `auto_damping_max` is 400. The configured values in Table C.2 provide fallback damping for the phases using inertia-scaled damping. With `auto_damping_min_from_manual` left at zero, those manual values act only as that fallback and impose no lower bound on a calculated coefficient.

Standard Cartesian pose hold instead used its configured matrices directly. All twelve reported null-space runs recorded `hold_auto_damping=0`, with $D_p = 50 I_3\,\mathrm{N\,s/m}$ and $D_R = 8 I_3\,\mathrm{N\,m\,s/rad}$. The hold-damping values in Table C.2 are therefore active values rather than fallbacks.

For set-up, the source damping blocks are calculated from TCP task inertia Λ and then shifted to the selected CoC p_c. This preserves symmetry and positive semidefiniteness. The factor $\zeta = 1$ is the directional coefficient used by the online damping calculation.

The coupled 6×6 law is enabled only during set-up, and only in Cases D to H. Case E used the surface-frame lever definition. The other shifted cases used tool-fixed centre displacements configured in end-effector coordinates. For the surface-fixed definition,

$$r_c = R_{\text{surface}}\, r_c^S, \qquad r_c^S = [\, r_{c,t_1},\; r_{c,t_2},\; r_{c,n}\,]^\top,$$

with the tested components listed in table 4.3. A tool-fixed lever remains constant in end-effector coordinates and rotates in the base frame with the end effector; a surface-fixed lever remains constant in surface coordinates. The controller transforms the selected definition to base coordinates during each set-up cycle.

C.4 Null-Space and Operating Parameters

ORANGE — sufficient. The selected mode, gains, probe settings, logging capacity, gripper state, and robot-side thresholds are stated explicitly.

The null-space settings are explained in Section 4.3.3. The final two rows of Table C.4 are Franka collision/reflex thresholds configured through the libfranka collision-behaviour interface. Robot-side monitoring applied these thresholds throughout the reported runs.

Table C.4: Null-space and operating parameters.

Parameter	Symbol / key	Value
Null-space mode	`nullspace_mode`	3 (projected damping and singular-value conditioning)
Null-space damping	d_{null}	2.0 N m s/rad
Sigma torque magnitude	k_σ	1.0 N m
Probe distance	α_{probe}	0.08 rad
Sigma difference deadband	δ_σ	10^{-6}
Relative SVD cutoff	ρ	10^{-4}
Recorded-data sampling / capacity	`log_every_n_cycles`, rows	every callback (1), 120000
Run duration / stop	`experiment_duration`	0 s: indefinite until `e+Enter`
Gripper grasp	width, speed, force	0.066 m, 0.050 m/s, 60 N
Tool-mount rotational play	observed pad-mount clearance	approximately $\pm 2^\circ$ about y_{EE}
Manual-guidance damping	`manual_guidance_damping`	0.5 N m s/rad
Joint-side collision threshold	acceleration/nominal	140 N m on each axis
Cartesian collision threshold	acceleration/nominal	240 N or N m on each channel

Appendix D

Supporting Results

Two of the items here resolve a summary statement of Chapter 5 into the settings and repetitions behind it. The third compares the primary response metric with the supporting one.

D.1 Conditions Classified as Tilted by TCP Height

Of the 55 tested parameter settings, 49 satisfied the TCP-height flatness criterion. Table D.1 lists the six that did not, with the excess TCP height above the flat-tool height.

Table D.1: Conditions classified as tilted by TCP height.

Condition	Excess height [mm]	$\Delta\theta_{\text{align}}$ [deg]	$\Delta\theta_{\text{set}}$ [deg]
Case B, 50 N m/rad about t_1	3.6	+2.38	+2.73
Case D, −20 mm at +10° about t_1	5.3	+0.70	+0.99
Case D, −40 mm at +10° about t_1	6.1	−0.06	+0.18
Case D, +20 mm at −10° about t_1	5.2	+4.43	−5.01
Case D, +40 mm at −10° about t_1	9.3	−0.13	−0.09
Case E, surface frame at +10° about t_1	5.9	−0.13	+0.13

The alignment angle can carry a common offset from the calibrated tool normal. Because the same calibration is used for all runs, this offset is common to the evaluation chain, but it does not necessarily appear as an identical additive angular offset in every orientation.

The primary signed set-up rotation does not use the calibrated tool normal; it is obtained from the measured end-effector motion and then resolved in the surface frame, which makes it the more direct quantity for comparing the controller-induced rotation between settings.

D.2 Per-Setting Spread of the Case-D Lever Positions

Figure D.1 shows the four Case-D command conditions on one pair of axes with the standard deviation over the repetitions of each setting. The chapter figure separates the two commanded axes for readability and plots the means alone; Figure D.1 is retained here because it is the only place the per-setting spread of these lever positions is shown.

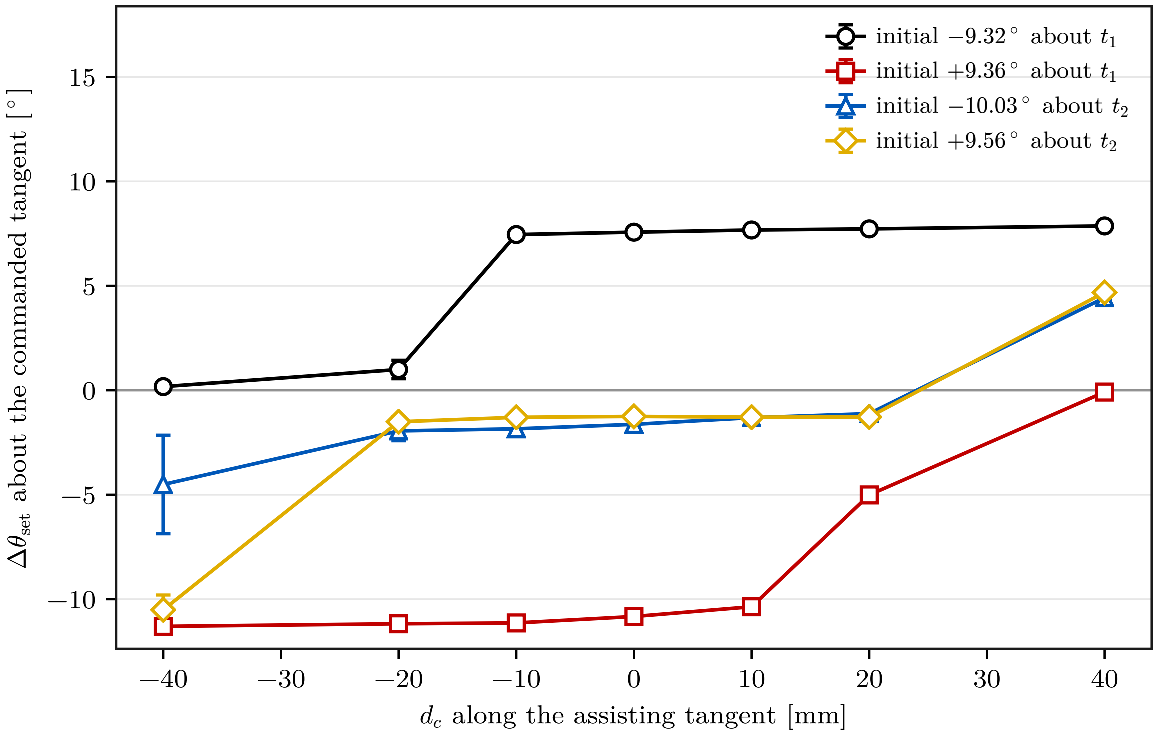


Figure D.1: Case-D set-up rotation with the spread over repetitions.

D.3 Consistency of the Two Angular Quantities

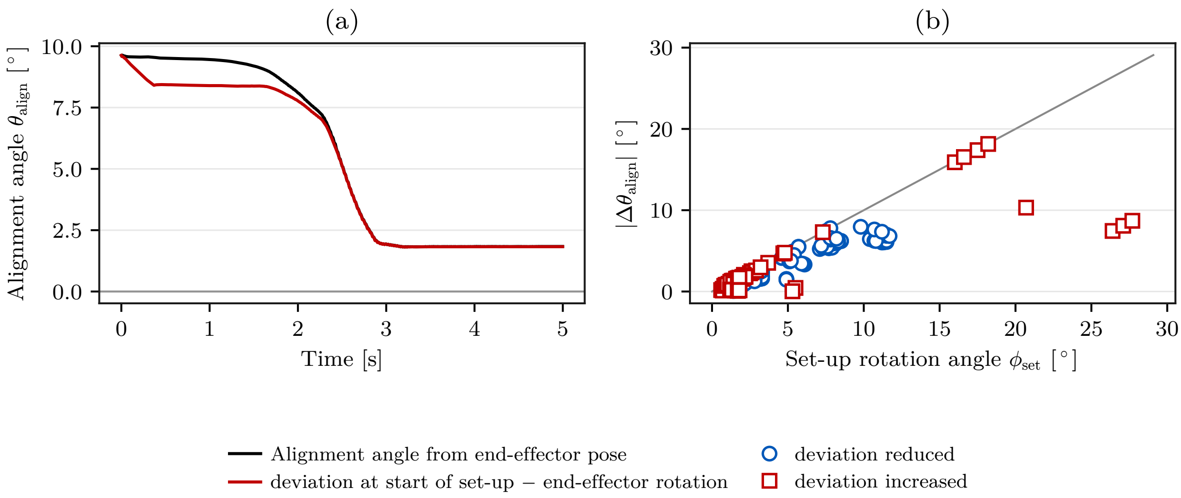


Figure D.2: Alignment angle against end-effector rotation.

Figure D.2 is a check on the choice of response metric. The signed set-up rotation $\Delta\theta_{set}$ is the quantity reported throughout Chapter 5, and the change in the pose-based alignment angle $\Delta\theta_{align}$ supports it. Defined differently, the two are not interchangeable.

Panel (a) plots both quantities for one representative trial. One is the alignment angle calculated from the end-effector pose; the other reconstructs that angle from the end-effector rotation since the start of set-up. Panel (b) reduces the comparison to one point per evaluated trial. Each point carries the magnitude of the change, and the marker distinguishes whether the calculated alignment angle decreased or increased during set-up. Across those trials the two magnitudes are broadly consistent, and the larger disagreements are confined to a small number of conditions.

Both quantities rest on the same assumed tool-to-gripper relation, so their agreement is internal to that assumption rather than an independent measurement of physical tool orientation. The approximately $\pm 2°$ of mounting play was measured unloaded and was not characterised as an uncertainty bound under contact load.